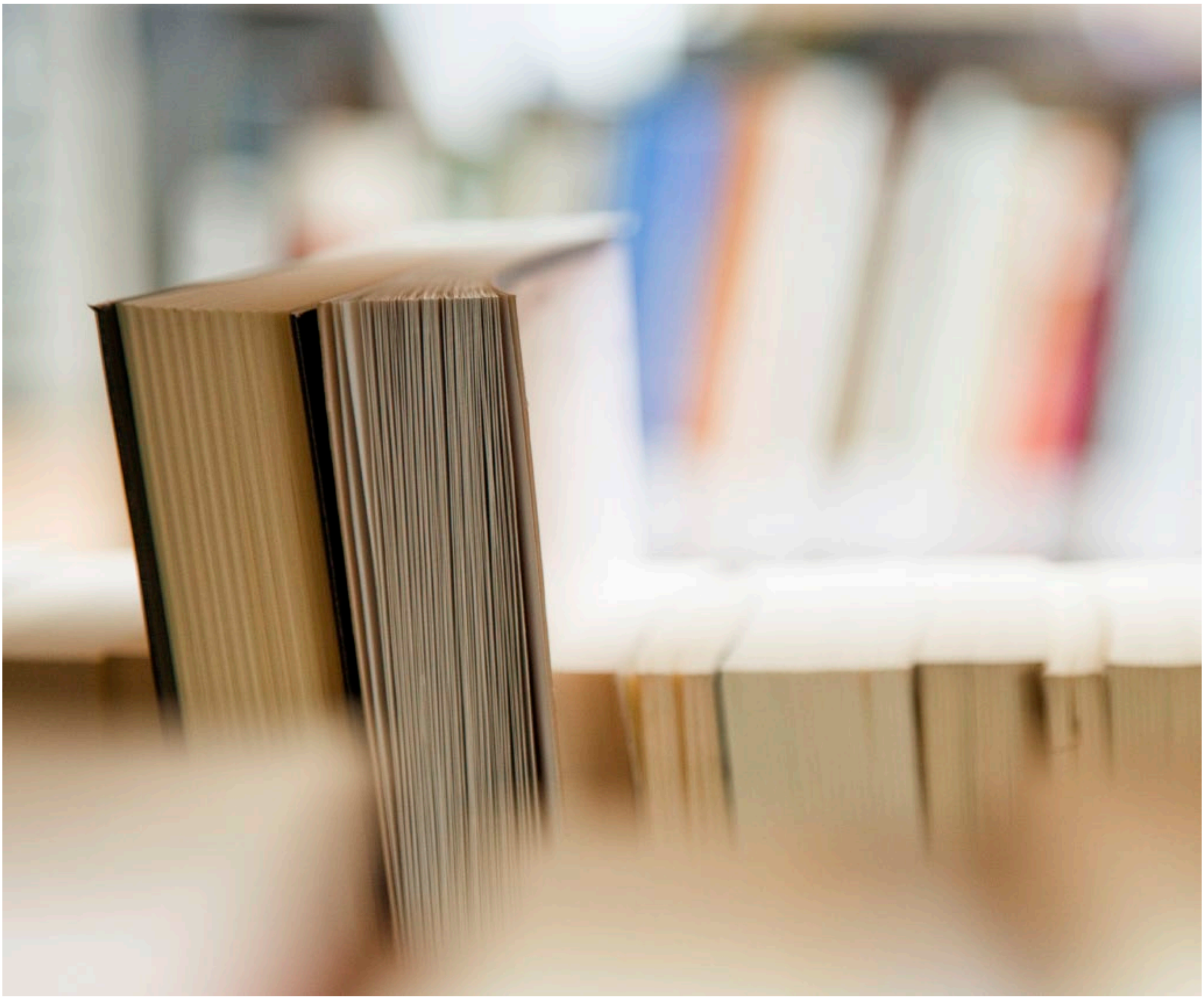

FIRE SAFETY

By Abdool Feroze Hoota CMIOSH

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Abbreviations:

- DSEAR - Dangerous Substances and Explosive Atmospheres Regulations 2002
- HSE - Health and Safety Executive
- HMO - House in Multiple Occupation
- PAT - Portable Appliance Testing
- PEEP - Personal Emergency Evacuation Plan

UNIT 1: INTRODUCTION TO FIRE SAFETY

1.1 What is Fire?

Fire is a chemical reaction that requires three elements to be present: a fuel source, heat, and oxygen. When these three elements combine in the right proportions, a fire can start. This is often represented using the "fire triangle" model.

The fuel is any material that can burn, such as wood, paper, oil, or gas. Heat provides the energy needed to ignite the fuel. Oxygen, usually from the air, sustains the chemical reaction. Without all three elements, a fire cannot begin or continue burning [1].

1.2 The Fire Triangle

The fire triangle is a simple model for understanding the necessary ingredients for most fires. The three sides of the triangle represent heat, fuel and oxygen. The triangle illustrates the interdependence of these ingredients in creating and sustaining fire. Remove any of the three elements, and the fire will go out.

Fuel examples include any flammable substance such as wood, paper, certain chemicals, gasoline or gases like propane. The heat can originate from an open flame, sparks, hot surfaces, the sun, or sometimes even static electricity. Oxygen is supplied by the air in the atmosphere [2].

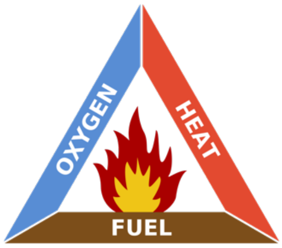

Figure 1: The Fire Triangle

Understanding the fire triangle is an important foundation for fire safety and prevention. By controlling or removing one or more of the three elements, fires can be extinguished or prevented from starting in the first place. This fundamental concept underpins strategies like removing ignition sources, proper storage of flammable materials, and depriving fires of oxygen using fire blankets or certain types of fire extinguishers.

1.3 Fundamentals of Fire Chemistry

Fire is fundamentally a chemical reaction known as combustion. Combustion occurs when a fuel rapidly reacts with oxygen, giving off energy in the form of heat and light. For combustion to occur, the fuel must be heated to its ignition temperature - the point at which it will ignite and continue burning.

Different materials have different ignition temperatures and burn at different rates. This explains why some substances ignite more easily and burn more intensely than others. For example, paper ignites around 233°C while wood ignites around 300°C [4].

As a material burns in air, the combustion reaction converts the fuel and oxygen into new substances, primarily carbon dioxide and water vapor, along with other potential by-products like carbon monoxide, soot, and ash, depending on the fuel composition and the completeness of the combustion.

Incomplete combustion, which occurs when there is insufficient oxygen, can result in additional toxic by-product and heavy smoke. This is an important consideration in fire safety, as the smoke and toxic gases are often more dangerous to people than the flames themselves [5].

1.4 What is Fire Safety?

Fire safety refers to the set of practices intended to reduce the destruction caused by fire. It involves the study of the behaviour, compartmentalisation, suppression and investigation of fire and its related emergencies, as well as the research and development, production, testing and application of mitigating systems [6].

Key aspects of fire safety include:

1. **Fire Prevention**: Measures taken to prevent fires from starting in the first place, such as controlling ignition sources and properly storing flammable materials.
2. **Fire Protection**: Systems and equipment designed to control and extinguish fires that do occur, such as fire alarms, sprinklers, and fire extinguishers.
3. **Means of Escape**: Ensuring that building occupants can safely evacuate in the event of a fire, through measures like fire exits, emergency lighting, and evacuation plans.
4. **Education and Training**: Ensuring that people understand fire risks and know how to respond appropriately to fire emergencies.
5. **Legislation and Regulations**: The laws and rules that govern fire safety requirements in different settings, such as building codes and workplace safety regulations [7].

Effective fire safety requires a comprehensive approach that addresses all of these aspects. The specifics of fire safety measures will vary depending on the setting (e.g., a workplace, a public building, a home), but the fundamental principles apply universally.

1.5 Typical Fire Risks

Fire risks are present in nearly all environments, but some common ones include:

1. **Electrical Hazards**: Overloaded circuits, damaged wires, and faulty electrical equipment can all generate heat and sparks that can start fires.
2. **Heating Equipment**: Space heaters, furnaces, boilers, and other heating devices can ignite nearby combustibles if they overheat or are placed too close to other objects.
3. **Cooking Equipment**: Stoves, ovens, and other cooking appliances are a leading cause of structure fires, particularly when left unattended or used improperly.
4. **Smoking Materials**: Lit cigarettes, cigars, and pipes can easily ignite flammable materials if disposed of carelessly.
5. **Candles and Open Flames**: Candles, lighters, matches, and other open flames are inherently risky and can quickly start a fire if left unattended or placed near combustibles.
6. **Flammable Liquids and Gases**: Gasoline, propane, solvents, and other flammable substances can ignite or explode if not stored and handled properly.
7. **Combustible Storage**: Large quantities of combustible materials like paper, cardboard, or fabrics can allow a fire to spread rapidly if ignited.
8. **Hot Work**: Welding, cutting, grinding, and other tasks that generate sparks or use open flames can easily start fires if proper precautions aren't taken.
9. **Arson**: The deliberate setting of fires is a significant risk in some settings [8].

The specific fire risks present will vary depending on the particular environment and activities involved. A thorough fire risk assessment is necessary to identify and mitigate the specific hazards in any given setting.

1.6 Prevention of Fire

Fire prevention is the most effective way to protect life and property from the dangers of fire. It involves a combination of engineering controls, safe work practices, and public education to eliminate or reduce the chances of a fire starting. Some key aspects of fire prevention include:

1. **Ignition Control**: Identifying and controlling potential sources of ignition, such as open flames, hot surfaces, sparks, and electrical hazards.
2. **Fuel Control**: Reducing the amount of flammable and combustible materials present, and ensuring that any necessary flammables are stored and handled safely.
3. **Safe Work Practices**: Implementing and enforcing rules and procedures to ensure that activities like cooking, hot work, and smoking are conducted safely.

4. **Maintenance**: Regularly inspecting, testing, and maintaining equipment and systems to ensure they are in safe working order and do not pose a fire risk.
5. **Education**: Providing training and information to people about fire risks and safe practices, so they can prevent fires through their own actions.
6. **Design and Construction**: Ensuring that buildings and structures are designed and built in a way that minimizes fire risks and supports safe evacuation if a fire does occur.
7. **Regulatory Compliance**: Ensuring that all fire safety laws, regulations, and standards are followed [9].

Effective fire prevention requires ongoing effort and vigilance. It's not a one-time activity, but a continuous process of identifying and mitigating risks. All members of a community have a role to play in fire prevention, from building managers and employees to residents and the general public.

1.7 Measures to Prevent Fires

There are numerous specific measures that can be taken to prevent fires, depending on the setting and the specific risks present. Some common fire prevention measures include:

1. **Electrical Safety**: Ensuring that electrical systems and equipment are properly designed, installed, used, and maintained to prevent overheating, sparks, and other fire hazards. This includes regular inspections and tests, repairing or replacing faulty equipment, and not overloading circuits [10].
2. **Cooking Safety**: Ensuring that cooking equipment is used properly and safely, such as never leaving cooking unattended, keeping combustibles away from heat sources, and having appropriate fire suppression systems in commercial kitchens [11].
3. **Smoking Policies**: Implementing and enforcing safe smoking policies, such as designating specific smoking areas away from flammable materials, providing appropriate ashtrays, and fully extinguishing smoking materials [12].
4. **Hot Work Permits**: Requiring a permit system for any hot work activities like welding, cutting, or grinding. The permit process ensures that appropriate precautions are taken, such as removing combustibles from the area, having fire extinguishers ready, and monitoring for sparks or smoldering after the work is done [13].
5. **Flammable Storage**: Storing flammable and combustible materials properly, such as in approved containers, in designated areas, and away from ignition sources. Also, limiting the quantities of flammables kept on hand to what is necessary for operations [14].
6. **Housekeeping**: Maintaining a clean and orderly environment, free of excess clutter and combustible waste. This includes regular cleaning, proper disposal of waste, and keeping exits and firefighting equipment clear and accessible [15].
7. **Education and Training**: Providing regular fire safety education and training to employees, residents, students, or other relevant parties. This includes information on fire risks, prevention measures, evacuation procedures, and how to use fire extinguishers [16].

These are just a few examples of the many potential fire prevention measures. The specific measures needed will depend on the results of a thorough fire risk assessment for the particular setting. The key is to systematically identify and control all potential fire hazards.

1.8 Fixed Fire Fighting Systems

Fixed firefighting systems are permanently installed in buildings or structures to detect and suppress fires. They are designed to activate automatically in the event of a fire, controlling or extinguishing the fire without the need for human intervention. Some common types of fixed firefighting systems include:

1. **Sprinkler Systems**: These consist of a network of water pipes and sprinkler heads that are activated by heat. When a fire occurs, the heat breaks the glass bulb or melts the fusible link in the sprinkler head, allowing water to flow out and suppress the fire [17].
2. **Gaseous Suppression Systems**: These use inert gases (like nitrogen or argon) or chemical agents (like FM-200) to displace oxygen or interfere with the chemical reaction of a fire. They are often used in areas where water damage from sprinklers would be problematic, such as in computer rooms or museums [18].
3. **Foam Systems**: These are designed for flammable liquid fires. They deliver a foam solution that spreads across the surface of the liquid, suppressing vapours and smothering the fire [19].

4. **Dry Chemical Systems**: These use a powder composed of very small particulates of chemicals such as sodium bicarbonate, potassium bicarbonate, or monoammonium phosphate. The powder is propelled by compressed gas and works by interrupting the chemical reaction of the fire [20].
5. **Water Mist Systems**: These use very fine water sprays to control or suppress fires. The small water droplets create a large surface area for heat absorption and also evaporate quickly, displacing oxygen [21].

The type of fixed firefighting system used in a particular setting will depend on factors like the type of fire risk present, the potential consequences of a fire, the need to protect sensitive equipment or materials, and building codes or insurance requirements.

Fixed systems are designed to activate quickly and automatically, providing an immediate response to a fire. However, they do require regular inspection, testing, and maintenance to ensure they will function properly when needed. They are an important part of a comprehensive fire safety strategy, but not a substitute for other measures like fire prevention, manual firefighting equipment, and safe evacuation procedures.

1.9 Automatic Water Sprinkler Systems

Automatic water sprinkler systems are one of the most common and effective types of fixed firefighting systems. They consist of a network of pipes filled with water under pressure, with sprinkler heads installed at regular intervals. The sprinkler heads are designed to open automatically when they detect a certain level of heat, allowing water to flow out and suppress the fire.

The key components of an automatic sprinkler system include:

1. **Water Supply**: This can be from the municipal water main, a gravity tank, a pressurized tank, or a pump.
2. **Piping**: The pipes are usually made of steel and are sized to deliver enough water flow and pressure to the sprinkler heads.
3. **Sprinkler Heads**: These are the devices that detect heat and release water. They contain a heat-sensitive element (usually a glass bulb filled with liquid or a fusible metal link) that breaks at a specific temperature, allowing water to flow [22].

There are several types of sprinkler systems, including:

1. **Wet Pipe Systems**: The pipes are constantly filled with water. This is the most common type.
2. **Dry Pipe Systems**: The pipes are filled with pressurized air or nitrogen. When a sprinkler head opens, the air escapes, allowing water to flow into the pipes and out of the sprinkler head. These are used in areas subject to freezing.
3. **Pre-Action Systems**: These are similar to dry pipe systems, but also require a separate detection system to activate. The detection system opens a valve that allows water into the pipes. These are used where accidental activation could cause significant damage, such as in museums or computer rooms [23].

Sprinkler systems are highly effective at controlling fires. They react quickly, often before the fire can grow large, and they apply water directly to the fire, minimizing water damage in unaffected areas. Studies have shown that sprinklers operate in 91% of fires large enough to activate them, and when they do operate, they are effective 96% of the time in controlling the fire [24].

However, sprinklers are not designed to completely extinguish fires. They are intended to control the fire, preventing it from growing and spreading, until the fire department arrives to fully extinguish it. Also, sprinklers don't operate in every fire. If the fire is too small to generate enough heat, or if the fire is in an area not protected by sprinklers (like an exterior area), the sprinklers may not activate.

Sprinkler systems are required by building codes in many types of structures, particularly in new commercial and high-occupancy residential buildings. They are a crucial part of the fire protection system in these types of properties.

1.10 Portable Fire Fighting Equipment

Portable fire extinguishers are a first line of defence against small fires. They are designed to be used by people with minimal training to control or extinguish fires in their early stages, before they grow too large for the extinguisher to handle.

There are several types of portable fire extinguishers, each designed for specific types of fires:

1. **Water Extinguishers**: These are suitable for Class A fires (ordinary combustibles like wood, paper, and fabric). They work by cooling the fuel below its ignition temperature.
2. **Foam Extinguishers**: These can be used on Class A and Class B fires (flammable liquids and gases). They work by smothering the fire and separating the fuel from the oxygen.
3. **Dry Powder Extinguishers**: These are suitable for Class A, B, and C fires (energized electrical equipment). They work by interrupting the chemical reaction of the fire.
4. **Carbon Dioxide (CO2) Extinguishers**: These are used for Class B and C fires. They work by displacing oxygen and cooling the fuel.
5. **Wet Chemical Extinguishers**: These are designed for Class K fires (cooking oils and fats). They work by creating a soapy foam that cools the fuel and prevents re-ignition [25].

Portable extinguishers are rated by the type and size of fire they can handle. The rating consists of a letter (the class of fire) and a number (the size of fire). For example, a 2A:10B:C rating means the extinguisher can handle a Class A fire of 2 square feet, a Class B fire of 10 square feet, and is suitable for use on Class C fires.

It's important to use the right type of extinguisher for the fire. Using the wrong type can be ineffective or even dangerous. For example, using a water extinguisher on an electrical fire could cause electrocution.

Barking and Dagenham College is in the process of transitioning to a new type of fire extinguisher, the Firexo 6 Litre Fire Extinguisher. This innovative extinguisher is suitable for all types of fires, including Class A, B, C, D, Electrical, F, and Lithium-ion Battery Fires.

The Firexo extinguisher uses a unique, globally patented technology that combines the power of water with the ability to cool, contain, and suppress fires involving solids, liquids, gases, metals, and electricity. It eliminates the confusion of selecting the right extinguisher for the type of fire, making it safer and easier to use in an emergency situation.

The Firexo extinguishing agent is also more environmentally friendly compared to traditional powder extinguishers. It is made from natural ingredients and is non-toxic and biodegradable.

As the college gradually replaces its existing extinguishers with the Firexo 6 Litre model, it will enhance the fire safety preparedness across the campus. This universal extinguisher will provide fast and effective fire suppression capabilities for any type of fire emergency that may arise.

1.11 Siting of Portable Extinguishing Equipment

The effectiveness of portable fire extinguishers depends not just on having the right type and size of extinguisher, but also on having them located properly. The siting of extinguishers should follow certain guidelines:

1. **Visibility**: Extinguishers should be clearly visible. If they are in cabinets or recesses, there should be clear signs indicating their location [26].
2. **Accessibility**: Extinguishers should be readily accessible, near normal paths of travel and near exits. They should not be blocked by equipment or storage [27].
3. **Distribution**: Extinguishers should be evenly distributed, so that the travel distance to an extinguisher is no more than 30 meters for Class A fires and 15 meters for Class B fires [28].
4. **Height**: Extinguishers should be mounted at a height where they can be easily reached. The handle or carrying device should be no more than 1.5 meters from the floor [29].
5. **Environment**: Extinguishers should be protected from physical damage and from environmental conditions that could impair their operation, like corrosion or freezing [30].
6. **Signage**: There should be clear signs indicating the location of extinguishers, particularly if they are not immediately visible [31].

7. **Risk Assessment**: The type, size, and location of extinguishers should be based on a risk assessment of the area. Areas with higher risk or specific hazards may need additional or specialized extinguishers [32].

Regular inspections should be conducted to ensure that extinguishers are in their designated locations, are accessible, and are in good operating condition. Staff should also be trained on where extinguishers are located and how to use them.

While portable extinguishers are an important part of a fire safety plan, they are not a substitute for other measures like fire prevention, automatic suppression systems, and safe evacuation procedures. They are intended as a first response to small fires, not for fighting large or advanced fires.

1.12 Fire Alarm and Detection Systems

Fire alarm and detection systems are designed to quickly identify a developing fire and alert building occupants and emergency responders. These systems can be activated manually (such as by pulling a fire alarm pull station) or automatically (such as by smoke detectors, heat detectors, or water flow in a sprinkler system).

The key components of a fire alarm system include:

1. **Initiating Devices**: These are the devices that detect fires and activate the alarm system. They include smoke detectors, heat detectors, flame detectors, and manual pull stations [33].
2. **Notification Appliances**: These are the devices that alert people to the fire. They include horns, sirens, bells, and strobe lights [34].
3. **Control Panel**: This is the brain of the system. It receives signals from the initiating devices and activates the notification appliances. It also sends signals to other systems like the sprinkler system, HVAC system, and elevator system [35].
4. **Power Supply**: The system must have a reliable power supply, usually with a battery backup in case of power failures [36].
5. **Wiring**: The components of the system are connected by a network of wires or fiber optic cables [37].

There are two main types of fire alarm systems:

1. **Conventional Systems**: In these systems, the building is divided into zones, and the control panel identifies the zone where the alarm originated but not the specific device [38].
2. **Addressable Systems**: In these systems, each device has a unique identifier, and the control panel can pinpoint the exact device that activated the alarm [39].

Fire alarm systems are required by building codes in most commercial and high-occupancy residential buildings. They must be designed, installed, and maintained in accordance with strict standards and regulations.

Regular testing and maintenance of fire alarm systems is critical to ensure they will function properly in an emergency. This includes annual inspections and tests by qualified technicians, as well as regular checks by building staff.

Fire alarm and detection systems are a crucial part of the overall fire safety strategy in a building. They provide early warning of a fire, allowing occupants to evacuate safely and firefighters to respond quickly. However, they are not a standalone solution. They must be integrated with other systems like sprinklers and ventilation systems, and must be backed up by well-practiced evacuation procedures.

This concludes Unit 1: Introduction to Fire Safety. The next unit will delve into the specifics of Fire Safety Legislation.

UNIT 2: FIRE SAFETY LEGISLATION

2.1 DSEAR Regulations

The Dangerous Substances and Explosive Atmospheres Regulations 2002 (DSEAR) are a set of UK regulations that require employers to control the risks to safety from fire, explosions and similar events in the workplace. These regulations were introduced to comply with the EU's ATEX directives [40].

DSEAR applies to all workplaces where dangerous substances are present, or could be present, in quantities that could pose a risk. This includes substances used directly in work activities (like solvents or fuels), substances generated by work activities (like dusts or fumes), and substances that occur naturally (like methane in mines) [41].

Under DSEAR, employers are required to:

1. **Assess Risks**: Carry out a risk assessment of any work activities involving dangerous substances [42].
2. **Eliminate or Reduce Risks**: Where possible, eliminate risks from dangerous substances. If elimination is not possible, reduce risks as much as is reasonably practicable [43].
3. **Control Risks**: Apply control measures to prevent fires, explosions, or similar events, and mitigate their effects if they do occur [44].
4. **Prepare Emergency Plans**: Prepare plans and procedures to deal with accidents, incidents and emergencies involving dangerous substances [45].
5. **Provide Information and Training**: Provide employees with information, instruction and training on the risks from dangerous substances and the control measures in place [46].
6. **Classify Areas**: Where necessary, classify places where explosive atmospheres may occur into zones and mark the zones where necessary [47].
7. **Use Suitable Equipment**: Ensure that equipment and protective systems are suitable for use in their designated zones [48].

The goal of DSEAR is to protect workers and others from the risks of fire, explosion and similar events arising from dangerous substances in the workplace. It is part of the broader system of health and safety regulations in the UK.

2.2 What are Dangerous Substances?

Under DSEAR, a dangerous substance is any substance or preparation that could create a risk to people from fires and explosions or similar events, such as the release of gases under pressure. This includes [49]:

1. **Substances or mixtures classified as explosive, oxidizing, extremely flammable, highly flammable, or flammable under the Classification, Labelling and Packaging Regulation (CLP)**.
2. **Substances and mixtures that are self-reactive or emit flammable gases in contact with water**.
3. **Chemically unstable gases**.
4. **Any dust, whether in the form of powder or fibrous materials, which can form an explosive mixture with air**.
5. **Any other substance, mixture or material that, because of its properties or the way it is used, could create a risk to the safety of people from fires and explosions**.

Examples of dangerous substances include petrol, liquefied petroleum gas (LPG), paints, varnishes, solvents, and dust from machining and sanding operations.

It's important to note that it's not just the properties of a substance that determine whether it's dangerous under DSEAR, but also the way it's used and the quantities involved. For example, a small quantity of a flammable solvent used in a well-ventilated area might not pose a significant risk, but a large quantity of the same solvent used in an enclosed space could create an explosive atmosphere.

Employers are responsible for identifying the dangerous substances in their workplace and assessing the risks they pose. This is the first step in complying with DSEAR and ensuring the safety of employees and others who may be affected.

2.3 What are the Requirements of DSEAR?

The Dangerous Substances and Explosive Atmospheres Regulations 2002 (DSEAR) place several requirements on employers to ensure the safety of their employees and others who may be affected by work with dangerous substances. The main requirements are:

1. **Risk Assessment**: Employers must carry out a suitable and sufficient assessment of the risks to employees and others from dangerous substances. The risk assessment should consider the hazardous properties of the substances, the way they are used or produced, and the potential for them to create fires, explosions, or similar events [50].
2. **Elimination or Reduction of Risks**: Where reasonably practicable, employers must eliminate or reduce the risks from dangerous substances. This could involve replacing a dangerous substance with a less hazardous one, or changing a process to reduce the quantity of dangerous substances produced or used [51].
3. **Control of Risks**: Where risks cannot be eliminated, employers must take measures to control them. This could include measures to prevent the release of dangerous substances, to control the release at source, to prevent the formation of an explosive atmosphere, and to mitigate the effects of a fire or explosion [52].
4. **Emergency Arrangements**: Employers must prepare emergency plans and procedures to deal with incidents involving dangerous substances. Employees must be provided with relevant information, instruction and training [53].
5. **Information and Training**: Employees must be provided with suitable and sufficient information, instruction and training on the risks from dangerous substances and the control measures in place [54].
6. **Classification of Hazardous Areas**: Employers must identify and classify areas where hazardous explosive atmospheres may occur. These areas must be marked with the appropriate warning signs [55].
7. **Selection of Equipment and Protective Systems**: Equipment and protective systems used in hazardous areas must be suitable for their intended use [56].
8. **Maintenance of Control Measures**: All control measures, including equipment and protective systems, must be maintained in an efficient state, in efficient working order, and in good repair [57].

Compliance with DSEAR is not a one-time event but an ongoing process. Employers must regularly review their risk assessments and control measures to ensure they remain effective. They must also investigate any accidents or incidents that occur and take necessary measures to prevent recurrence.

Failure to comply with DSEAR can result in enforcement action by the Health and Safety Executive (HSE), including prosecution in serious cases. More importantly, non-compliance can put the lives of employees and others at risk.

2.4 Where Does DSEAR Apply?

DSEAR applies to virtually all workplaces in the UK where dangerous substances are present, or could be present, in quantities that could pose a risk. This includes [58]:

1. **Industrial premises** such as factories, warehouses, and chemical plants where dangerous substances are used or produced.
2. **Commercial premises** such as shops, offices, and hotels where dangerous substances may be present, for example, in cleaning products or fuel for backup generators.
3. **Construction sites** where dangerous substances such as fuels, paints, and solvents are used.
4. **Offshore installations** where dangerous substances such as oil and gas are extracted and processed.
5. **Hospitals and other healthcare facilities** where dangerous substances such as oxygen, anaesthetic gases, and disinfectants are used.
6. **Schools and colleges** where dangerous substances may be used in science laboratories and art studios.
7. **Agriculture** where dangerous substances such as fertilizers, pesticides, and fuels are used.
8. **Transport facilities** such as airports, docks, and railway stations where dangerous substances may be present in cargo or fuel.
9. **Waste facilities** where dangerous substances may be present in the waste being handled.
10. **Any other workplace** where dangerous substances are present or produced.

There are a few exemptions from DSEAR, such as for substances that only present a slight risk to health, or that are already covered by other specific legislation (like radioactive substances or explosives). However, these exemptions are limited and most workplaces where dangerous substances are present will be subject to DSEAR.

It's important to note that DSEAR applies not just to the use of dangerous substances, but also to their storage, production, and disposal. It also applies to work activities that could lead to the production of dangerous substances, such as dust from machining operations.

Employers are responsible for determining whether DSEAR applies to their workplace and for complying with its requirements if it does. If in doubt, employers should seek advice from a competent health and safety professional or from the Health and Safety Executive (HSE).

2.5 Fire Extinguishers

Fire extinguishers are a crucial piece of fire safety equipment and are required in most workplaces under the Regulatory Reform (Fire Safety) Order 2005 [59]. They provide a first line of defence against small fires, allowing trained individuals to control or extinguish a fire in its early stages before it grows too large.

The type and number of fire extinguishers required will depend on the size and use of the premises, the materials present, and the findings of the fire risk assessment. The main types of fire extinguishers include [60]:

1. **Water**: Suitable for Class A fires involving solid materials such as wood, paper, and textiles. Not suitable for use on live electrical equipment or flammable liquids.
2. **Foam**: Suitable for Class A and B fires involving flammable liquids such as petrol, diesel, and oil.
3. **Powder**: Suitable for Class A, B, and C fires involving gases. Can also be used on live electrical equipment.
4. **Carbon Dioxide (CO2)**: Suitable for Class B fires and electrical fires. Not suitable for use in confined spaces due to the risk of asphyxiation.
5. **Wet Chemical**: Specifically designed for Class F fires involving cooking oils and fats.

Employers must ensure that fire extinguishers are [61]:

1. **Suitable** for the type and size of fire that may occur.
2. **Located in appropriate positions**, easily accessible, and not obstructed by other equipment or storage.
3. **Properly maintained** and serviced by a competent person at least annually.
4. **Accompanied by appropriate signage** indicating their location and type.
5. **Staff are trained** in their use.

The Firexo 6 Litre Fire Extinguisher, which Barking and Dagenham College is transitioning to, is a unique solution that can be used on all types of fires. This eliminates the need for multiple types of extinguishers and reduces confusion in an emergency situation.

2.6 Fire Safety Signs

Fire safety signs are required to provide information to employees and others about the fire precautions in the workplace. They are covered by the Health and Safety (Safety Signs and Signals) Regulations 1996 [62].

There are several types of fire safety signs [63]:

1. **Fire Action Notices**: These provide instructions on what to do in case of fire. They should be posted in conspicuous positions and be easy to read.
2. **Fire Exit Signs**: These indicate the route to the nearest fire exit. They should be clearly visible and, where necessary, illuminated.
3. **Fire Door Signs**: These indicate the location of fire doors and provide instructions on keeping them closed or unobstructed.
4. **Fire Extinguisher Signs**: These indicate the location and type of fire extinguishers.
5. **Warning Signs**: These warn of specific hazards, such as flammable materials or electrical risk.

All fire safety signs should be [64]:

1. **Clear and legible**.
2. **Easily understood** (using pictograms where necessary).
3. **Constructed of durable materials**.
4. **Securely fixed** and properly maintained.
5. **Positioned at an appropriate height** and location for easy visibility.

Employers must ensure that all necessary fire safety signs are provided in the workplace and that employees understand their meaning. This is particularly important in larger premises or where there are significant numbers of non-English speakers.

2.7 Fire Alarm Systems

Fire alarm systems are designed to quickly detect a fire and alert occupants, allowing them to evacuate safely. They are required in most workplaces under the Regulatory Reform (Fire Safety) Order 2005 [65].

The type of fire alarm system required will depend on the size and use of the premises, the materials present, and the findings of the fire risk assessment. The main types of fire alarm systems are [66]:

1. **Conventional Systems**: The premises are divided into zones, and the fire alarm panel indicates in which zone the fire has been detected.
2. **Addressable Systems**: Each detector and call point has its own unique address. The fire alarm panel can pinpoint exactly which detector or call point has been activated.
3. **Wireless Systems**: These work similarly to conventional or addressable systems but without the need for wiring.

Employers must ensure that fire alarm systems are [67]:

1. **Suitable for the premises** and the risk of fire.
2. **Properly designed, installed, and commissioned** by a competent person.
3. **Tested weekly** to ensure they are working properly.
4. **Serviced at least every six months** by a competent person.
5. **Not obstructed** or covered.
6. **Staff are trained** in their use and know how to respond when the alarm sounds.

Employers must also have a procedure in place for the safe evacuation of the premises in case of fire. This should include the designation of fire wardens or marshals to assist in the evacuation and to check that all areas have been cleared.

2.8 Emergency Lighting

Emergency lighting is designed to illuminate escape routes and help people find firefighting equipment in the event of a power failure during a fire. It is required in most workplaces under the Regulatory Reform (Fire Safety) Order 2005 [68].

Emergency lighting should be [69]:

1. **Provided in all escape routes**, open areas larger than 60m2, and in all areas where illumination is necessary for safety.
2. **Adequate to allow safe evacuation** of the premises.
3. **Powered by a source independent** from normal lighting circuits, such as a backup battery or generator.
4. **Tested monthly** (a short functional test) and annually (a full duration test).
5. **Properly maintained** and repaired immediately if found to be faulty.

Employers must ensure that emergency lighting is provided where necessary and that it is properly maintained and tested. They must also ensure that escape routes are kept clear and unobstructed at all times.

2.9 Fire Safety Training

Employers are required under the Regulatory Reform (Fire Safety) Order 2005 to provide employees with adequate fire safety training [70]. This training should be provided on induction and repeated periodically, usually annually.

Fire safety training should cover [71]:

1. **The fire risks in the premises**.
2. **The fire safety measures in the premises**, including the location and use of fire extinguishers, fire alarms, and emergency lighting.
3. **The action to take in case of fire**, including how to raise the alarm and evacuate safely.
4. **The identity of the fire wardens or marshals**.
5. **Any special procedures**, such as assisting disabled persons or shutting down critical equipment.

Additional training should be provided for fire wardens or marshals, including [72]:

1. **Detailed knowledge** of the fire safety measures in the premises.
2. **How to use fire extinguishers** and other firefighting equipment.
3. **How to conduct a sweep** of the premises to ensure everyone has evacuated.
4. **How to liaise with the fire and rescue service** when they arrive.

Employers must keep records of all fire safety training provided, including the date, content, and names of those who attended. They must also ensure that employees understand the training and know what to do in case of fire.

Effective fire safety training is crucial in ensuring that employees can respond quickly and safely in the event of a fire. It can help to prevent panic, minimize confusion, and ultimately save lives.

This concludes Unit 2: Fire Safety Legislation. The next unit will cover Fire Risk Assessment.

UNIT 3: FIRE RISK ASSESSMENT

3.1 What is a Fire Risk Assessment?

A fire risk assessment is a systematic examination of a premises, the activities carried out there and the likelihood that a fire could start and cause harm to those in and around the premises [73].

It's a legal requirement under the Regulatory Reform (Fire Safety) Order 2005 for virtually all non-domestic premises in England and Wales, including workplaces, commercial properties, and public buildings [74].

The purpose of a fire risk assessment is to [75]:

1. **Identify the fire hazards**.
2. **Identify people at risk**.
3. **Evaluate, remove or reduce the risks**.
4. **Record the findings, prepare an emergency plan and provide training**.
5. **Review and update the fire risk assessment regularly**.

The fire risk assessment should be carried out by a competent person, which is defined as someone with enough training and experience or knowledge and other qualities to enable them to properly assist in undertaking the preventive and protective measures [76].

For simple, low-risk premises, this could be someone within the organisation who has received appropriate training. For larger, more complex, or high-risk premises, it may be necessary to engage a specialist fire risk assessor.

The findings of the fire risk assessment should be recorded if the organization has five or more employees, if the premises are licensed, or if an alterations notice is in force [77]. However, it's good practice to record the findings regardless of the size of the organization or premises.

The fire risk assessment should be reviewed regularly and updated if there are any significant changes to the premises, the activities carried out there, or the number of people present.

3.2 Identifying Fire Hazards

The first step in a fire risk assessment is to identify potential sources of ignition, fuel, and oxygen. These are the three elements needed for a fire to start and spread [78].

Sources of Ignition

These are things that can start a fire, such as [79]:

1. **Smoking materials**, such as cigarettes, matches, and lighters.
2. **Electrical equipment**, such as overloaded sockets, faulty wiring, and portable heaters.
3. **Cooking equipment**, such as ovens, toasters, and microwaves.
4. **Heating equipment**, such as boilers, furnaces, and portable heaters.
5. **Naked flames**, such as candles, gas fires, and Bunsen burners.
6. **Friction**, such as from mechanical equipment or processes.
7. **Sparks**, such as from welding or grinding.
8. **Chemical reactions**, such as from mixing incompatible substances.

Sources of Fuel

These are materials that can burn, such as [80]:

1. **Paper and cardboard**, such as files, boxes, and packaging.
2. **Wood**, such as furniture, fixtures, and fittings.
3. **Textiles**, such as curtains, carpets, and clothing.
4. **Flammable liquids**, such as petrol, diesel, and solvents.
5. **Flammable gases**, such as propane and butane.
6. **Plastics**, such as packaging, furniture, and fittings.
7. **Rubber**, such as tyres and conveyor belts.
8. **Waste materials**, such as rubbish and recycling.

Sources of Oxygen

Oxygen is present in the air, but additional sources can make a fire burn more intensely, such as [81]:

1. **Oxidising materials**, such as oxygen cylinders and oxidising chemicals.
2. **Ventilation systems**, such as air conditioning and extraction systems.
3. **Open windows and doors**.

The fire risk assessor should walk around the premises and identify any potential sources of ignition, fuel, and oxygen. They should also consider any processes or activities that might introduce additional risks, such as hot work, cooking, or the storage of flammable materials.

3.3 Identifying People at Risk

The next step in a fire risk assessment is to identify who might be harmed if a fire starts. This includes people who are in the premises and those who might be affected if a fire spreads to neighbouring premises [82].

People in the Premises

This includes [83]:

1. **Employees**, including full-time, part-time, and temporary staff.
2. **Visitors**, such as customers, clients, and contractors.
3. **Members of the public**, such as in shops, museums, and libraries.
4. **Vulnerable persons**, such as children, elderly, disabled, or those with language difficulties.
5. **People working alone or in isolated areas**.
6. **People working outside normal hours**, such as cleaners and security staff.

People in Neighbouring Premises

This includes [84]:

1. **Occupants of adjacent buildings**, such as in terraced properties or shared office blocks.
2. **Passers-by**, such as pedestrians and road users.

The fire risk assessor should consider the characteristics and behaviours of the people who might be at risk, such as [85]:

1. **Their familiarity with the premises** and the fire safety measures in place.
2. **Their ability to detect and respond to a fire**, such as those with hearing or visual impairments.
3. **Their ability to evacuate safely and quickly**, such as those with mobility impairments or young children.
4. **Their likelihood to take appropriate action** in case of fire, such as those who are under the influence of alcohol or drugs.

The fire risk assessor should also consider the numbers of people who might be present at different times of the day and night, and in different parts of the premises.

3.4 Evaluate, Remove, Reduce and Protect from Risk

Once the fire hazards and the people at risk have been identified, the next step is to evaluate the level of risk and decide whether the existing fire safety measures are adequate or if more needs to be done [86].

Evaluating the Risk

The fire risk assessor should consider [87]:

1. **The likelihood of a fire starting**, based on the identified sources of ignition and fuel.
2. **The consequences if a fire were to start**, based on the people who might be harmed and the potential for the fire to spread.
3. **The adequacy of the existing fire safety measures**, such as fire alarms, fire extinguishers, emergency lighting, and escape routes.

Removing or Reducing the Risks

Where possible, the risks should be removed completely. Where this is not possible, they should be reduced as far as is reasonably practicable [88]. This can be done by:

1. **Removing sources of ignition**, such as banning smoking, replacing faulty electrical equipment, and ensuring cooking equipment is properly maintained.
2. **Removing or reducing sources of fuel**, such as keeping the workplace clean and tidy, storing flammable materials properly, and using non-flammable substitutes where possible.
3. **Removing or reducing sources of oxygen**, such as closing windows and doors, controlling ventilation systems, and not storing oxidising materials.
4. **Replacing highly flammable materials** with less flammable ones.
5. **Separating flammable materials** from sources of ignition and oxygen.
6. **Reducing the number of people in the premises**, particularly in high-risk areas.

Protecting from Risk

Where the risks cannot be removed or reduced any further, measures should be taken to protect people from the effects of a fire [89]. This can include:

1. **Providing fire detection and warning systems**, such as smoke alarms and fire alarms.
2. **Providing firefighting equipment**, such as fire extinguishers and fire blankets.
3. **Providing safe escape routes**, such as fire exits, emergency lighting, and clear signage.
4. **Providing fire resisting construction**, such as fire doors, fire walls, and fire-resistant materials.
5. **Providing fire safety information and training** to employees and others.

The fire risk assessor should document their findings and the measures taken to remove, reduce, or protect from the risks. They should also set a date for reviewing the assessment, particularly if there are any significant changes planned.

3.5 Checklist: Evaluate, Remove, Reduce and Protect from Risks

To help ensure that all fire risks have been properly evaluated and appropriate measures have been taken, the fire risk assessor can use a checklist like the one below [90]:

Evaluating the Risks

- Have all potential sources of ignition been identified?
- Have all potential sources of fuel been identified?
- Have all potential sources of oxygen been identified?
- Have all people who might be at risk been identified?
- Have the risks been evaluated considering the likelihood and consequences of a fire?
- Are the existing fire safety measures adequate?

Removing or Reducing the Risks

- Can any sources of ignition be removed or replaced?
- Can any sources of fuel be removed or reduced?
- Can any sources of oxygen be removed or reduced?
- Can any flammable materials be replaced with less flammable ones?
- Can any flammable materials be separated from sources of ignition and oxygen?
- Can the number of people in the premises be reduced, particularly in high-risk areas?

Protecting from Risk

- Is there an adequate fire detection and warning system?
- Is there adequate firefighting equipment?
- Are there adequate escape routes?
- Are the escape routes clearly signed and illuminated?
- Are there adequate fire doors and fire-resistant construction?
- Have employees and others been provided with fire safety information and training?

Recording, Planning and Training

- Have the findings been recorded?
- Has an emergency plan been prepared?
- Have employees and others been informed of the risks and what to do in case of fire?
- Has a date been set for reviewing the assessment?

This checklist is not exhaustive and should be adapted to suit the specific characteristics and risks of the premises being assessed. However, it provides a useful starting point for ensuring that all key aspects of fire safety have been considered.

3.6 Record, Plan, Inform, Instruct and Train

Once the fire risk assessment has been completed, the findings should be recorded, an emergency plan should be prepared, and employees and others should be informed, instructed, and trained [91].

Recording the Findings

If the organisation has five or more employees, the premises are licensed, or an alterations notice is in force, the significant findings of the fire risk assessment must be recorded [92]. This should include:

1. **The fire hazards that have been identified**.
2. **The people who might be at risk**.
3. **The measures that are in place, or will be put in place, to remove, reduce, or protect from the risks**.
4. **Any group of people identified as being especially at risk**.

Even if it's not legally required, it's good practice to record the findings of the fire risk assessment.

Emergency Plans

An emergency plan should be prepared, tailored to the specific characteristics and risks of the premises [93]. This should include:

1. How people will be warned if there is a fire.
2. What staff and others should do if they discover a fire.
3. How the evacuation of the premises should be carried out.
4. Where people should assemble after they have left the premises.
5. Procedures for checking whether the premises have been evacuated.
6. Identification of key escape routes.
7. Arrangements for fighting the fire.
8. The duties and identity of staff who have specific responsibilities in case of fire.
9. Arrangements for the safe evacuation of people identified as being especially at risk.
10. Contingency plans for when life safety systems, such as fire detection and alarm systems or sprinklers, are out of order.
11. How the fire and rescue service and any other necessary services will be called.
12. Procedures for meeting the fire and rescue service on their arrival and notifying them of any special risks.

The emergency plan should be kept in the workplace and be available to employees, their representatives (where appointed), and the enforcing authority.

Inform, Instruct and Train

Employees and others working in the premises should be informed about the risks that have been identified and the measures that have been taken to remove, reduce, or protect from them [94]. This should include:

1. **The emergency procedures**.
2. **The identity of the people who have been nominated with responsibilities for fire safety**.
3. **Any special arrangements that have been made for people with disabilities**.

Employees should be given adequate fire safety instruction and training when they are first employed and on being exposed to new or increased risks [95]. This should include:

1. **The fire risks in the premises**.
2. **The fire safety measures in the premises**.
3. **What to do in case of fire, including how to raise the alarm**.
4. **What to do upon hearing the fire alarm**.
5. **The location and use of fire extinguishers and other firefighting equipment**.
6. **The location of escape routes and fire exits**.
7. **The importance of keeping fire doors closed**.

Fire safety instruction and training should be repeated periodically, particularly if there are any changes to the fire risk assessment or the emergency procedures.

3.7 Fire Safety Training

Fire safety training is a crucial aspect of ensuring that employees and others know what to do in case of fire and how to prevent fires from starting [96].

Who Should Receive Training?

All employees should receive fire safety training, including [97]:

1. **New employees, as part of their induction**.
2. **Existing employees, periodically and when there are changes to the fire risks or procedures**.
3. **Employees with special responsibilities, such as fire wardens or those who assist people with disabilities**.
4. **Other people who might be present, such as contractors, temporary workers, and volunteers**.

What Should the Training Cover?

The training should be tailored to the specific risks and characteristics of the premises, but should generally include [98]:

1. **The fire risks in the premises**.
2. **The fire safety measures in the premises, including the location and use of fire extinguishers, fire alarms, and escape routes**.
3. **The emergency procedures, including how to raise the alarm and what to do upon hearing the alarm**.
4. **The importance of good housekeeping and fire prevention**.
5. **The importance of reporting fire hazards**.
6. **How to assist people with disabilities or special needs**.

How Should the Training be Delivered?

The training should be delivered in a way that is appropriate for the audience and the risks involved [99]. This could include:

1. **Face-to-face training sessions**.
2. **Online training modules**.
3. **Practical demonstrations and drills**.
4. **Video or multimedia presentations**.
5. **Handbooks or leaflets**.

The training should be delivered by someone who is competent in fire safety and who has an understanding of the specific risks and procedures in the premises.

Records of Training

Records should be kept of all fire safety training that is delivered [100]. These should include:

1. **The date of the training**.
2. **The name of the person who delivered the training**.
3. **The names of the people who received the training**.
4. **The content of the training**.

These records can be used to demonstrate compliance with legal requirements and to ensure that everyone has received the necessary training.

Fire safety training should not be a one-time event, but an ongoing process that is regularly refreshed and updated. By ensuring that employees and others are well-informed and well-prepared, the risk of fire and the consequences if a fire does occur can be significantly reduced.

This concludes Unit 3: Fire Risk Assessment. The next unit will cover Fire Safety Measures and Fire Prevention.

UNIT 4: FIRE SAFETY MEASURES & FIRE PREVENTION

4.1 Fire Risks and Preventative Measures

Fire safety measures and fire prevention are about taking steps to reduce the likelihood of a fire occurring and to mitigate the consequences if a fire does occur [101]. These measures are based on the findings of the fire risk assessment and should be tailored to the specific risks and characteristics of the premises.

Common Fire Risks

Some of the most common fire risks in workplaces include [102]:

1. **Electrical equipment and wiring**, such as overloaded sockets, damaged cables, and faulty appliances.
2. **Smoking**, particularly if smoking is allowed in the workplace or if there are inadequate facilities for safe disposal of smoking materials.
3. **Cooking equipment**, such as ovens, hot plates, and microwaves, particularly if left unattended or not properly cleaned and maintained.
4. **Heating equipment**, such as boilers, furnaces, and portable heaters, particularly if they are faulty, misused, or left too close to flammable materials.
5. **Flammable materials**, such as paper, cardboard, textiles, flammable liquids and gases, particularly if they are not stored or used safely.
6. **Accumulation of waste**, such as paper, packaging, and general rubbish, particularly if it is allowed to build up and is not disposed of regularly.
7. **Arson**, which can be a significant risk if the premises are vulnerable to unauthorized access or if there are inadequate security measures in place.

Preventative Measures

The specific preventative measures will depend on the findings of the fire risk assessment, but some general principles include [103]:

1. **Good housekeeping**, such as keeping the workplace clean and tidy, removing waste regularly, and maintaining equipment and machinery.
2. **Safe storage**, such as keeping flammable materials in fire-resistant cabinets, separating incompatible substances, and not overloading electrical sockets.
3. **Safe handling and use**, such as using equipment and substances in accordance with manufacturer's instructions, not leaving cooking or heating equipment unattended, and enforcing no smoking policies.
4. **Maintenance and testing**, such as regular inspections and servicing of electrical and gas equipment, fire alarms, emergency lighting, and firefighting equipment.
5. **Staff training and awareness**, such as providing fire safety training to all staff, encouraging the reporting of fire hazards, and promoting a culture of fire safety.
6. **Security measures**, such as controlling access to the premises, securing doors and windows, and installing intruder alarms and CCTV.
7. **Fire detection and warning systems**, such as smoke alarms, heat detectors, and fire alarms, to provide early warning in case of fire.
8. **Firefighting equipment**, such as fire extinguishers, fire blankets, and sprinkler systems, to allow small fires to be tackled quickly.
9. **Safe means of escape**, such as keeping escape routes clear, providing emergency lighting and signage, and ensuring that fire doors are not wedged open.

These preventative measures should be kept under regular review and should be updated whenever there are changes to the workplace, the activities carried out, or the number and type of people present.

4.2 Housekeeping

Good housekeeping is one of the most important and effective fire prevention measures [104]. It involves keeping the workplace clean, tidy, and free from unnecessary combustible materials and ignition sources.

Benefits of Good Housekeeping

Good housekeeping can [105]:

1. **Reduce the risk of fire starting**, by removing potential fuel and ignition sources.

2. **Reduce the spread of fire**, by limiting the amount of combustible materials and ensuring that escape routes and firefighting equipment are not obstructed.
3. **Facilitate the early detection of fire**, by ensuring that fire detection and warning systems are not obscured or obstructed.
4. **Facilitate the safe evacuation of the premises**, by ensuring that escape routes are clear and that emergency lighting and signage are visible.
5. **Facilitate the work of the fire and rescue service**, by providing clear access and reducing the risk of fire spread and structural collapse.

Housekeeping Measures

Some specific housekeeping measures include [106]:

1. **Regular cleaning and tidying**, to remove dust, debris, and other combustible materials.
2. **Prompt disposal of waste**, using suitable receptacles and ensuring that waste is removed from the premises regularly.
3. **Safe storage of combustible materials**, away from ignition sources and in fire-resistant containers or rooms where necessary.
4. **Maintenance of equipment and machinery**, to prevent the accumulation of dust, grease, and other combustible materials and to ensure that they are in safe working order.
5. **Keeping escape routes and firefighting equipment clear**, by not storing goods or materials in corridors, stairways, or in front of fire extinguishers or fire alarm call points.
6. **Keeping fire doors closed**, to prevent the spread of fire and smoke.
7. **Controlling the use of electrical equipment**, by not overloading sockets, using extension cords safely, and switching off and unplugging equipment when not in use.
8. **Enforcing no smoking policies**, and providing suitable facilities for the safe disposal of smoking materials where smoking is permitted.

Housekeeping should be an ongoing activity, with all staff taking responsibility for keeping their own work areas clean and tidy and reporting any hazards or deficiencies. Regular inspections should also be carried out to ensure that housekeeping standards are being maintained.

4.3 Storage

The safe storage of materials, particularly those that are combustible or flammable, is an important aspect of fire prevention [107].

Risks of Poor Storage

Poor storage practices can increase the risk of fire by [108]:

1. **Providing a source of fuel**, particularly if large quantities of combustible materials are stored together.
2. **Allowing fire to spread rapidly**, if combustible materials are not separated or contained.
3. **Creating ignition sources**, if flammable liquids or gases are not stored correctly or are allowed to leak.
4. **Impeding escape routes and firefighting equipment**, if storage obstructs corridors, stairways, or access to fire extinguishers and fire alarm call points.

Safe Storage Principles

The principles of safe storage include [109]:

1. **Separation of incompatible materials**, such as keeping oxidizing agents away from flammable liquids or gases.
2. **Containment of combustible or flammable materials**, by using fire-resistant containers, cabinets, or rooms.
3. **Ventilation of storage areas**, particularly where flammable liquids or gases are stored, to prevent the build-up of vapours.
4. **Temperature control**, to prevent the overheating of stored materials that could lead to ignition.
5. **Secure storage**, to prevent unauthorized access and reduce the risk of arson.

6. **Regular inspections**, to ensure that storage arrangements remain safe and to identify any leaks, spills, or damage.

Specific Storage Measures

Some specific storage measures include [110]:

1. **Flammable liquids**: Should be stored in a separate, fire-resistant room or cabinet, with a bund or drip tray to contain any leaks. The quantity of flammable liquids stored should be kept to a minimum and should not exceed 50 litres unless in an approved container.
2. **Flammable gases**: Should be stored in a separate, well-ventilated area, away from ignition sources and oxidizing agents. Cylinders should be secured to prevent falling and should be regularly checked for leaks.
3. **Combustible materials**: Should be stored in a separate room or area, away from ignition sources. The quantity of combustible materials stored should be kept to a minimum and should be regularly removed from the premises.
4. **Oxidizing agents**: Should be stored in a separate room or area, away from flammable liquids, gases, and combustible materials.
5. **Electrical equipment**: Should be stored in a cool, dry place, away from flammable materials. Electrical storage areas should not be used for general storage.

All staff should be aware of the storage arrangements and should be trained in the safe handling and use of the materials stored. Smoking and the use of naked flames should be prohibited in storage areas.

4.4 Dangerous Substances: Storage, Display and Use

Dangerous substances are materials or preparations that have the potential to cause harm to people, property, or the environment due to their properties, such as flammability, explosivity, or toxicity [111].

Examples of Dangerous Substances

Some common examples of dangerous substances in the workplace include [112]:

1. **Flammable liquids**, such as petrol, solvents, and paints.
2. **Flammable gases**, such as propane, butane, and acetylene.
3. **Oxidizing agents**, such as oxygen, chlorine, and hydrogen peroxide.
4. **Compressed gases**, such as air, nitrogen, and carbon dioxide.
5. **Corrosive substances**, such as acids and alkalis.
6. **Toxic substances**, such as pesticides, biocides, and carcinogens.

Risks of Dangerous Substances

Dangerous substances can present risks of [113]:

1. **Fire or explosion**, if flammable or explosive substances are not stored, used, or handled correctly.
2. **Toxic effects**, if toxic substances are inhaled, ingested, or come into contact with skin or eyes.
3. **Corrosive effects**, if corrosive substances come into contact with skin, eyes, or other materials.
4. **Asphyxiation**, if large quantities of gases are released in confined spaces.
5. **Environmental pollution**, if dangerous substances are spilled or released into the environment.

Control Measures

The measures required to control the risks from dangerous substances will depend on the properties of the substance and the way it is used, but some general principles include [114]:

1. **Substitution**, by replacing dangerous substances with less harmful alternatives where possible.
2. **Reduction**, by minimizing the quantities of dangerous substances used or stored.

3. **Isolation**, by storing and using dangerous substances in separate, well-ventilated areas, away from incompatible materials and ignition sources.
4. **Containment**, by using closed systems, bunds, or drip trays to prevent spills and releases.
5. **Ventilation**, by providing adequate ventilation to prevent the build-up of flammable or toxic vapours.
6. **Personal protective equipment (PPE)**, by providing suitable PPE, such as gloves, goggles, and respirators, and ensuring that it is used correctly.
7. **Information, instruction, and training**, by providing employees with information about the hazards and risks of the dangerous substances they work with and training in safe handling and emergency procedures.
8. **Emergency procedures**, by having procedures in place to deal with spills, leaks, or releases of dangerous substances, including the provision of spill kits and the training of staff.

The storage, display, and use of dangerous substances should be carefully controlled and should be based on a risk assessment. The risk assessment should consider the properties of the substance, the way it is used, and the potential for fire, explosion, or other harmful effects.

4.5 Equipment and Machinery

Equipment and machinery can present a range of fire risks, depending on their type, use, and condition [115].

Fire Risks from Equipment and Machinery

Some of the ways in which equipment and machinery can cause or contribute to fires include [116]:

1. **Overheating**, if equipment is faulty, overloaded, or not maintained properly, leading to ignition of the equipment itself or nearby combustible materials.
2. **Sparks or heat**, from friction, grinding, welding, or other hot work processes, which can ignite flammable vapours, gases, or combustible materials.
3. **Electrical faults**, such as short circuits, overloaded circuits, or damaged cables, which can cause sparks or overheating.
4. **Leaks or spills**, of flammable liquids or gases from equipment or machinery, which can create a fire or explosion hazard.
5. **Dust or debris accumulation**, in or around equipment or machinery, which can be ignited by heat or sparks.

Control Measures

The measures required to control the fire risks from equipment and machinery will depend on the type of equipment and the way it is used, but some general principles include [117]:

1. **Selection**, by ensuring that equipment and machinery is suitable for its intended use and location, and that it meets relevant safety standards.
2. **Installation**, by ensuring that equipment and machinery is installed correctly, in accordance with manufacturer's instructions, and that it is commissioned and tested before use.
3. **Maintenance**, by carrying out regular inspections, servicing, and repairs of equipment and machinery, in accordance with manufacturer's recommendations, and keeping records of maintenance.
4. **Training**, by providing employees with training in the safe operation and maintenance of equipment and machinery, including the procedures for reporting faults and emergencies.
5. **Housekeeping**, by keeping equipment and machinery clean and free from dust, debris, and flammable materials, and ensuring that ventilation openings are not obstructed.
6. **Fire protection**, by providing suitable fire detection and protection systems, such as fire alarms, extinguishers, or sprinklers, in areas where equipment and machinery is used.
7. **Emergency procedures**, by having procedures in place to deal with equipment or machinery fires, including the safe shutdown of equipment, the evacuation of the area, and the calling of the fire and rescue service.

The use of equipment and machinery should be based on a risk assessment, which should consider the fire risks required to control them. The risk assessment should be reviewed regularly and whenever there are changes to the equipment, the way it is used, or the environment in which it is located.

4.6 Electrical Safety

Electricity is a common cause of workplace fires, and electrical equipment and installations can present a range of fire risks [118].

Fire Risks from Electricity

Some of the ways in which electricity can cause or contribute to fires include [119]:

1. **Overheating**, if electrical equipment or wiring is overloaded, faulty, or not maintained properly, leading to ignition of the equipment or nearby combustible materials.
2. **Sparks or arcing**, from loose connections, damaged insulation, or faulty switches, which can ignite flammable vapours, gases, or combustible materials.
3. **Short circuits**, where electrical current takes an unintended path, generating heat and sparks, which can ignite surrounding materials.
4. **Static electricity**, where the build-up of electrical charge on materials or equipment can cause sparks, which can ignite flammable vapours or gases.

Control Measures

The measures required to control the fire risks from electricity will depend on the type of electrical equipment and installations and the way they are used, but some general principles include [120]:

1. **Design and installation**, by ensuring that electrical installations are designed, installed, and commissioned by competent persons, in accordance with relevant standards and regulations.
2. **Inspection and testing**, by carrying out regular inspections and tests of electrical installations and equipment, in accordance with relevant standards and regulations, and keeping records of inspections and tests.
3. **Maintenance**, by carrying out regular maintenance of electrical equipment and installations, in accordance with manufacturer's recommendations, and ensuring that any faults or damage are repaired promptly.
4. **Portable appliance testing (PAT)**, by carrying out regular tests of portable electrical appliances, to ensure that they are safe to use and free from defects.
5. **Use**, by ensuring that electrical equipment is used in accordance with manufacturer's instructions, and that it is not overloaded, damaged, or modified.
6. **Environments**, by ensuring that electrical equipment is suitable for the environment in which it is used, such as in wet or dusty conditions, and that it is protected from damage.
7. **Training**, by providing employees with training in the safe use of electrical equipment, including the procedures for reporting faults and emergencies.

The use of electrical equipment and installations should be based on a risk assessment, which should consider the fire risks and the measures required to control them. The risk assessment should be reviewed regularly and whenever there are changes to the equipment, the way it is used, or the environment in which it is located.

4.7 Smoking

Smoking is a significant cause of workplace fires, particularly where smoking is allowed in the workplace or where there are inadequate facilities for the safe disposal of smoking materials [121].

Fire Risks from Smoking

Some of the ways in which smoking can cause or contribute to fires include [122]:

1. **Discarded cigarettes**, if not properly extinguished, can ignite combustible materials such as paper, textiles, or dry vegetation.
2. **Matches or lighters**, if not used safely or if left where they can be accessed by children or vulnerable people, can cause ignition of combustible materials.
3. **Ashtrays**, if not emptied regularly or if made of combustible materials, can ignite, particularly if they contain paper or other combustible waste.

4. **Smoking in prohibited areas**, such as in storage areas for flammable liquids or gases, can ignite flammable vapours or gases.

Control Measures

The measures required to control the fire risks from smoking will depend on the smoking policy of the workplace and the facilities provided, but some general principles include [123]:

1. **Smoking policy**, by having a clear policy on smoking in the workplace, which is communicated to all employees and visitors, and enforced consistently.
2. **Designated smoking areas**, by providing designated smoking areas, which are away from buildings, combustible materials, and flammable storage areas, and which have suitable facilities for the safe disposal of smoking materials.
3. **No smoking signs**, by displaying no smoking signs in all areas where smoking is prohibited, including in vehicles and outdoor areas.
4. **Smoking breaks**, by ensuring that employees who smoke take smoking breaks in designated smoking areas and do not smoke in prohibited areas or while working.
5. **Safe disposal**, by providing suitable receptacles for the safe disposal of smoking materials, such as metal ashtrays or sand buckets, which are emptied regularly and safely.
6. **Fire protection**, by providing suitable fire detection and protection systems, such as smoke alarms or sprinklers, in areas where smoking is permitted.
7. **Training**, by providing employees with training on the smoking policy and the fire risks associated with smoking, including the procedures for the safe disposal of smoking materials.

The smoking policy and facilities should be based on a risk assessment, which should consider the fire risks and the measures required to control them. The risk assessment should be reviewed regularly and whenever there are changes to the smoking policy, the facilities provided, or tÁhe environment in which smoking takes place.

4.8 Managing Building Work and Alterations

Building work and alterations can introduce new fire risks or can impact on existing fire safety measures, so it is important that they are managed carefully [124].

Fire Risks from Building Work and Alterations

Some of the ways in which building work and alterations can cause or contribute to fires include [125]:

1. **Ignition sources**, such as hot work (e.g., welding, cutting, or grinding), electrical equipment, or smoking, which can ignite combustible materials or flammable vapours.
2. **Combustible materials**, such as packaging, waste, or building materials, which can provide fuel for a fire.
3. **Flammable liquids or gases**, such as solvents, adhesives, or propane, which can ignite or explode if not stored or used safely.
4. **Compartmentation**, if fire-resisting walls, floors, or doors are breached or compromised, allowing fire and smoke to spread.
5. **Escape routes**, if they are blocked, obstructed, or altered, making it difficult for people to evacuate safely in the event of a fire.
6. **Fire detection and protection systems**, if they are damaged, disconnected, or impaired, reducing their effectiveness in detecting or suppressing a fire.

Control Measures

The measures required to control the fire risks from building work and alterations will depend on the nature and extent of the work, but some general principles include [126]:

1. **Planning**, by involving the fire safety manager or advisor in the planning of any building work or alterations, to identify the fire risks and the measures required to control them.
2. **Risk assessment**, by carrying out a fire risk assessment before, during, and after the building work or alterations, to identify the fire risks and the measures required to control them.

3. **Contractor management**, by appointing competent contractors, who have the necessary skills, knowledge, and experience to carry out the work safely, and by providing them with information about the fire risks and the fire safety measures in the building.
4. **Permits to work**, by using a permit to work system for high-risk activities, such as hot work, to ensure that the risks are assessed and controlled and that the work is authorized and monitored.
5. **Segregation**, by separating the building work or alterations from the rest of the building, using physical barriers, fire-resisting construction, or by scheduling the work outside of normal working hours.
6. **Fire watch**, by providing a fire watch during and after high-risk activities, such as hot work, to detect and respond to any fires that may occur.
7. **Temporary fire safety measures**, by providing temporary fire detection and protection systems, escape routes, and firefighting equipment, where the existing measures are impaired or compromised by the building work or alterations.

The management of building work and alterations should be based on a fire risk assessment and should involve the cooperation and coordination of all parties involved, including the client, the designer, the contractor, and the fire safety manager or advisor.

4.9 Existing Layout and Construction

The existing layout and construction of a building can have a significant impact on the fire risks and the measures required to control them [127].

Fire Risks from Existing Layout and Construction

Some of the ways in which the existing layout and construction can influence the fire risks include [128]:

1. **Means of escape**, if the number, location, or capacity of escape routes is inadequate for the number of people in the building, or if the escape routes are too long, complex, or pass-through high-risk areas.
2. **Compartmentation**, if the building is not divided into fire-resisting compartments, allowing fire and smoke to spread rapidly, or if the compartmentation is compromised by openings, penetrations, or poor maintenance.
3. **Fire loading**, if the building contains a high level of combustible materials, such as in storage areas, or if the combustible materials are not separated from ignition sources or escape routes.
4. **Structural fire protection**, if the building's structure is not adequately protected from the effects of fire, such as by fire-resisting construction or by the use of non-combustible materials.
5. **Facilities for firefighting**, if the building does not have adequate facilities for the fire and rescue service, such as access roads, water supplies, or firefighting shafts.

Control Measures

The measures required to control the fire risks from the existing layout and construction will depend on the specific features of the building, but some general principles include [129]:

1. **Fire risk assessment**, by carrying out a fire risk assessment to identify the fire risks and the measures required to control them, taking into account the existing layout and construction of the building.
2. **Upgrading**, by upgrading the existing layout and construction to improve the fire safety of the building, such as by providing additional escape routes, improving the compartmentation, or increasing the structural fire protection.
3. **Compensatory measures**, by providing compensatory measures where the existing layout and construction cannot be upgraded, such as by providing additional fire detection and protection systems, or by limiting the number of people in the building.
4. **Management**, by managing the fire risks through effective housekeeping, maintenance, and staff training, and by having robust emergency plans and procedures in place.
5. **Cooperation with the fire and rescue service**, by providing the fire and rescue service with information about the layout and construction of the building, and by ensuring that the facilities for firefighting are maintained and accessible.

The control of fire risks from the existing layout and construction should be based on a fire risk assessment and should take into account the use of the building, the number and type of people in the building, and the potential for fire spread and firefighting operations. The fire risk assessment should be reviewed regularly and whenever there are changes to the building, its use, or its occupancy.

4.10 Particular Hazards in Corridors and Stairways used as Escape Routes

Corridors and stairways that form part of the escape routes from a building can present particular fire hazards that need to be controlled [130].

Fire Hazards in Corridors and Stairways

Some of the particular fire hazards that can occur in corridors and stairways include [131]:

1. **Combustible materials**, such as furniture, equipment, or waste, which can provide fuel for a fire and can obstruct the escape routes.
2. **Ignition sources**, such as electrical equipment, lighting, or smoking, which can start a fire in the escape routes.
3. **Smoke and heat**, which can quickly fill the escape routes, making it difficult for people to see and breathe, and which can cause injury or death.
4. **Obstructions**, such as locked doors, blocked corridors, or items stored in stairways, which can prevent or delay escape.
5. **Slips and trips**, caused by uneven or slippery surfaces, poor lighting, or obstacles, which can cause falls and injuries during escape.

Control Measures

The measures required to control the fire hazards in corridors and stairways will depend on the specific layout and use of the building, but some general principles include [132]:

1. **Keep escape routes clear**, by ensuring that corridors and stairways are kept free from combustible materials, ignition sources, and obstructions, and by having a policy for the management of escape routes.
2. **Fire-resisting construction**, by ensuring that the walls, floors, and ceilings of corridors and stairways are constructed of fire-resisting materials, and that any openings, such as doors or vents, are protected by fire-resisting doors or shutters.
3. **Compartmentation**, by dividing the building into fire-resisting compartments, to prevent the spread of fire and smoke between different parts of the building, and to protect the escape routes.
4. **Fire detection and warning systems**, by providing automatic fire detection and warning systems in the corridors and stairways, to give early warning of a fire and to enable people to escape safely.
5. **Emergency lighting**, by providing emergency lighting in the corridors and stairways, to enable people to see the escape routes and the exit signs in the event of a power failure.
6. **Signage**, by providing clear and unambiguous signs to indicate the escape routes and the location of the final exits, and to provide instructions on what to do in the event of a fire.
7. **Maintenance**, by regularly inspecting and maintaining the corridors and stairways, to ensure that they are in good condition and free from hazards, and by carrying out repairs or improvements as necessary.

The control of fire hazards in corridors and stairways should be based on a fire risk assessment and should take into account the number and type of people using the escape routes, the distance and direction of travel, and the potential for fire and smoke spread. The fire risk assessment should be reviewed regularly and whenever there are changes to the building, its use, or its occupancy.

4.11 Insulated Core Panels

Insulated core panels (ICPs) are composite panels made of two metal faces bonded to an insulating core, which are commonly used in the construction of buildings, particularly for walls and ceilings [133].

Fire Risks from Insulated Core Panels

Some of the fire risks associated with ICPs include [134]:

1. **Combustibility**, as some types of ICP cores, such as those made of polyurethane or polystyrene, are combustible and can contribute to the rapid spread of fire.
2. **Delamination**, where the metal faces of the panels can separate from the core under the influence of heat, allowing the fire to spread through the cavity and to other parts of the building.
3. **Smoke and toxic gases**, as the burning of some types of ICP cores can produce large quantities of dense smoke and toxic gases, which can impede escape and cause injury or death.
4. **Hidden fire spread**, as a fire can develop unseen within the panels or in the cavities behind them, making it difficult to detect and control.
5. **Structural failure**, as the panels can lose their strength and integrity when exposed to fire, leading to collapse and the spread of fire to other parts of the building.

Control Measures

The measures required to control the fire risks from ICPs will depend on the type and location of the panels, but some general principles include [135]:

1. **Risk assessment**, by carrying out a fire risk assessment to identify the fire risks associated with the ICPs and the measures required to control them, taking into account the type and location of the panels, and the potential for fire spread and structural failure.
2. **Specification**, by specifying ICPs that have been tested and certified to the relevant fire performance standards, such as those for fire resistance, reaction to fire, and fire propagation, and by ensuring that the panels are installed in accordance with the manufacturer's instructions and the relevant building codes and standards.
3. **Compartmentation**, by using ICPs as part of a fire-resisting compartment wall or ceiling, to prevent the spread of fire between different parts of the building, and by ensuring that the panels are properly sealed and protected at the edges and joints.
4. **Fire stopping**, by providing fire stopping around any penetrations or openings in the panels, such as for services or ventilation, to maintain the fire resistance of the panels and to prevent the spread of fire and smoke.
5. **Fire detection and suppression**, by providing automatic fire detection and suppression systems, such as sprinklers or fire alarms, in the areas where ICPs are used, to provide early warning of a fire and to control its spread.
6. **Maintenance**, by regularly inspecting and maintaining the ICPs, to ensure that they are in good condition and free from damage or deterioration, and by carrying out repairs or replacements as necessary.

The control of fire risks from ICPs should be based on a fire risk assessment and should take into account the specific properties and performance of the panels, as well as the overall fire safety strategy for the building. The fire risk assessment should be reviewed regularly and whenever there are changes to the building, its use, or its occupancy.

4.12 Restricting the Spread of Fire and Smoke

Restricting the spread of fire and smoke is a key element of fire safety in buildings, as it can help to protect escape routes, to prevent damage to property and assets, and to facilitate firefighting operations [136].

Compartmentation

One of the main ways of restricting the spread of fire and smoke is through compartmentation, which involves dividing the building into fire-resisting compartments, separated by fire-resisting walls, floors, and doors [137].

The objectives of compartmentation are [138]:

1. **To prevent the rapid spread of fire and smoke** from the compartment of origin to other parts of the building, giving people more time to escape and the fire and rescue service more time to intervene.
2. **To protect escape routes**, such as corridors and stairways, from the effects of fire and smoke, enabling people to escape safely.
3. **To subdivide the building** into manageable areas for firefighting and rescue operations, and to limit the damage caused by fire and smoke.

The fire resistance of the compartment walls, floors, and doors should be appropriate for the fire risk and the size of the compartment, as determined by the fire risk assessment and the relevant building codes and standards.

Fire Doors

Fire doors are an essential element of compartmentation, as they provide a barrier to the spread of fire and smoke when closed, while allowing people to move between compartments when open [139].

The main requirements for fire doors are [140]:

1. **Fire resistance**, as the doors should be able to withstand the passage of fire and smoke for a specified period of time, as determined by the fire risk assessment and the relevant standards.
2. **Smoke control**, as the doors should be fitted with seals or brushes to prevent the passage of smoke around the edges, and with a self-closing device to ensure that they close automatically when not in use.
3. **Ease of operation**, as the doors should be easy to open and close, and should not be obstructed or wedged open, unless they are fitted with an automatic release mechanism that closes the door in the event of a fire.
4. **Maintenance**, as the doors should be regularly inspected and maintained to ensure that they are in good condition and functioning correctly, and should be repaired or replaced if damaged or defective.

Smoke Control

Smoke control is another important aspect of restricting the spread of fire and smoke, as smoke can be more dangerous than the fire itself, causing injury, death, and damage to property [141].

The main methods of smoke control are [142]:

1. **Smoke containment**, which involves creating smoke-tight barriers, such as walls, floors, and doors, to prevent the spread of smoke from the area of origin to other parts of the building.
2. **Smoke extraction**, which involves using mechanical or natural ventilation to remove smoke from the building, either through dedicated smoke shafts or through the normal ventilation system.
3. **Pressurization**, which involves using fans to create a positive pressure in the escape routes and a negative pressure in the fire compartment, to prevent the spread of smoke and to make it easier for people to escape.

The choice of smoke control method will depend on the fire risk assessment, the layout and use of the building, and the relevant building codes and standards.

Conclusion

Restricting the spread of fire and smoke is a critical aspect of fire safety in buildings, which requires a combination of passive and active measures, such as compartmentation, fire doors, and smoke control. These measures should be based on a fire risk assessment and should be regularly inspected and maintained to ensure their effectiveness. The fire risk assessment should be reviewed regularly and updated whenever there are changes to the building, its use, or its occupancy.

4.13 Arson

Arson is a serious crime that involves the deliberate setting of fires to damage or destroy property. It is a significant cause of fire in many types of buildings, including schools, hospitals, offices, and industrial premises [143].

Arson Prevention

The prevention of arson involves a combination of physical, technical, and organisational measures, based on a fire risk assessment that considers the likelihood and potential consequences of an arson attack [144].

Some of the key measures for preventing arson include:

1. **Perimeter security**, such as fencing, gates, and barriers, to prevent unauthorized access to the site and to deter potential arsonists.
2. **Access control**, such as locks, alarms, and CCTV, to prevent unauthorized entry to the building and to detect and deter intruders.
3. **Lighting**, both internal and external, to deter potential arsonists and to aid in the detection and identification of intruders.
4. **Housekeeping**, to reduce the number of combustible materials that could be used as fuel for an arson attack, and to ensure that waste and other materials are stored securely and disposed of regularly.
5. **Staff training and awareness**, to ensure that employees are aware of the risk of arson and know how to prevent, detect, and respond to suspicious activities or incidents.
6. **Liaison with the police and fire services**, to share information about potential threats and to agree on response plans and protocols in the event of an arson attack.

Arson Response

In the event of an arson attack, the priority is to ensure the safety of people in the building and to minimize the damage caused by the fire. This requires a well-planned and rehearsed emergency response, based on the fire risk assessment and the specific circumstances of the incident [145].

Some of the key elements of an effective arson response include:

1. **Early detection and warning**, through the use of automatic fire detection systems, such as smoke detectors and heat detectors, and through the vigilance of staff and occupants.
2. **Evacuation**, to ensure that all persons in the building are able to escape safely and quickly, using the designated escape routes and following the established procedures.
3. **Firefighting**, using portable fire extinguishers and fixed installations, such as sprinklers or hose reels, to control and extinguish the fire, where safe and appropriate to do so.
4. **Calling the emergency services**, to ensure that the fire and rescue service and the police are alerted and can respond quickly and effectively to the incident.
5. **Preserving evidence**, to assist with the investigation of the incident and the identification and prosecution of the offenders, by not disturbing the scene and by recording any relevant information or observations.

Conclusion

Arson is a serious and potentially devastating crime that requires a proactive and comprehensive approach to prevention and response. This approach should be based on a thorough fire risk assessment and should involve a range of physical, technical, and organizational measures, as well as effective emergency planning and liaison with the relevant authorities. The risk of arson should be regularly reviewed and the measures in place should be tested and updated as necessary to ensure their ongoing effectiveness.

This concludes Unit 4: Fire Safety Measures & Fire Prevention. The next unit will cover Fire Detection and Warning Systems.

UNIT 5: FIRE DETECTION AND WARNING SYSTEMS

5.1 People with Hearing Difficulties

Fire safety systems in buildings need to cater to all occupants, including those with hearing difficulties who may not be able to hear audible fire alarms. It's crucial that provisions are made to ensure they can be alerted in case of a fire [146].

Alerting Methods for People with Hearing Difficulties

There are several methods that can be used to alert people with hearing difficulties in case of a fire [147]:

1. **Visual Alarms**: These are devices that emit a bright flashing light when the fire alarm is activated. They should be positioned in a way that ensures they can be seen from all areas of the building.
2. **Vibrating Devices**: These can be worn on the body (like a pager) or placed under a pillow. They vibrate strongly when the fire alarm is activated.

3. **Paging Systems**: These send text messages to a designated pager or mobile phone when the fire alarm is activated.
4. **Buddy System**: This involves assigning a colleague or "buddy" to alert the person with hearing difficulties in case of a fire alarm.

Considerations for Alerting Systems

When implementing alerting systems for people with hearing difficulties, several factors should be considered [148]:

1. **Coverage**: The system should cover all areas of the building where the person with hearing difficulties might be.
2. **Reliability**: The system should be regularly tested to ensure it will function correctly in case of a fire.
3. **Clarity**: Visual alarms should be distinct from other types of lighting to avoid confusion.
4. **Personal Needs**: The type of system used should be based on the individual's needs and preferences.
5. **Training**: People with hearing difficulties should be trained on what to do when they receive an alert.

Maintenance and Testing

Like all fire safety systems, those for alerting people with hearing difficulties should be regularly maintained and tested to ensure they will work effectively in case of a fire [149].

This should include:

1. **Regular Testing**: The system should be tested weekly or monthly (depending on the type of system) to check it's functioning correctly.
2. **Battery Checks**: For systems that use batteries, these should be regularly checked and replaced as needed.
3. **Repair and Replacement**: Any faults should be promptly repaired, and outdated systems should be replaced.
4. **Record Keeping**: Records should be kept of all tests, maintenance, and repairs.

Providing appropriate alerting systems for people with hearing difficulties is a vital part of an inclusive fire safety strategy. It ensures that everyone can be safely alerted and evacuated in case of a fire, regardless of their hearing ability.

5.2 Voice Alarms

Voice alarms, also known as voice evacuation systems or voice alarm devices (VADs), are systems that use recorded or live voice messages to alert people in a building in case of a fire or other emergency [150].

Benefits of Voice Alarms

Voice alarms have several advantages over traditional bell or siren alarms [151]:

1. **Clear Instructions**: Voice alarms can provide clear, specific instructions on what to do in case of an emergency, such as which exit to use or where to assemble. This can help prevent confusion and panic.
2. **Phased Evacuation**: In large or complex buildings, voice alarms can be used to implement phased evacuation, where different parts of the building are evacuated in a controlled sequence. This can help prevent congestion and ensure the most at-risk areas are evacuated first.
3. **Reduced Alarm Fatigue**: People can become desensitized to frequent bell or siren alarms, especially in buildings where false alarms are common. Voice alarms are more noticeable and are less likely to be ignored.
4. **Accessibility**: Voice alarms can be beneficial for people with hearing difficulties, as the voice message may be easier to understand than a bell or siren.

Designing a Voice Alarm System

When designing a voice alarm system, several factors need to be considered [152]:

1. **Audibility**: The system should be loud enough to be heard clearly above any background noise in all parts of the building.
2. **Intelligibility**: The voice messages should be clear and easy to understand. The use of a professional voice actor and high-quality recordings can help ensure this.
3. **Content**: The messages should be concise, clear, and provide specific instructions. They should be tailored to the specific building and the types of emergencies that could occur.
4. **Zones**: In larger buildings, the system should be divided into zones so that different messages can be played in different parts of the building, depending on the location and type of emergency.
5. **Backup Power**: The system should have a backup power supply to ensure it will continue to function in case of a power outage.

Maintenance and Testing

Voice alarm systems should be regularly maintained and tested to ensure they will work effectively in case of an emergency [153]. This should include:

1. **Weekly Tests**: A short test should be conducted each week to check the system is functioning correctly.
2. **Full Tests**: A full test of the system, including all speakers and backup power, should be conducted every six months.
3. **Servicing**: The system should be serviced by a competent person at least annually.
4. **Record Keeping**: Records should be kept of all tests, maintenance, and repairs.

Voice alarms can be a highly effective way to alert and instruct people in case of a fire or other emergency. However, they need to be carefully designed, maintained, and tested to ensure they will perform their critical function when needed.

5.3 Schematic Plan

A schematic plan, in the context of fire safety, is a simplified diagram that shows the layout of a building and the location of key fire safety systems and equipment [154]. It provides a quick and easy reference for occupants, fire wardens, and firefighters.

Information on a Schematic Plan

A fire safety schematic plan should include [155]:

1. **Building Layout**: A simplified floor plan of the building, showing the location of rooms, corridors, stairways, and exits.
2. **Fire Alarm Panel**: The location of the main fire alarm control panel.
3. **Detectors**: The location of smoke detectors, heat detectors, and manual call points.
4. **Sounders**: The location of fire alarm sounders or bells.
5. **Extinguishers**: The location and type of fire extinguishers.
6. **Hose Reels**: The location of hose reels, if present.
7. **Dry/Wet Risers**: The location of dry or wet riser inlets for firefighting.
8. **Gas/Electricity Shutoffs**: The location of gas and electricity shutoff points.
9. **Evacuation Chair**: The location of evacuation chairs for people with mobility impairments.
10. **Assembly Point**: The location of the designated assembly point outside the building.

Uses of a Schematic Plan

A fire safety schematic plan has several important uses [156]:

1. **Orientation**: It helps occupants familiarize themselves with the layout of the building and the location of fire safety equipment.
2. **Evacuation**: In case of a fire, it can help occupants identify the nearest exit and the route to the assembly point.
3. **Fire Warden Duties**: It assists fire wardens in their sweep of the building during an evacuation.
4. **Firefighting**: It provides firefighters with a quick overview of the building layout and the location of key equipment when they arrive on the scene.

Displaying a Schematic Plan

Schematic plans should be displayed prominently at key locations in the building, such as [157]:

1. **Fire Alarm Panel**: A copy should be kept next to the main fire alarm control panel.
2. **Entrances**: At the main entrance and any secondary entrances.
3. **Firefighting Lift**: In the lobby of the firefighting lift, if present.
4. **Each Floor**: At strategic points on each floor, such as near stairways or exits.

The plans should be oriented so that they match the direction the viewer is facing, and they should be clearly legible and easy to understand.

Updating a Schematic Plan

A schematic plan should be updated whenever there are significant changes to the building layout or fire safety systems [158]. This might include:

1. **Renovations**: If walls are added, removed, or moved.
2. **System Updates**: If new fire safety systems are installed or existing ones are moved.
3. **Change of Use**: If the use of a room or area changes significantly.

The date of the last update should be clearly marked on the plan.

A well-designed and maintained schematic plan is a valuable tool in fire safety. It helps ensure that occupants, fire wardens, and firefighters have the information they need to respond effectively in case of a fire.

5.4 Manual Call Points

Manual call points, also known as break glass points or manual pull stations, are devices that allow people to manually raise the alarm in case of a fire [159]. They are an important part of a comprehensive fire detection and warning system.

Design of Manual Call Points

Manual call points are designed to be simple and easy to use. They typically consist of a small, red box mounted on the wall, with a button or handle that can be pushed or pulled to activate the alarm [160].

Some key design features include:

1. **Visible and Accessible**: They should be clearly visible and easily accessible, usually mounted at a height of around 1.4m from the floor.
2. **Simple Operation**: They should be easy to operate, usually by pushing a button or pulling down a handle.
3. **Clear Instructions**: They should have clear, concise instructions for use, such as "In case of fire, break glass" or "Pull down handle".
4. **Tamper-proof**: They should have a mechanism to prevent accidental activation, such as a small glass panel that must be broken to reach the button or handle.
5. **Identifiable**: When activated, they should send a signal to the fire alarm control panel indicating their location.

Location of Manual Call Points

Manual call points should be located strategically throughout a building to ensure that one can be reached quickly from any point [161]. Typical locations include:

1. **Exit Routes**: Near exits and along escape routes.
2. **High Risk Areas**: In areas where a fire is more likely to start, such as kitchens or workshops.
3. **Large Rooms**: In large rooms or areas where a fire might not be immediately detected by automatic sensors.

4. **Regularly Spaced**: At regular intervals in corridors and on each floor, so that no one needs to travel more than 30m to reach one.

Testing and Maintenance

Manual call points should be regularly tested and maintained to ensure they will work effectively in case of a fire [162]. This should include:

1. **Weekly Tests**: A different call point should be tested each week to ensure they are all functioning correctly.
2. **Cleaning**: Call points should be kept free from dust, dirt, and obstructions.
3. **Repair**: Any faults should be promptly reported and repaired.
4. **Record Keeping**: Records should be kept of all tests, maintenance, and repairs.

Use in Case of Fire

If someone discovers a fire, they should immediately activate the nearest manual call point [163]. This will sound the alarm throughout the building and alert others to the need to evacuate. The person should then follow the building's evacuation procedure.

It's important to note that manual call points should only be used in case of a real fire. Misuse or malicious activation of a call point is a criminal offense in many jurisdictions.

Manual call points are a vital backup to automatic fire detection systems. They allow people to quickly raise the alarm if they discover a fire, ensuring a rapid response and a safe evacuation.

5.5 Automatic Fire Detection

Automatic fire detection systems are designed to detect fires and raise the alarm without the need for human intervention. They are a crucial part of modern fire safety, providing early warning of a fire and allowing for a quick response [164].

Types of Automatic Fire Detectors

There are several types of automatic fire detectors, each designed to detect different aspects of a fire [165]:

1. **Smoke Detectors**: These detect the smoke produced by a fire. There are two main types:
    - Ionization detectors, which are good at detecting fast-flaming fires.
    - Optical detectors, which are better at detecting slow-burning, smouldering fires.
2. **Heat Detectors**: These detect the heat produced by a fire. They're less sensitive than smoke detectors but are useful in areas where smoke detectors might generate false alarms, such as kitchens.
3. **Flame Detectors**: These detect the infrared or ultraviolet radiation emitted by flames. They're useful in areas where a fire might develop very quickly, such as fuel storage areas.
4. **Carbon Monoxide Detectors**: These detect the carbon monoxide gas produced by a fire. They're useful in areas where a fire might smoulder for a long time before producing significant smoke or heat, such as in bedrooms.
5. **Multi-sensor Detectors**: These combine two or more types of sensors, such as smoke and heat, to provide more reliable detection and reduce false alarms.

Design of Automatic Fire Detection Systems

When designing an automatic fire detection system, several factors need to be considered [166]:

1. **Risk Assessment**: The type, number, and location of detectors should be based on a thorough risk assessment of the building, considering factors such as the use of the building, the fire load, and the occupants.
2. **Coverage**: Detectors should be placed to provide adequate coverage of all areas of the building, with no "blind spots".

3. **Zoning**: The system should be divided into zones, so that the location of a fire can be quickly identified.
4. **Alarm Signalling**: The system should be connected to sounders or a voice alarm system to alert occupants in case of a fire.
5. **Fire Panel**: The detectors should be connected to a central fire alarm control panel, which can monitor the status of the system and provide information to firefighters.
6. **Standards**: The system should be designed, installed, and maintained in accordance with relevant standards, such as BS 5839 in the UK.

Maintenance and Testing

Automatic fire detection systems should be regularly maintained and tested to ensure they will work effectively in case of a fire [167]. This should include:

1. **Regular Tests**: The system should be tested weekly to ensure it is functioning correctly.
2. **Servicing**: The system should be serviced by a competent person at least every six months.
3. **Cleaning**: Detectors should be kept clean and free from dust and obstructions.
4. **Repair**: Any faults should be promptly reported and repaired.
5. **Record Keeping**: Records should be kept of all tests, maintenance, and repairs.

Automatic fire detection systems are a key part of modern fire safety. They provide early warning of a fire, allowing for a quick response and a safe evacuation. However, they need to be carefully designed, maintained, and tested to ensure they will perform their critical function when needed.

5.6 Reducing False Alarms

False alarms, where a fire alarm system is activated without a real fire, are a significant problem. They cause disruption, waste resources, and can lead to complacency about real alarms [168]. Reducing false alarms is an important part of effective fire safety management.

Causes of False Alarms

There are many potential causes of false alarms [169]:

1. **Dust and Insects**: Dust and insects can enter smoke detectors and cause them to activate.
2. **Steam and Vapor**: Steam from bathrooms or kitchens, or vapor from certain industrial processes, can set off smoke detectors.
3. **Cooking Fumes**: Smoke from cooking can activate nearby detectors.
4. **Smoking**: Cigarette smoke can set off detectors, especially if smoking occurs near a detector.
5. **Aerosols**: Spray from aerosol cans, such as deodorants or cleaning products, can cause detectors to activate.
6. **Malicious Activation**: Manual call points can be maliciously activated by people as a prank or vandalism.
7. **Testing and Maintenance**: Improper testing or maintenance of the system can cause false alarms.

Strategies for Reducing False Alarms

There are several strategies that can be used to reduce the occurrence of false alarms [170]:

1. **Appropriate Detector Selection**: Choosing the right type of detector for each area, based on the potential fire risks and the likely causes of false alarms.
2. **Proper Installation**: Ensuring detectors are installed in the right locations, away from potential sources of false alarms.
3. **Regular Cleaning**: Regularly cleaning detectors to prevent the buildup of dust and dirt.
4. **Cooking Controls**: Using heat detectors instead of smoke detectors near kitchens, and ensuring good ventilation to remove cooking fumes.
5. **Smoking Policies**: Implementing and enforcing clear policies on smoking, including designating smoking areas away from detectors.
6. **Staff Training**: Training staff on the proper use of the fire alarm system, including how to avoid accidentally causing false alarms.

7. **Protective Covers**: Using protective covers on manual call points to prevent accidental or malicious activation.
8. **Investigating Causes**: Thoroughly investigating the cause of any false alarm and taking action to prevent a recurrence.

Managing False Alarms

Even with prevention strategies in place, some false alarms may still occur. It's important to have procedures in place to manage these incidents [171]:

1. **Investigate**: Every false alarm should be investigated to determine the cause.
2. **Record**: Details of the false alarm, including the cause if known, should be recorded.
3. **Report**: False alarms should be reported to the fire and rescue service if they were called out.
4. **Review**: Regular reviews of false alarms should be conducted to identify patterns or recurring issues that need to be addressed.

False alarms are a serious issue that can undermine the effectiveness of a fire warning system. By understanding the causes of false alarms, implementing prevention strategies, and managing incidents effectively, the number of false alarms can be minimized. This helps ensure that when the alarm does sound, people will react appropriately.

5.7 Staged Fire Alarms

Staged fire alarms, also known as phased evacuation or zone evacuation, are a method of evacuating a building in stages rather than all at once [172]. This approach can be useful in large, complex buildings where a total evacuation might be difficult or dangerous.

How Staged Fire Alarms Work

In a staged alarm system, the building is divided into several zones [173]. When a fire is detected, the alarm will initially sound only in the zone where the fire is located and in adjacent zones that might be immediately affected.

Occupants in these zones will be instructed to evacuate immediately, while those in other zones will be alerted that an incident is occurring but told to stand by for further instructions.

If the fire spreads or cannot be quickly controlled, the evacuation will be progressively extended to other zones until the entire building is evacuated.

Benefits of Staged Fire Alarms

Staged fire alarms have several potential benefits [174]:

1. **Reduced Congestion**: By evacuating only the immediately affected areas first, congestion on escape routes can be reduced, making evacuation safer and faster.
2. **Phased Response**: Staged alarms allow for a phased response to a fire. For example, the initial stages might focus on evacuating people, while later stages involve firefighting.
3. **Reduced Disruption**: In large buildings, a full evacuation can be highly disruptive. Staged alarms can minimize this disruption by only evacuating areas directly under threat.
4. **Targeted Assistance**: Staged alarms can help focus assistance efforts, such as helping people with mobility issues, on the areas that need it most.

Considerations for Staged Fire Alarms

Staged fire alarms are not suitable for all buildings. They are most appropriate for large, complex buildings with multiple compartments or fire zones, such as high-rise office buildings, hospitals, or shopping centers [175].

Some key considerations for staged fire alarms include:

1. **Clear Procedures**: There must be clear, well-understood procedures in place for how the staged evacuation will be managed and communicated.
2. **Robust Detection**: The fire detection system must be reliable and able to quickly identify the location of a fire.
3. **Communication**: There must be effective means of communicating with occupants, such as a voice alarm system, to provide instructions.
4. **Staff Training**: Staff, particularly fire wardens, must be well-trained in the staged evacuation procedures.
5. **Regular Drills**: Regular drills should be conducted to ensure that everyone understands the staged evacuation process.
6. **Flexibility**: The system should be flexible enough to allow for a full evacuation if necessary, such as if a fire spreads quickly.

Regulatory Requirements

The use of staged fire alarms must comply with relevant fire safety regulations and standards [176]. In the UK, for example, the design of a staged alarm system would need to be agreed upon with the local fire and rescue authority and would need to meet the requirements of the Regulatory Reform (Fire Safety) Order 2005 and relevant British Standards.

Staged fire alarms can be an effective way to manage evacuation in large, complex buildings. However, they require careful planning, robust systems, and regular training and drills to be effective. The decision to use a staged alarm system should be based on a thorough risk assessment and consultation with fire safety professionals.

5.8 Testing and Maintenance

Regular testing and maintenance of fire detection and warning systems is essential to ensure that they will function correctly in the event of a fire. It's a legal requirement for most non-domestic premises under the Regulatory Reform (Fire Safety) Order 2005 [177].

Testing Regime

The exact testing regime will depend on the type of system and the requirements of the manufacturer, but a typical regime might include [178]:

1. **Daily**: Check the panel for any fault indicators or alarms.
2. **Weekly**: Test a different manual call point each week, so that all are tested over a period of time. Check that the alarm sounds and the panel receives the signal.
3. **Monthly**: Test the system's backup batteries.
4. **Six-Monthly**: A competent person should carry out a more extensive test and inspection of the system.
5. **Annually**: The system should be tested and serviced by a competent contractor.

Maintenance

In addition to regular testing, fire detection and warning systems should also be subject to regular maintenance [179]:

1. **Cleaning**: Detectors should be periodically cleaned to prevent dust or dirt accumulation which could affect their sensitivity.
2. **Battery Replacement**: Backup batteries should be replaced in accordance with the manufacturer's recommendations, usually every 3-5 years.
3. **Repair**: Any faults or damages should be promptly repaired by a competent person.
4. **Replacement**: Detectors and other components should be replaced at the end of their recommended lifespan, even if they appear to be functioning correctly.

Records

It's important to keep records of all testing and maintenance carried out on the fire detection and warning system [180]. These records should include:

1. **Date**: The date the test or maintenance was carried out.
2. **Action**: What was done (e.g., which call point was tested, what maintenance was performed).
3. **Result**: The result of the test or maintenance (e.g., if the system functioned correctly, any faults found).
4. **Person**: The name of the person who carried out the test or maintenance.

These records can be kept in a logbook or electronically. They should be kept safe and be available for inspection by the fire and rescue service or other enforcing authority.

Competence

Testing and maintenance should only be carried out by competent persons [181]. This means people who have the necessary knowledge, skills, and experience to carry out the task safely and effectively.

For simple tests like weekly call point tests, this could be a trained member of staff. For more complex tasks like servicing or repairs, it should be a qualified fire alarm technician.

Importance

Regular testing and maintenance of fire detection and warning systems is crucial because [182]:

1. **Ensures Functionality**: It ensures that the system will function correctly in the event of a fire, providing early warning and allowing for safe evacuation.
2. **Identifies Faults**: It allows for the identification and repair of any faults before they can impact the system's performance.
3. **Prolongs Lifespan**: It helps prolong the lifespan of the system by keeping it in good working condition.
4. **Legal Compliance**: It's a legal requirement under fire safety legislation.

Neglecting the testing and maintenance of fire detection and warning systems can put lives at risk and lead to legal consequences for the responsible person. It should be a top priority in any fire safety management plan.

5.9 Guaranteed Power Supply

A reliable power supply is critical for fire detection and warning systems. If the power fails, the system must continue to function to ensure the safety of the building's occupants. This is why a guaranteed power supply is a legal requirement for these systems [183].

Primary Power Supply

The primary power supply for a fire detection and warning system is typically the building's main electricity supply [184]. This is usually sufficient to power the system under normal conditions.

However, the main power supply can fail for various reasons, such as a power outage, an electrical fault, or damage to the supply lines. Therefore, a backup power supply is essential.

Backup Power Supply

The backup power supply for a fire detection and warning system is typically provided by batteries [185]. These batteries will automatically take over if the main power supply fails.

The batteries used must be of sufficient capacity to power the system for a specified period, which is usually at least 24 hours in normal condition followed by 30 minutes in alarm condition [186]. This gives enough time for the main power supply to be restored or for the building to be safely evacuated.

The exact capacity of the batteries will depend on the size and complexity of the system. It must be calculated by a competent person based on the system's power consumption.

Battery Types

The most common types of batteries used for fire alarm system backup power are [187]:

1. **Sealed Lead-Acid**: These are the most widely used due to their reliability, long life, and relatively low cost. They require minimal maintenance but are quite heavy.
2. **Nickel-Cadmium**: These have a longer life and can tolerate more discharge cycles than lead-acid batteries. However, they are more expensive and require more complex charging systems.
3. **Lithium-Ion**: These are lighter and have a higher energy density than lead-acid batteries. They also have a longer life and require less maintenance. However, they are significantly more expensive.

The choice of battery will depend on factors such as the system's power requirements, the available space, and the budget.

Battery Maintenance

Backup batteries must be properly maintained to ensure they will function correctly when needed [188]. This includes:

1. **Regular Testing**: The batteries should be tested as part of the system's regular testing regime to ensure they are holding a charge and can power the system.
2. **Charging**: The batteries should be kept charged using an appropriate charging system. Overcharging or undercharging can reduce the batteries' life and capacity.
3. **Replacement**: The batteries should be replaced at the end of their recommended life, even if they appear to be functioning correctly. The lifespan will depend on the type of battery and the conditions of use.

Other Power Supply Considerations

In addition to the backup batteries, there are other considerations for ensuring a reliable power supply for fire detection and warning systems [189]:

1. **Surge Protection**: The system should be protected from electrical surges that could damage the components or cause false alarms.
2. **Electrical Isolation**: The system's power supply should be electrically isolated from other systems to prevent interference or damage.
3. **Monitoring**: The system should monitor the condition of the power supply and provide a warning if there are any issues, such as a loss of main power or a low battery charge.
4. **Maintenance**: All aspects of the power supply system should be regularly inspected and maintained by a competent person.

Importance

A reliable, guaranteed power supply is vital for fire detection and warning systems because [190]:

1. **Continuous Operation**: It ensures that the system will continue to function even if the main power fails, providing continuous protection.
2. **Early Warning**: It allows for early warning of a fire, even during a power outage, giving occupants more time to evacuate safely.
3. **Legal Compliance**: It's a legal requirement under fire safety legislation.

Without a guaranteed power supply, a fire detection and warning system would be rendered useless in the event of a power failure. This could have catastrophic consequences if a fire were to occur during this time. Therefore, ensuring a robust and reliable power supply should be a critical part of any fire safety strategy.

5.10 New and Altered Systems

When a new fire detection and warning system is installed, or an existing system is altered, it's important to ensure that it meets all the necessary standards and requirements. This is to ensure that the system will function effectively in the event of a fire and provide adequate protection for the building's occupants.

Design and Installation

The design and installation of a new or altered fire detection and warning system should be carried out by a competent person [191]. This is someone with the necessary knowledge, training, and experience to design and install the system in accordance with the relevant standards and regulations.

In the UK, this typically means a company or individual that is certified by a recognized third-party certification scheme, such as the Fire Industry Association (FIA) or the British Approvals for Fire Equipment (BAFE) [192].

The design of the system should be based on a thorough risk assessment of the building, taking into account factors such as [193]:

1. **The use and occupancy of the building**.
2. **The fire hazards present**.
3. **The size and layout of the building**.
4. **The need for early warning in case of a fire**.
5. **The requirements of the local building regulations and fire safety legislation**.

The installation should be carried out in accordance with the design and the manufacturer's instructions. It should also follow the relevant standards, such as BS 5839-1 for system design, installation, commissioning, and maintenance [194].

Commissioning

After installation or alteration, the fire detection and warning system should be commissioned to ensure that it functions as intended [195]. Commissioning involves a series of tests and checks to verify that:

1. **All components are installed correctly and function properly**.
2. **The system performs in accordance with the design specification**.
3. **The system interfaces correctly with other systems, such as the fire alarm panel, the evacuation system, and the fire suppression system**.
4. **The system provides adequate coverage and detection for the building**.

Commissioning should be carried out by a competent person, typically the installer, in accordance with the relevant standards, such as BS 5839-1 [196].

Certification

After commissioning, the installer should provide a certificate of compliance [197]. This certifies that the system has been designed, installed, and commissioned in accordance with the relevant standards and regulations.

The certificate should include details of:

1. **The system designs**.
2. **The components used**.
3. **The installation work carried out**.
4. **The commissioning tests performed**.
5. **Any variations from the standard or deviations from the design**.

The certificate is an important document that demonstrates the system's compliance and should be kept safe by the responsible person.

Handover

After certification, the system should be formally handed over to the responsible person [198]. This involves providing:

1. **The certificate of compliance**.
2. **The operating and maintenance instructions**.
3. **The logbook for recording tests and maintenance**.
4. **Training for the responsible person and other relevant staff**.

The handover is an opportunity for the installer to explain the system's operation and maintenance requirements and for the responsible person to ask any questions.

Ongoing Responsibilities

After handover, the responsible person has ongoing duties to ensure that the fire detection and warning system remains effective [199]. These include:

1. **Carrying out regular tests and maintenance in accordance with the instructions**.
2. **Keeping records of all tests, maintenance, and events in the logbook**.
3. **Arranging for any faults or defects to be rectified promptly by a competent person**.
4. **Ensuring that any changes to the building or its use are assessed for their impact on the system**.
5. **Providing training and information to staff and other relevant persons**.

These ongoing responsibilities are crucial to ensure that the fire detection and warning system continues to provide effective protection throughout its lifespan.

This concludes Unit 5: Fire Detection and Warning Systems. The next unit will cover Firefighting Equipment and Facilities.

UNIT 6: FIREFIGHTING EQUIPMENT AND FACILITIES

6.1 Portable Firefighting Equipment

Portable firefighting equipment, such as fire extinguishers and fire blankets, are an important part of a building's fire safety measures. They provide a first line of defence against small fires, allowing trained occupants to control or extinguish a fire in its early stages before it can grow and spread [200].

Types of Portable Firefighting Equipment

There are several types of portable firefighting equipment, each designed for use on specific classes of fire [201]:

1. **Water Extinguishers**: These are suitable for Class A fires involving solid materials such as wood, paper, and textiles. They work by cooling the burning material.
2. **Foam Extinguishers**: These can be used on Class A and B fires involving flammable liquids such as petrol, diesel, and oil. They work by smothering the fire and sealing vapors.
3. **Powder Extinguishers**: These are suitable for Class A, B, and C fires involving gases. They can also be used on live electrical equipment. They work by interrupting the chemical reaction of the fire.
4. **Carbon Dioxide (CO2) Extinguishers**: These are used for Class B fires and electrical fires. They work by displacing oxygen and cooling the fire.
5. **Wet Chemical Extinguishers**: These are specifically designed for Class F fires involving cooking oils and fats. They work by creating a soapy foam layer on the surface of the burning oil.
6. **Fire Blankets**: These are used to smother small fires or wrap around a person whose clothing is on fire.

As previously discussed, Barking and Dagenham College is transitioning to Firexo extinguishers, which can be used on all classes of fire. This simplifies the selection process and reduces confusion in an emergency.

Selection and Positioning

The selection and positioning of portable firefighting equipment should be based on a fire risk assessment of the building [202]. This should consider factors such as:

1. **The types of fire hazards present**.
2. **The size and layout of the building**.
3. **The occupancy and activities carried out**.
4. **The availability of trained personnel**.

As a general rule, portable firefighting equipment should be positioned [203]:

1. **Conspicuously**, so that it can be easily seen.
2. **Accessibly**, so that it can be quickly reached in an emergency.
3. **Near exits**, so that users have a clear escape route.
4. **Close to specific risks**, such as cooking appliances or electrical equipment.

The equipment should be mounted on wall brackets or placed in cabinets, with clear signage indicating its location and type.

Maintenance and Testing

Portable firefighting equipment must be regularly maintained and tested to ensure it will function correctly when needed [204]. This should include:

1. **Monthly visual checks** to ensure the equipment is in place, visible, and undamaged.
2. **Annual servicing** by a competent person in accordance with the manufacturer's instructions.
3. **Periodic discharge testing** (usually every 5 years) to test the operation of the equipment.

Records of all maintenance and testing should be kept in a logbook.

Training

Occupants who are expected to use portable firefighting equipment must be trained in its use [205]. This training should cover:

1. **The types of fire and which extinguishers to use**.
2. **How to operate the equipment safely**.
3. **The limitations of the equipment**.
4. **When to evacuate instead of fighting the fire**.

The training should be refreshed periodically and whenever new equipment or hazards are introduced.

Limitations

It's important to recognize the limitations of portable firefighting equipment [206]:

1. **Capacity**: Extinguishers only contain a limited amount of extinguishing agent and may not be sufficient for larger fires.
2. **Range**: The jet of an extinguisher has a limited range, typically 2-3 meters, so users must be close to the fire.
3. **Duration**: Extinguishers typically only discharge for a short period (usually less than 1 minute), so quick and decisive action is necessary.
4. **Safety**: Fighting a fire can be dangerous, and users must always put their own safety first. If the fire is too large, or if there is any doubt, occupants should evacuate immediately.

For these reasons, portable firefighting equipment should only be used on small, contained fires. They are not a substitute for professional firefighting services.

6.2 Number and Type of Extinguishers

The number and type of fire extinguishers provided in a building should be based on the fire risk assessment. This assessment should consider the size and use of the building, the nature of the contents and occupants, the fire hazards present, and the distances to be travelled to reach an extinguisher [207].

Number of Extinguishers

The number of extinguishers required will depend on the size and layout of the building. As a general rule [208]:

1. **Class A fires**: One 13A rated extinguisher for every 200 square meters of floor space.
2. **Class B fires**: One 21B rated extinguisher for every 100 square meters of floor space.
3. **Class C fires**: Determined by the size and nature of the risk.
4. **Class D fires**: Determined by the size and nature of the risk.
5. **Electrical fires**: One CO2 extinguisher for every 100 square meters of floor space containing electrical apparatus.

These are minimum requirements, and more extinguishers may be necessary depending on the fire risks present.

Type of Extinguishers

The type of extinguishers provided should be appropriate for the class of fire risk present [209]:

1. **Class A (Solids)**: Water, foam, or multi-purpose dry powder extinguishers.
2. **Class B (Liquids)**: Foam, dry powder, or CO2 extinguishers.
3. **Class C (Gases)**: Dry powder extinguishers.
4. **Class D (Metals)**: Specialist dry powder extinguishers.
5. **Electrical**: CO2 or dry powder extinguishers.
6. **Class F (Cooking Oils)**: Wet chemical extinguishers.

In many cases, multi-purpose extinguishers that can cover several classes of fire (like the Firexo extinguishers being introduced at Barking and Dagenham College) can be used to simplify the provision.

Distribution of Extinguishers

Extinguishers should be evenly distributed throughout the building, with [210]:

1. **Sufficient numbers** to ensure that no one needs to travel more than 30 meters to reach an extinguisher.
2. **Positioning** near exits, in conspicuous positions, on escape routes, and close to specific risks.
3. **Wall mounting** or placement in cabinets, with clear signage.
4. **Grouping** of different types of extinguishers where multiple classes of fire risk are present.

Specific Risks

Additional extinguishers may be necessary for specific fire risks, such as [211]:

1. **Server rooms or electrical switchrooms**: CO2 extinguishers.
2. **Kitchens**: Wet chemical extinguishers for cooking oil fires, plus a fire blanket.
3. **Workshops or laboratories**: Appropriate extinguishers for the materials used or stored.
4. **Fuel storage areas**: Foam or dry powder extinguishers.

These specific risks should be identified in the fire risk assessment and appropriate provision made.

Competent Advice

The selection, provision, and positioning of fire extinguishers should be based on the advice of a competent person, such as a fire risk assessor or a fire protection specialist [212]. They can provide expert guidance on the types and numbers of extinguishers required based on the specific risks and layout of the building.

It's important to note that fire extinguishers are only one part of a comprehensive fire safety strategy. They should be used in conjunction with other measures such as fire detection and alarm systems, compartmentation, escape routes, and staff training. The provision of extinguishers should be regularly reviewed as part of the ongoing fire risk assessment process.

6.3 Foam Extinguishers (Cream)

Foam fire extinguishers, identified by their cream colour coding, are a type of portable firefighting equipment suitable for use on Class A and Class B fires. They are particularly effective on liquid fires, such as those involving petrol, diesel, or oil [213].

How They Work

Foam extinguishers work by discharging a foam solution that spreads across the surface of the burning liquid. The foam creates a seal over the surface of the liquid, preventing flammable vapours from reaching the air and starving the fire of oxygen [214].

The foam also has a cooling effect, which helps to prevent re-ignition. Some foam extinguishers also contain an aqueous film-forming foam (AFFF) solution, which creates a thin film over the surface of the liquid, providing additional protection against re-ignition [215].

Advantages

Foam extinguishers have several advantages [216]:

1. **Effectiveness on liquid fires**: The smothering and sealing action of the foam is particularly effective on liquid fires.
2. **Prevents re-ignition**: The cooling effect and the AFFF film (where present) help to prevent re-ignition of the fire.
3. **Safe on electrical equipment**: Foam extinguishers are safe to use on electrical equipment, as long as the equipment is switched off and isolated.

Disadvantages

However, foam extinguishers also have some disadvantages [217]:

1. **Limited effectiveness on other classes of fire**: While they can be used on Class A fires (solids), they are not as effective as water extinguishers for this purpose.
2. **Contamination**: The foam can cause contamination and damage to sensitive equipment or materials.
3. **Cleaning**: The foam residue must be cleaned up after use, which can be a time-consuming process.

Using a Foam Extinguisher

To use a foam extinguisher [218]:

1. **Ensure it is the correct type** for the fire (Class A or B).
2. **Pull the safety pin** to break the tamper seal.
3. **Aim the nozzle** at the base of the fire.
4. **Squeeze the handles together** to discharge the foam.
5. **Sweep the nozzle** from side to side across the base of the fire until it is extinguished.

It's important to remember that foam extinguishers, like all portable firefighting equipment, have limitations. They are only suitable for small, contained fires and should not be used on larger fires or where there is a risk of the fire spreading rapidly.

Maintenance and Testing

Foam extinguishers should be subject to regular maintenance and testing to ensure they will function correctly when needed [219]. This includes:

1. **Monthly visual inspections** to check for any signs of damage, corrosion, or leakage.
2. **Annual servicing** by a competent person in accordance with the manufacturer's instructions.
3. **Periodic discharge testing** (usually every 5 years) to test the operation of the extinguisher.

Any extinguishers that fail the inspection or testing should be replaced or repaired immediately.

Training

Staff who are expected to use foam extinguishers should receive training in their use [220]. This should cover:

1. **The types of fire** that foam extinguishers can be used on.
2. **How to operate** the extinguisher safely and effectively.
3. **The limitations** of the extinguisher.
4. **When to evacuate** instead of fighting the fire.

The training should be refreshed periodically and whenever new equipment is introduced.

Positioning

Foam extinguishers should be positioned in areas where there is a risk of Class A or B fires, such as [221]:

1. **Garages or workshops** where flammable liquids are used or stored.
2. **Fuel storage areas**.
3. **Plant rooms** containing oil-filled machinery.

They should be wall-mounted or placed in cabinets, with clear signage indicating their location and type.

Foam extinguishers are a valuable part of a building's firefighting provision, particularly where there is a risk of liquid fires. However, they should always be used in accordance with the manufacturer's instructions and only by trained personnel. They are not a substitute for evacuation in the event of a larger or uncontrollable fire.

6.4 Powder Extinguishers (Blue)

Powder fire extinguishers, identified by their blue colour coding, are a versatile type of portable firefighting equipment. They can be used on Class A, B, and C fires, and are safe to use on live electrical equipment [222].

Types of Powder Extinguisher

There are two main types of powder extinguisher [223]:

1. **ABC Powder**: This is a multi-purpose extinguisher that can be used on Class A (solids), Class B (liquids), and Class C (gases) fires. It contains a mixture of monoammonium phosphate and ammonium sulfate.
2. **BC Powder**: This type is suitable for use on Class B (liquids) and Class C (gases) fires only. It contains sodium bicarbonate or potassium bicarbonate.

Both types of powder extinguisher are pressurized with nitrogen gas, which propels the powder out of the nozzle when the extinguisher is activated.

How They Work

Powder extinguishers work by interrupting the chemical reaction that takes place when fuels burn. The powder acts as a thermal blanket, smothering the fire and absorbing heat [224].

When the powder is discharged, it also creates a barrier between the fuel and the oxygen in the air. This starves the fire of one of the elements it needs to continue burning [225].

Advantages

Powder extinguishers have several advantages [226]:

1. **Versatility**: ABC powder extinguishers can be used on a wide range of fire types.
2. **Effectiveness**: Powder extinguishers are very effective at extinguishing fires quickly.
3. **Safe on electrical equipment**: Powder extinguishers can be used safely on live electrical equipment (although the equipment should be isolated if possible).

Disadvantages

However, powder extinguishers also have some significant disadvantages [227]:

1. **Messy**: The powder can create a significant mess, which can be difficult to clean up.
2. **Damaging**: The powder can be damaging to sensitive equipment and machinery.
3. **Visibility**: The powder can create a thick cloud, which can reduce visibility and make it difficult to see if the fire has been fully extinguished.
4. **Inhalation risk**: The powder can be an inhalation hazard, particularly in confined spaces.

Using a Powder Extinguisher

To use a powder extinguisher [228]:

1. **Ensure it is the correct type** for the fire.
2. **Pull the safety pin** to break the tamper seal.
3. **Aim the nozzle** at the base of the fire.
4. **Squeeze the handles together** to discharge the powder.
5. **Sweep the nozzle** from side to side across the base of the fire until it is extinguished.

It's important to ensure that the fire has been fully extinguished and is not smouldering before leaving the area.

Maintenance and Testing

Like all fire extinguishers, powder extinguishers should be subject to regular maintenance and testing [229]. This includes:

1. **Monthly visual inspections** to check for any signs of damage, corrosion, or leakage.
2. **Annual servicing** by a competent person in accordance with the manufacturer's instructions.
3. **Periodic discharge testing** (usually every 5 years) to test the operation of the extinguisher.

Powder extinguishers should be replaced or recharged immediately after use, even if they have not been fully discharged.

Training

Staff who are expected to use powder extinguishers should receive training in their use [230]. This should cover:

1. **The types of fire** that powder extinguishers can be used on.
2. **How to operate** the extinguisher safely and effectively.
3. **The limitations** of the extinguisher, including the potential mess and damage it can cause.
4. **When to evacuate** instead of fighting the fire.

The training should be refreshed periodically and whenever new equipment is introduced.

Positioning

Powder extinguishers should be positioned in areas where there is a risk of Class A, B, or C fires, or where there is electrical equipment [231]. This could include:

1. **Workshops or laboratories**.
2. **Plant rooms or boiler rooms**.
3. **Garages or storage areas**.

They should be wall-mounted or placed in cabinets, with clear signage indicating their location and type.

Powder extinguishers are a valuable part of a building's firefighting provision due to their versatility and effectiveness. However, they should be used with caution due to the potential mess and damage they can cause. As with all firefighting equipment, they should only be used by trained personnel and are not a substitute for evacuation in the event of a larger or uncontrollable fire.

6.5 Carbon Dioxide Extinguishers (Black)

Carbon dioxide (CO2) fire extinguishers, identified by their black colour coding, are a type of clean agent extinguisher. They are primarily used for Class B (liquid) and electrical fires, but can also be used on Class A (solid) fires in certain circumstances [232].

How They Work

CO2 extinguishers work by displacing the oxygen that a fire needs to burn. When the CO2 is discharged, it expands rapidly, cooling and smothering the fire [233].

CO2 is a non-conductive and clean agent, which means it does not leave any residue and is safe to use on electrical equipment (although the equipment should be isolated if possible) [234].

Advantages

CO2 extinguishers have several advantages [235]:

1. **Clean**: They do not leave any residue, which makes them ideal for use on sensitive equipment or in areas where contamination must be avoided.
2. **Non-conductive**: They are safe to use on live electrical equipment.
3. **No thermal shock**: Unlike water extinguishers, CO2 extinguishers do not cause thermal shock damage to electrical equipment.
4. **No pressure build-up**: CO2 extinguishers do not create a pressure build-up in the fire area, which can be a risk with some other types of extinguishers.

Disadvantages

However, CO2 extinguishers also have some disadvantages [236]:

1. **Limited effectiveness**: They are only effective on relatively small fires and may not be able to extinguish larger fires.
2. **No cooling effect**: Unlike water or foam extinguishers, CO2 extinguishers do not provide a cooling effect, which means there is a risk of re-ignition.
3. **Asphyxiation risk**: CO2 can cause asphyxiation in confined spaces or if inhaled in large quantities.
4. **Noise**: The discharge of a CO2 extinguisher can be very loud, which can be startling.

Using a CO2 Extinguisher

To use a CO2 extinguisher [237]:

1. **Ensure it is the correct type** for the fire (Class B or electrical).

2. **Pull the safety pin** to break the tamper seal.
3. **Aim the horn** at the base of the fire.
4. **Squeeze the handles together** to discharge the CO2.
5. **Sweep the horn** from side to side across the base of the fire until it is extinguished.

It's important to ensure that the fire has been fully extinguished and is not smouldering before leaving the area. If the fire re-ignites, the area should be evacuated immediately.

Maintenance and Testing

CO2 extinguishers should be subject to regular maintenance and testing [238]. This includes:

1. **Monthly visual inspections** to check for any signs of damage, corrosion, or leakage.
2. **Annual servicing** by a competent person in accordance with the manufacturer's instructions.
3. **Periodic weighing** to check the CO2 content (usually every 6-12 months).

CO2 extinguishers should be replaced or recharged immediately if they have been used or if they fail the weight check.

Training

Staff who are expected to use CO2 extinguishers should receive training in their use [239]. This should cover:

1. **The types of fire** that CO2 extinguishers can be used on.
2. **How to operate** the extinguisher safely and effectively.
3. **The limitations** of the extinguisher, including the risk of re-ignition and asphyxiation.
4. **When to evacuate** instead of fighting the fire.

The training should be refreshed periodically and whenever new equipment is introduced.

Positioning

CO2 extinguishers should be positioned in areas where there is a risk of Class B or electrical fires [240]. This could include:

1. **Server rooms or electrical switch rooms**.
2. **Laboratories or workshops with sensitive equipment**.
3. **Printing or reprographics areas**.

They should be wall-mounted or placed in cabinets, with clear signage indicating their location and type.

CO2 extinguishers are a valuable part of a building's firefighting provision for Class B and electrical fires. However, they have limitations and should only be used by trained personnel. They are not suitable for large fires or for use in confined spaces due to the risk of asphyxiation. As with all firefighting equipment, they are not a substitute for evacuation in the event of an uncontrollable fire.

6.6 Fixed Firefighting Installations

Fixed firefighting installations are systems that are permanently installed in a building to detect and suppress fires. They are designed to activate automatically in the event of a fire, providing an immediate response and helping to control the fire until the fire and rescue service arrives [241].

Types of Fixed Firefighting Installations

There are several types of fixed firefighting installations [242]:

1. **Sprinkler Systems**: These are the most common type of fixed installation. They consist of a network of water pipes and sprinkler heads that are activated by heat. When a fire occurs, the sprinklers in the affected area will open, discharging water onto the fire.
2. **Gaseous Suppression Systems**: These systems use inert gases (such as argon or nitrogen) or chemical agents (such as FM-200) to suppress fires. They work by displacing the oxygen that the fire needs to burn or by interrupting the chemical reaction of the fire. They are often used in areas where water damage from sprinklers would be unacceptable, such as in server rooms or museums.
3. **Water Mist Systems**: These systems use fine water sprays to control or suppress fires. The small water droplets create a large surface area for heat absorption and also evaporate quickly, displacing oxygen from the fire.
4. **Foam Systems**: These systems are designed for Class B (liquid) fires. They deliver a foam solution that spreads across the surface of the burning liquid, smothering the fire and preventing re-ignition.
5. **Powder Systems**: These systems use a fine powder (usually a mixture of monoammonium phosphate and ammonium sulphate) to suppress fires. They work in a similar way to portable powder extinguishers, interrupting the chemical reaction of the fire.

Design and Installation

The design and installation of fixed firefighting systems is a complex process that must take into account many factors, including [243]:

1. **The fire risk assessment**: The type and extent of the system should be based on a thorough assessment of the fire risks in the building.
2. **Building layout and use**: The system must be designed to provide adequate coverage for the specific layout and use of the building.
3. **Regulatory requirements**: The system must comply with relevant building regulations and fire safety legislation.
4. **Interaction with other systems**: The system must be compatible with and properly interfaced with other building systems, such as the fire alarm system, ventilation system, and elevator controls.

The design and installation should be carried out by competent professionals in accordance with relevant standards, such as BS EN 12845 for sprinkler systems [244].

Maintenance and Testing

Fixed firefighting installations must be regularly maintained and tested to ensure they will function correctly in the event of a fire [245]. This includes:

1. **Weekly tests**: A different section of the system should be tested each week to check for any faults.
2. **Monthly tests**: The system should be tested as a whole each month.
3. **Annual servicing**: The system should be fully serviced by a competent person each year.

Records of all tests and servicing should be kept in the fire logbook.

Staff Training

Staff should be trained in the purpose and operation of the fixed firefighting installations in their building [246]. This should include:

1. **How the system works** and what it is designed to do.
2. **How to interpret** any warnings or signals from the system.
3. **What to do** if the system activates, including evacuation procedures.
4. **Not to interfere** with the system or to hang items from sprinkler heads.

The training should be refreshed periodically and whenever changes are made to the system.

Limitations

While fixed firefighting installations provide a high level of fire protection, they do have limitations [247]:

1. **They are not a substitute for other fire safety measures**, such as passive fire protection, fire compartmentation, and safe evacuation procedures.
2. **They may not be suitable for all types of fire**, depending on the system. For example, sprinklers are not suitable for fires involving certain chemicals or metals.
3. **They can be damaged or impaired**, for example by building works or by incorrect maintenance. Regular testing and maintenance are critical to ensure the system remains operational.

Despite these limitations, fixed firefighting installations are a crucial part of the fire safety strategy in many buildings. They can control or suppress a fire in its early stages, preventing it from spreading and giving occupants more time to evacuate safely. They are a valuable supplement to, but not a replacement for, other fire safety measures and procedures.

6.7 Other Facilities (Including those for Firefighters)

In addition to portable and fixed firefighting equipment, buildings may have other facilities that are used for firefighting or that assist firefighters in their operations. These facilities are an important part of the overall fire safety provision in the building.

Dry and Wet Rising Mains

Dry and wet rising mains are vertical pipes installed in tall buildings to provide a water supply for firefighting on upper floors [248].

1. **Dry Rising Mains**: These are empty pipes that can be connected to a water source by firefighters when they arrive. They have inlet connections at ground level and outlet valves on each floor.
2. **Wet Rising Mains**: These are pipes that are permanently filled with water. They have outlet valves on each floor and are pressurized by a pump.

Rising mains allow firefighters to quickly establish a water supply on upper floors without having to carry hoses up the stairs. They are required in buildings over 18m tall in England and Wales [249].

Firefighting Lifts

Firefighting lifts are elevators that are designed to be used by firefighters during an emergency [250]. They have special features such as:

1. **Dual power supplies** to ensure they can still operate if the main power fails.
2. **Fire resistant doors and controls**.
3. **Intercoms** for communication with the fire command center.
4. **Specific firefighting controls** that override normal operation.

Firefighting lifts allow firefighters to quickly reach upper floors and to transport equipment. They are required in buildings over 18m tall in England and Wales [251].

Firefighting Lobbies and Shafts

Firefighting lobbies and shafts are protected areas that provide access to the firefighting stairs and lifts [252].

1. **Firefighting Lobbies**: These are fire-resisting enclosures that provide a safe space for firefighters to set up operations on each floor. They have fire-resisting doors and are kept clear of combustibles.
2. **Firefighting Shafts**: These are vertical enclosures that contain the firefighting stairs, lifts, and rising mains. They are constructed to have a high level of fire resistance to protect the firefighting facilities.

Firefighting lobbies and shafts are designed to provide a safe and protected route for firefighters to access all parts of the building.

Smoke Control Systems

Smoke control systems are designed to control the movement of smoke in a building during a fire [253]. They can operate in different ways, such as:

1. **Pressurisation Systems**: These create positive pressure in firefighting shafts and stairwells to keep them clear of smoke.
2. **Smoke Ventilation Systems**: These remove smoke from corridors or lobbies to keep escape routes clear.
3. **Smoke Extraction Systems**: These remove smoke directly from the fire area to improve visibility and reduce damage.

Smoke control systems are a complex area of fire engineering and must be carefully designed to work with the building's layout and other fire safety systems.

Hydrants and Water Supplies

Buildings must have adequate water supplies for firefighting. This is usually provided by fire hydrants, which can be [254]:

1. **Public Hydrants**: These are water mains in the street that can be accessed by firefighters.
2. **Private Hydrants**: These are hydrants on the building's own water supply, usually located around the perimeter of the building.

The location and specification of hydrants must be agreed with the local fire and rescue service to ensure they are sufficient for the building's size and risk profile.

Firefighter Access

Buildings must have provisions for firefighter access, including [255]:

1. **Access Roads**: These allow fire appliances to get close to the building. They must be of sufficient width and load-bearing capacity.
2. **Hard Standing Areas**: These are paved areas where fire appliances can be safely parked and operated.
3. **Access Points**: These are designated entrances for firefighters, usually at ground level and on upper floors via the firefighting lobbies.

The access provisions must be agreed with the local fire and rescue service and kept clear at all times.

Information for Firefighters

Buildings should have information available for firefighters, including [256]:

1. **Fire Safety Plan**: This shows the layout of the building, the location of firefighting facilities, and any special risks.
2. **Premises Information Box**: This is a secure box containing keys, floor plans, and other information for firefighters.
3. **Firefighter Switch**: This is a switch that allows firefighters to take control of the building's systems, such as the lifts and ventilation.

The information must be kept up to date and readily accessible to firefighters in an emergency.

Staff Training

Staff should be trained in the location and use of firefighting facilities, including [257]:

1. **Rising Mains**: How to identify and operate the outlet valves.
2. **Firefighting Lifts**: How to recognize and use the special controls.
3. **Smoke Control Systems**: The purpose and operation of the systems.

4. **Access Points**: The location of firefighter access points and how to guide firefighters to them.

The training should be refreshed periodically and whenever changes are made to the facilities.

The provision of firefighting facilities must be based on a comprehensive risk assessment and designed in consultation with the local fire and rescue service. They must be regularly inspected, tested, and maintained to ensure they are always ready for use. While these facilities are primarily for the use of professional firefighters, it's important that staff understand their purpose and how to interact with them safely in an emergency situation.

6.8 Access for Fire Engines and Firefighters

Ensuring that fire engines and firefighters have quick and unobstructed access to a building is a critical part of fire safety. Delays in reaching a fire can allow it to grow and spread, making it more difficult and dangerous to extinguish.

Access Roads

Access roads are designed to allow fire engines to get close to a building. They must meet certain specifications to ensure they are suitable for fire engines [258]:

1. **Width**: Access roads should be at least 3.7m wide between kerbs. This allows fire engines to pass each other if necessary.
2. **Height Clearance**: There should be a minimum clear height of 3.7m to allow for the passage of fire engines.
3. **Turning Radius**: Access roads should have a minimum turning radius of 6.55m to the center line. This allows fire engines to navigate corners and turnings.
4. **Load Bearing Capacity**: Access roads and any bridges or ramps must be able to support a minimum weight of 12.5 tonnes. This ensures they can bear the weight of a fire engine.
5. **Gradient**: Access roads should have a maximum gradient of 1 in 8 (12.5%) for a maximum distance of 9m. This ensures fire engines can safely navigate inclines.

The specific requirements may vary depending on the local fire and rescue service and the building's risk profile. The access roads should be kept clear of obstructions at all times and clearly marked as fire access routes.

Hard Standing Areas

Hard standing areas are paved areas where fire engines can be safely parked and operated. They should be [259]:

1. **Adjacent to the Building**: Hard standing areas should be as close as possible to the building, ideally within 18m of an entrance usable by firefighters.
2. **Sufficient Size**: The hard standing should be large enough to accommodate at least one fire engine. A typical fire engine is about 8m long and 3m wide.
3. **Load Bearing**: The hard standing should be able to support the weight of a fire engine (at least 12.5 tonnes).
4. **Clearly Marked**: The hard standing should be clearly marked as a fire engine parking area and kept clear of obstructions at all times.

The location and specification of hard standing areas should be agreed with the local fire and rescue service.

Access Points

Access points are the designated entrances for firefighters to use in an emergency. They should be [260]:

1. **Clearly Identifiable**: Access points should be clearly marked with a firefighter symbol and the words "Fire Exit Keep Clear".
2. **Easily Openable**: Access points should be easily openable from the outside. This may involve a key lock that can be opened by firefighters.

3. **Unobstructed**: Access points and the routes to them should be kept clear of obstructions at all times.
4. **Close to Firefighting Facilities**: Access points should be close to firefighting facilities such as dry rising mains and firefighting lifts.

Access points should be provided at ground level and on upper floors (via firefighting lobbies) in tall buildings.

Premises Information Box

A premises information box is a secure box located at the designated firefighter entrance point. It contains information and equipment to assist firefighters, such as [261]:

1. **Building Plans**: These show the layout of the building, including the location of firefighting facilities and any special risks.
2. **Keys**: Keys to access the building and any locked firefighting facilities.
3. **Firefighter Switch**: A key to operate the firefighter switch, which allows firefighters to take control of the building's systems.
4. **Hazardous Materials Information**: Information about any hazardous materials stored in the building.

The premises information box must be kept up to date and should be checked regularly to ensure all contents are present and in working order.

Liaison with Fire and Rescue Service

The provision of access for fire engines and firefighters should be planned in liaison with the local fire and rescue service [262]. They can provide expert advice on the specific requirements for the building based on its size, layout, and risk profile.

It's important to involve the fire and rescue service early in the design process for new buildings and to consult them on any changes to access arrangements in existing buildings.

The access provisions must be regularly inspected and maintained to ensure they are always available and fit for use. This includes ensuring that access roads and hard standing areas are kept clear, that access points are unobstructed and openable, and that premises information boxes are up to date.

Providing clear and unobstructed access for fire engines and firefighters is a vital component of a building's fire safety strategy. It ensures that in the event of a fire, the fire and rescue service can quickly reach the building and begin their firefighting and rescue operations without delay.

6.9 Smoke Control Systems

Smoke control systems are designed to control the movement of smoke within a building in the event of a fire. They are an important part of a building's fire safety strategy, as smoke is a major cause of death and injury in fires and can also hinder evacuation and firefighting operations.

Objectives of Smoke Control

The main objectives of smoke control are [263]:

1. **Life Safety**: To maintain escape routes free from smoke, allowing occupants to safely evacuate the building.
2. **Firefighting Access**: To assist firefighting operations by keeping access routes clear of smoke and improving visibility.
3. **Property Protection**: To reduce smoke damage to the building and its contents.

Smoke control systems work in conjunction with other fire safety measures, such as compartmentation and fire suppression systems, to achieve these objectives.

Types of Smoke Control Systems

There are several types of smoke control systems, each with its own method of operation [264]:

1. **Natural Ventilation**: This uses the buoyancy of hot smoke to drive it out of the building through high-level vents. Cool fresh air enters the building at low level to replace the expelled smoke.
2. **Powered Ventilation**: This uses fans to create a pressure differential that drives smoke out of the building. The system can be designed to extract smoke from specific areas or to pressurize escape routes to keep them clear of smoke.
3. **Pressurization**: This uses fans to create positive pressure in escape routes and firefighting shafts, preventing smoke from entering these areas.
4. **Smoke Curtains**: These are barriers that drop down from the ceiling to contain smoke in a specific area, preventing it from spreading to other parts of the building.

The choice of smoke control system will depend on the specific characteristics and risk profile of the building.

Design of Smoke Control Systems

The design of smoke control systems is a complex area of fire engineering. It must take into account a range of factors, including [265]:

1. **Building Layout**: The size, shape, and internal layout of the building, including the location of escape routes and compartment boundaries.
2. **Fire Risk**: The potential locations and characteristics of fires that could occur in the building.
3. **Occupancy**: The number, distribution, and characteristics of the people in the building, including their ability to evacuate independently.
4. **Ventilation**: The natural and mechanical ventilation of the building, including the location and size of windows, doors, and shafts.
5. **Interaction with Other Systems**: The interaction of the smoke control system with other systems such as the fire alarm, sprinklers, and HVAC.

The design of the smoke control system must be based on a comprehensive risk assessment and must be modelled and tested to ensure it will perform as intended in different fire scenarios.

Operation of Smoke Control Systems

Smoke control systems can be operated in different ways [266]:

1. **Automatic Activation**: The system activates automatically in response to a signal from the fire alarm system.
2. **Manual Activation**: The system can be activated manually by a trained operator from a control panel.
3. **Phased Activation**: The system can be activated in phases, with different parts of the system operating at different times based on the location and progression of the fire.

The operation of the smoke control system must be coordinated with the building's evacuation procedures and firefighting operations.

Maintenance and Testing

Smoke control systems must be regularly maintained and tested to ensure they will operate effectively in the event of a fire [267]:

1. **Weekly Tests**: The system should be visually inspected each week to check for any obvious faults or damage.
2. **Monthly Tests**: The system should be activated each month to check it operates correctly.
3. **Annual Tests**: The system should be fully tested and serviced by a competent person each year.

Records of all tests and maintenance should be kept in the building's fire logbook.

Interaction with Firefighting Operations

Smoke control systems can assist firefighting operations by [268]:

1. **Improving Visibility**: By removing smoke from the fire area, smoke control systems can improve visibility for firefighters, allowing them to locate and extinguish the fire more quickly.
2. **Protecting Escape Routes**: By pressurizing escape routes or extracting smoke from them, smoke control systems can help keep these routes clear for firefighters to use.
3. **Containing Smoke**: By using smoke curtains or extraction to contain smoke in certain areas, smoke control systems can help prevent the spread of smoke to other parts of the building.

Firefighters should be familiar with the operation of the smoke control system in the building and should be able to override the system if necessary, using the firefighter's switch.

Smoke control systems are a complex but crucial part of a building's fire safety strategy. They require careful design, regular maintenance, and close coordination with other fire safety systems and procedures. When designed and operated correctly, they can play a vital role in saving lives and assisting firefighting operations in the event of a fire.

This concludes Unit 6: Firefighting Equipment and Facilities. The next unit will cover Emergency Escape Routes.

UNIT 7: EMERGENCY ESCAPE ROUTES

7.1 The Type and Number of People Using the Premises

The design of emergency escape routes must take into account the type and number of people likely to be in the building. This is to ensure that the escape routes are sufficient to allow all occupants to evacuate safely in the event of a fire.

Occupancy Levels

The occupancy level of a building is the maximum number of people expected to be in the building at any one time. This is determined by the use of the building and the floor area [269].

For example:

1. **Office**: 6 square meters per person
2. **Retail**: 2 square meters per person
3. **Assembly Halls**: 0.5 square meters per person

These figures are from the UK Building Regulations Approved Document B and are used to calculate the minimum required width and capacity of escape routes.

Occupancy Type

The type of occupants in a building can affect their ability to evacuate quickly and safely. Factors to consider include [270]:

1. **Age**: Children and elderly people may move more slowly and require assistance.
2. **Mobility**: People with disabilities or impaired mobility may need more time and space to evacuate.
3. **Familiarity**: Visitors or customers may be less familiar with the building layout and escape routes than regular occupants.
4. **Sleeping Risk**: Occupants of residential buildings or hotels may be asleep and take longer to respond to an alarm.
5. **Vulnerability**: Some occupants, such as patients in a hospital or residents in a care home, may be particularly vulnerable due to age, illness, or dependency.

The escape routes must be designed to accommodate the specific needs of the expected occupants. This may involve providing additional assistance, wider corridors and doors, or refuge areas for people who cannot use stairs.

Calculation of Escape Route Capacity

The capacity of an escape route is the number of people that can pass through it in a given time. This is determined by the width of the route and the rate of travel [271].

The minimum required width of an escape route is:

1. **Corridors**: 1200mm for up to 110 people
2. **Doors**: 750mm for up to 110 people
3. **Stairs**: 1000mm for up to 190 people

These figures are from the UK Building Regulations Approved Document B. They are based on a rate of travel of 40 people per minute per unit width (520mm).

The number and distribution of escape routes must be sufficient to ensure that all occupants can reach a place of safety within a reasonable time, typically 2-3 minutes for small buildings and 5-8 minutes for larger buildings [272].

Merging Flows

Where escape routes merge, such as at the top of stairs or at final exits, the combined width must be sufficient to accommodate the merging flows [273].

For example, if two stairs with a combined capacity of 380 people merge into a final exit corridor, that corridor must be wide enough to accommodate 380 people (i.e., at least 1900mm wide).

Discounting of Escape Routes

To ensure there are always sufficient escape routes available, the design must allow for the possibility that one route may be unavailable due to fire or smoke [274].

For buildings with more than one floor and more than one staircase:

1. **Storeys with an occupancy of over 60** should have at least two escape stairs. The loss of one stair should not prevent the occupants from reaching a place of safety.
2. **Storeys with an occupancy of 60 or less** can have a single escape stair if the travel distance to the stair is within limits.

Travel distance limits ensure that occupants can reach an escape route quickly. They vary depending on the risk level of the building and whether there is a single direction of escape or multiple directions.

Management of Escape Routes

The management of the building must ensure that escape routes are always available and usable [275]:

1. **Obstructions**: Escape routes must be kept clear of obstructions and combustible materials.
2. **Doors**: Fire doors must be kept closed and fitted with self-closing devices.
3. **Signage**: Escape routes must be clearly signed with directional signs and emergency exit signs.

4. **Lighting**: Adequate lighting, including emergency lighting, must be provided to illuminate escape routes.
5. **Maintenance**: All elements of the escape routes, including doors, signs, and lighting, must be regularly inspected and maintained.

Any deficiencies that could affect the ability of occupants to evacuate safely must be promptly addressed.

Information and Training

Occupants must be provided with information and training on the escape routes and evacuation procedures [276]:

1. **Fire Action Notices**: These should be displayed prominently, detailing the action to take in case of fire.
2. **Fire Drills**: Regular fire drills should be conducted to familiarize occupants with the escape routes and procedures.
3. **Personal Emergency Evacuation Plans (PEEPs)**: These should be developed for individuals who may need assistance to evacuate.
4. **Staff Training**: Staff should be trained in their roles and responsibilities in an evacuation, including directing occupants to escape routes and assisting those needing help.

Effective communication and training ensure that all occupants know what to do in an emergency and can use the escape routes quickly and safely.

The design and management of escape routes is a critical part of a building's fire safety strategy. It must be based on a thorough assessment of the occupancy profile and be continually reviewed to ensure it remains adequate as the use of the building changes over time. The ultimate goal is to ensure that all occupants can evacuate to a place of safety in the event of a fire.

7.2 The Age and Construction of the Premises

The age and construction of a building can have a significant impact on the design and adequacy of its escape routes. Older buildings may have been constructed to different standards and may have features that can impede safe evacuation.

Pre-20th Century Buildings

Buildings constructed before the 20th century, particularly those with historical or architectural significance, can present unique challenges for fire safety [277]:

1. **Timber Construction**: Many old buildings have timber frames, floors, and stairs, which can contribute to the spread of fire.
2. **Narrow Passages**: Older buildings often have narrow corridors, stairways, and doorways that can impede evacuation and access for firefighters.
3. **Unprotected Openings**: The lack of fire-resisting doors, walls, and floors can allow smoke and fire to spread quickly.
4. **Insufficient Exits**: Older buildings may not have enough exits or the exits may not be adequately separated.
5. **Altered Layout**: The original layout may have been altered over time, creating dead ends or confusing routes.

When upgrading fire safety in historical buildings, care must be taken to balance the needs of preservation with the safety of occupants. This may involve innovative solutions such as reversible modifications, wireless alarm systems, or the use of fire-retardant treatments on wooden elements.

20th Century Buildings

Buildings constructed in the 20th century may have been built to varying standards depending on the building regulations in force at the time:

1. **Pre-1970s**: Buildings from this era may lack features such as fire compartmentation, protected escape routes, and adequate smoke ventilation [278].
2. **1970s-1990s**: Buildings from this period may have some fire safety features but may not meet current standards. For example, they may have narrow stairs or long travel distances [279].

3. **Post-1990s**: Buildings constructed after the 1990s are likely to have been designed to modern fire safety standards, with features such as fire-resisting construction, protected escape routes, and adequate exit capacity [280].

When assessing the adequacy of escape routes in 20th century buildings, it's important to consider the specific regulations that applied at the time of construction and any subsequent alterations or changes of use.

Modern Buildings

Modern buildings are designed to current fire safety standards, which include specific requirements for escape routes [281]:

1. **Fire Resistance**: Escape routes must be protected by fire-resisting construction to prevent the spread of fire and smoke.
2. **Compartmentation**: The building must be divided into fire compartments to limit the spread of fire and smoke.
3. **Travel Distances**: The distance from any point to the nearest escape route must be within prescribed limits.
4. **Exit Capacity**: The number and width of escape routes must be sufficient for the occupancy of the building.
5. **Smoke Control**: There must be means to ventilate smoke from escape routes, such as automatic smoke vents.

Despite being designed to modern standards, escape routes in modern buildings can still be compromised by poor management practices, such as blocked exit doors, obstructed corridors, or inadequate maintenance of fire safety systems.

Alterations and Changes of Use

When a building undergoes alterations or a change of use, the escape routes must be reassessed to ensure they remain adequate [282]:

1. **Material Alterations**: Alterations that affect the structure, fire resistance, or layout of the building may require upgrades to the escape routes.
2. **Change of Use**: If the use of the building changes, such as from office to residential, the escape routes must be reassessed based on the new occupancy profile.
3. **Increase in Occupancy**: If the alterations result in an increase in occupancy, the capacity of the escape routes must be rechecked.
4. **Reduction in Escape Routes**: Any alterations that reduce the number or width of escape routes, such as the removal of a staircase, must be carefully assessed.

Building regulations typically require that any alterations meet the standards for new buildings as far as is reasonably practicable. Where this is not possible, compensatory measures may be required, such as additional fire protection or management procedures.

Fire Risk Assessment

Regardless of the age or construction of the building, a comprehensive fire risk assessment is essential to ensure the adequacy of the escape routes [283]. This should consider:

1. **The construction and condition** of the building, including the fire resistance of walls, floors, and doors.
2. **The layout and complexity** of the escape routes, including travel distances and any potential obstructions.
3. **The occupancy profile**, including the number, type, and distribution of people in the building.
4. **The fire hazards present**, including sources of ignition, fuel, and oxygen.
5. **The fire safety systems in place**, including fire detection, alarm, suppression, and smoke control.

The fire risk assessment should be regularly reviewed and updated to reflect any changes in the building, its use, or its occupancy. Any deficiencies in the escape routes identified by the assessment must be addressed promptly.

The age and construction of a building can present both challenges and opportunities for the design and management of escape routes. While older buildings may have inherent limitations, a thorough understanding of their construction and a creative approach to fire safety can help ensure the safety of occupants. In all cases, regular fire risk assessments and proactive management are key to maintaining the effectiveness of escape routes over the life of the building.

7.3 The Number of Escape Routes and Exits

The number of escapes routes and exits required in a building is determined by several factors, including the occupancy level, the travel distances, and the need to provide alternative routes in case one becomes blocked by fire or smoke.

Number of Escape Routes

The number of escape routes required depends on the size and complexity of the building [284]:

1. **Single Escape Routes**: In some small buildings, a single escape route may be acceptable if the travel distance is within limits and the route is protected from fire and smoke.
2. **Two Escape Routes**: Most buildings will require at least two escape routes, so that if one route becomes unavailable, occupants can use the other.
3. **Multiple Escape Routes**: Large, complex buildings may require multiple escape routes to ensure that all occupants can evacuate safely within the required time.

The exact number of escape routes required will be determined by the fire risk assessment and the relevant building regulations.

Capacity of Escape Routes

The capacity of each escape route must be sufficient to accommodate the number of people expected to use it in an emergency [285]. This is determined by the width of the route and the rate of travel.

For example, a staircase with a width of 1000mm can accommodate up to 190 people. If the occupancy of the floor served by the staircase exceeds 190, either the width of the staircase must be increased or additional staircases must be provided.

The capacity of the escape routes must take into account the possibility that one route may be unavailable due to fire or smoke. The remaining routes must be sufficient to evacuate all occupants safely.

Distribution of Escape Routes

Escape routes must be distributed around the building to provide alternative means of escape from all areas [286]. This is to ensure that if one route becomes blocked, occupants can use another route to reach safety.

The distribution of escape routes should ensure that [287]:

1. **Travel Distances**: The distance from any point to the nearest escape route is within acceptable limits.
2. **Dead Ends**: There are no dead ends where occupants could become trapped if their only escape route is blocked.
3. **Alternative Routes**: There are alternative routes from all areas, so that occupants have a choice of routes in case one becomes unavailable.
4. **Separation**: Escape routes are adequately separated by fire-resisting construction to prevent a fire in one area from affecting the use of routes in another area.

The specific layout and distribution of escape routes will depend on the design of the building and the findings of the fire risk assessment.

Final Exits

Escape routes must lead to a final exit that discharges directly to a place of safety outside the building [288]. The number and width of final exits must be sufficient to accommodate the number of people using the escape routes.

The final exits should be [289]:

1. **Easily Identifiable**: Clearly marked with emergency exit signs.
2. **Easily Openable**: Readily openable from the inside without the use of a key.
3. **Unobstructed**: Leading directly to a place of safety without any obstruction or hazard.
4. **Adequately Separated**: Sufficiently separated from each other to minimize the risk of a fire affecting more than one exit.
5. **Suitably Located**: Positioned to avoid occupants having to pass through a high-risk area to reach safety.

The suitability and sufficiency of the final exits should be regularly assessed as part of the ongoing fire risk assessment process.

Calculating the Number of Escape Routes and Exits

The calculation of the required number of escape routes and exits is a complex process that must take into account multiple factors [290]:

1. **Occupancy Level**: The number of people expected to be in the building at any one time.
2. **Floor Area**: The size and layout of each floor of the building.
3. **Travel Distances**: The distance from any point to the nearest escape route.
4. **Stair Capacity**: The number of people that can use a staircase in a given time.
5. **Merging Flows**: The effect of escape routes merging at staircases or final exits.
6. **Discounting**: The need to discount one escape route to allow for the possibility of it being unavailable.

These factors are used in conjunction with the relevant building regulations and fire safety standards to determine the required number and distribution of escape routes and exits.

Professional Advice

Due to the complexity of the calculations and the importance of getting it right, it's recommended to seek the advice of a qualified fire safety professional when designing or assessing escape routes [291].

They can provide expert guidance on the application of the regulations and standards, and can use advanced modelling techniques to simulate evacuation scenarios and test the adequacy of the escape routes.

Providing an adequate number of well-designed and well-maintained escape routes and exits is a critical component of a building's fire safety strategy. It ensures that all occupants can evacuate quickly and safely in the event of a fire, regardless of where the fire starts or how it develops. Regular fire risk assessments and proactive management are essential to ensure that the escape routes remain fit for purpose over the lifetime of the building.

7.4 The Management of Escape Routes

Having an adequate number of well-designed escape routes is only part of the fire safety equation. Equally important is how these routes are managed on a day-to-day basis to ensure they are always available and fit for use in an emergency.

Keeping Escape Routes Clear

One of the most fundamental aspects of managing escape routes is keeping them clear of obstructions and combustible materials [292]. This includes:

1. **Corridors and Stairways**: These should be kept free of any furniture, equipment, or stored items that could impede evacuation or provide fuel for a fire.
2. **Fire Doors**: The self-closing mechanisms on fire doors should be regularly checked and the doors should never be wedged open.
3. **Final Exits**: The routes to final exits, and the exits themselves, should be kept clear. This may require regular checks to remove any build-up of waste or stored items outside the building.

Any obstructions identified should be removed immediately and the importance of keeping escape routes clear should be regularly reinforced to all building occupants.

Signage and Lighting

Escape routes must be clearly signed and adequately lit to assist occupants in evacuating safely [293]:

1. **Emergency Exit Signs**: These should be provided at all key decision points and changes of direction to guide occupants to the nearest escape route.
2. **Directional Signs**: These should be provided along escape routes to confirm the direction of travel to the final exit.
3. **Emergency Lighting**: Adequate emergency lighting must be provided to illuminate escape routes in the event of a power failure. This includes both primary illumination of the routes and secondary lighting of exit signs.

The signage and lighting should be regularly checked to ensure it is in good condition and functioning correctly. Any defects should be repaired immediately.

Fire Detection and Alarm Systems

An effective fire detection and alarm system is crucial for alerting occupants to the need to evacuate and activating any automatic fire protection measures [294]:

1. **Detectors**: The type and location of detectors should be appropriate for the fire hazards present and should provide early warning of a fire.
2. **Alarms**: The alarm system should provide an audible warning throughout the building, with additional visual or tactile alarms where necessary for occupants with hearing impairments.
3. **Control Panel**: The fire alarm control panel should be located in a secure but accessible location and should be regularly checked for any faults or alarms.

The fire detection and alarm system should be regularly tested and maintained by a competent person to ensure it will function correctly in an emergency.

Evacuation Plans and Procedures

Building occupants must be provided with clear information and instruction on what to do in the event of a fire [295]:

1. **Fire Action Notices**: These should be displayed prominently throughout the building, detailing the action to take on discovering a fire or hearing the alarm.
2. **Evacuation Plans**: Floor plans showing the location of escape routes and fire safety equipment should be provided in key locations.
3. **Fire Drills**: Regular fire drills should be conducted to familiarize occupants with the evacuation procedures and identify any issues with the escape routes.
4. **Personal Emergency Evacuation Plans (PEEPs)**: These should be developed for any occupants who may need assistance to evacuate, detailing the specific arrangements for their safe evacuation.

The evacuation plans and procedures should be regularly reviewed and updated to ensure they remain effective and accommodate any changes in the building or its occupancy.

Staff Training

Staff play a crucial role in managing escape routes and assisting occupants to evacuate safely. They should receive regular training on [296]:

1. **Fire Safety Awareness**: The fire hazards present in the building and the importance of good fire safety practices.
2. **Evacuation Procedures**: Their specific roles and responsibilities in an evacuation, including sweeping areas, guiding occupants to escape routes, and assisting those needing help.
3. **Fire Fighting Equipment**: The location and use of portable firefighting equipment, such as extinguishers and fire blankets.
4. **Fire Safety Systems**: The operation of fire safety systems, such as the fire alarm panel, emergency lighting, and smoke control.

Staff training should be refreshed regularly and whenever there are changes to the building, its occupancy, or the fire safety arrangements.

Maintenance and Testing

All elements of the escape routes and associated fire safety systems must be regularly inspected, tested, and maintained to ensure they will function correctly in an emergency [297]:

1. **Escape Route Inspections**: Daily checks to ensure escape routes are clear and weekly checks of signage, lighting, and fire doors.
2. **Fire Detection and Alarm Systems**: Weekly tests of manual call points and regular servicing by a competent contractor.
3. **Emergency Lighting**: Monthly functional tests and annual full discharge tests.
4. **Fire Fighting Equipment**: Monthly visual checks and annual servicing by a competent contractor.
5. **Smoke Control Systems**: Weekly tests of manual controls and annual servicing by a competent contractor.

Any defects identified during inspections or tests should be rectified immediately. Records of all inspections, tests, and maintenance should be kept in the building's fire safety logbook.

Ongoing Review

The management of escape routes is not a one-time activity but an ongoing process that requires continual vigilance and proactive intervention. The fire risk assessment should be regularly reviewed to ensure that the escape routes remain adequate for the building's use and occupancy [298].

Any changes to the building, its contents, or its occupancy should trigger a review of the escape routes. This could include alterations to the layout, a change of use, an increase in occupancy, or the introduction of new fire hazards.

Where deficiencies are identified, either through the risk assessment process or from practical experience, these should be addressed promptly. This may require physical alterations to the building, updates to the fire safety systems, or changes to the management procedures.

Effective management of escape routes is vital to ensure that they will be available and usable in an emergency. It requires a combination of good design, robust systems, clear procedures, and well-trained people. Most importantly, it requires a proactive and ongoing commitment from building managers and occupants alike to prioritize fire safety and maintain the integrity of the escape routes at all times.

7.5 Emergency Evacuation of Persons with Mobility Impairment

Persons with mobility impairments face particular challenges when evacuating a building in an emergency. They may be unable to use stairs, move quickly, or navigate obstacles. It's vital that building managers have robust plans in place to ensure their safe evacuation.

Personal Emergency Evacuation Plans (PEEPs)

The cornerstone of evacuating persons with mobility impairments is the Personal Emergency Evacuation Plan (PEEP) [299]. A PEEP is a bespoke plan developed for an individual who may need assistance to evacuate.

The PEEP should be developed in consultation with the individual and should consider [300]:

1. **The Nature of Their Impairment**: This could include mobility issues, visual or hearing impairments, cognitive disabilities, or temporary conditions such as pregnancy or injury.
2. **The Assistance They Require**: This could range from guidance to the escape routes to physical assistance to transfer from a wheelchair to an evacuation chair.
3. **The Evacuation Options Available**: This could include using lifts (if safe to do so), evacuation chairs on stairways, or horizontal evacuation to a refuge area.
4. **The Personnel Assigned to Assist**: Specific staff members should be designated and trained to provide the necessary assistance.
5. **The Evacuation Procedure**: The step-by-step actions to be taken in an emergency, including how to raise the alarm, where to go, and what to do if the primary escape route is blocked.

PEEPs should be regularly reviewed and updated to ensure they remain effective. They should be practiced during fire drills to familiarize both the individual and the assisting staff with the procedure.

Evacuation Lifts

In some buildings, lifts may be designated for use during an evacuation [301]. These are typically firefighter lifts that have additional protection, such as a separate power supply and fire-resistant construction.

Evacuation lifts can be a useful means of evacuating persons with mobility impairments, particularly from higher floors. However, they should only be used in accordance with the building's evacuation procedures and under the direction of trained staff.

Evacuation Chairs

Evacuation chairs are specially designed chairs used to transport individuals downstairs in an emergency [302]. They typically have wheels, tracks, or skids that allow them to be maneuverer on stairways, and they may have features such as restraints and head support.

Evacuation chairs should be:

1. **Located strategically** near stairways or in refuge areas.
2. **Regularly inspected and maintained** to ensure they are in good working order.
3. **Operated by trained personnel** who are familiar with their use and the building's evacuation procedures.

Individuals who may need to use an evacuation chair should be familiarized with the equipment and the procedure as part of their PEEP.

Refuge Areas

Refuge areas are designated spaces where persons with mobility impairments can safely wait for assistance to evacuate [303]. They are typically located in fire-protected stairways or other areas that offer a level of protection from fire and smoke.

Refuge areas should be:

1. **Clearly marked** with appropriate signage.

2. **Equipped with communication systems** to allow occupants to contact the building's emergency response team.
3. **Large enough** to accommodate the expected number of occupants and their mobility aids.
4. **Included in the building's evacuation procedures**, with clear guidance on when and how to use them.

The use of refuge areas should be a last resort, with the primary aim being to evacuate all occupants out of the building. They should not be used as a substitute for effective evacuation planning and management.

Horizontal Evacuation

In some situations, it may be safer or more practical to move occupants horizontally to another part of the building rather than attempting a vertical evacuation [304]. This is known as horizontal evacuation.

Horizontal evacuation can be appropriate in buildings with multiple fire compartments, where occupants can be moved from the affected compartment to an adjacent one that is protected from fire and smoke.

For persons with mobility impairments, horizontal evacuation can provide a quicker and safer alternative to using stairs. However, it should only be used where [305]:

1. **The building's design allows** for adequate fire separation between compartments.
2. **The evacuation procedures** include clear guidance on when and how to use horizontal evacuation.
3. **The staff are trained** in the procedures and can confidently guide occupants to the appropriate area.

Horizontal evacuation should not be seen as a complete solution, as there may still be a need to evacuate the building completely if the fire spreads or if instructed to do so by the emergency services.

Training and Drills

Effective evacuation of persons with mobility impairments relies on well-trained staff and well-practiced procedures. All staff should receive training on [306]:

1. **Disability Awareness**: Understanding the diverse range of disabilities and how they can impact on an individual's ability to evacuate.
2. **Evacuation Procedures**: The specific procedures for evacuating persons with mobility impairments, including the use of equipment like evacuation chairs.
3. **Communication Skills**: How to communicate clearly and calmly with individuals who may be distressed or anxious during an evacuation.

Regular drills should be conducted to practice the evacuation procedures, including the use of PEEPs. These drills should involve all staff and, where possible, the individuals with mobility impairments themselves.

Conclusion

Evacuating persons with mobility impairments requires careful planning, appropriate equipment, and well-trained staff. The key is to have a person-centered approach, with each individual's needs assessed and planned for through the PEEP process.

Building managers must ensure that the necessary equipment, such as evacuation lifts and chairs, is available, maintained, and staff are trained in its use. Refuge areas and horizontal evacuation routes should be incorporated into the building's design and evacuation procedures where appropriate.

Most importantly, the evacuation of persons with mobility impairments should be a key consideration in all fire risk assessments and evacuation planning. It should not be an afterthought, but an integral part of the building's fire safety management.

By prioritizing the safety of all occupants, regardless of their mobility, building managers can ensure that everyone has the best possible chance of evacuating safely in an emergency.

7.6 General Principles

The preceding sections have detailed many specific aspects of emergency escape routes, from their number and location to their management and use by persons with mobility impairments. However, there are also some general principles that underpin the effective provision and use of escape routes.

Alternative Escape Routes

A fundamental principle is that there should always be alternative means of escape available [307]. This means that if one escape route is blocked or unavailable due to fire, smoke, or other obstruction, occupants can still evacuate safely using another route.

The number and distribution of alternative escape routes will depend on the size, complexity, and occupancy of the building. In general [308]:

1. **Small, simple buildings** may be able to have a single escape route if the travel distance is short and the route is protected from fire and smoke.
2. **Larger, more complex buildings** will require multiple escape routes to ensure that all occupants can evacuate safely within a reasonable time.
3. **High-occupancy buildings**, such as theatres or shopping malls, will require multiple escape routes of sufficient width to accommodate the large number of people.

Alternative escape routes should be as independent of each other as possible to minimize the risk of a single fire affecting more than one route. They should also be distributed around the building to provide escape options from all areas.

Progressive Horizontal Evacuation

In some large, complex buildings, particularly those providing care such as hospitals, a strategy of progressive horizontal evacuation may be used [309]. This involves moving occupants from the immediate area of the fire to an adjacent fire compartment on the same level, rather than immediately evacuating them out of the building.

Progressive horizontal evacuation can be an effective strategy where:

1. **The building's design** provides adequate fire separation between compartments.
2. **The occupants** are unable to evacuate quickly due to medical conditions or treatments.
3. **The risks of vertical evacuation**, such as using stairs or lifts, outweigh the benefits.

However, progressive horizontal evacuation should only be used as part of a comprehensive evacuation strategy that also includes plans for full evacuation if necessary. It requires robust compartmentation, well-trained staff, and close coordination with the fire and rescue service.

Evacuation Time

A key factor in the design and management of escape routes is the time it takes for all occupants to evacuate safely. This is known as the Required Safe Egress Time (RSET) [310].

The RSET is influenced by several factors:

1. **Detection Time**: How long it takes to detect the fire and raise the alarm.
2. **Pre-movement Time**: How long it takes occupants to respond to the alarm and start evacuating. This can be affected by factors such as familiarity with the building, the clarity of alarm signals and instructions, and the occupants' physical and mental abilities.
3. **Travel Time**: How long it takes occupants to move from their location to a place of safety, considering factors such as travel distance, route complexity, crowd density, and the presence of stairs or narrow passages.

The RSET must be less than the Available Safe Egress Time (ASET), which is the time from ignition until conditions become untenable for evacuation [311]. The ASET is determined by factors such as the building's fire protection measures, the fire growth rate, and the ventilation conditions.

To ensure safe evacuation, designers and managers must ensure that the RSET is as short as possible and always less than the ASET. This may involve measures such as improving fire detection and alarm systems, simplifying escape routes, providing clear signage and instructions, and training staff and occupants on evacuation procedures.

Occupant Behaviour

The behaviour of occupants during an evacuation can significantly impact the effectiveness of escape routes. People do not always behave in the way that designers or managers expect, and this can lead to delays, congestion, or even failure to evacuate [312].

Some common behavioural issues include:

1. **Delayed response** to alarms, often due to a belief that it may be a false alarm or a drill.
2. **Seeking confirmation** of the need to evacuate from others, leading to a "herd mentality".
3. **Attempting to gather belongings** or complete tasks before evacuating.
4. **Using familiar routes** rather than designated escape routes.
5. **Assisting others**, potentially slowing their own evacuation.

Building designers and managers must consider these potential behaviours in their evacuation planning. This may involve measures such as:

1. **Providing clear, unambiguous alarm signals** and instructions.
2. **Training occupants** on the importance of immediate evacuation.
3. **Designating staff** to guide and direct occupants during an evacuation.
4. **Ensuring escape routes are intuitive** and align with occupants' natural wayfinding tendencies.
5. **Planning for the evacuation of persons requiring assistance**, so that they do not unduly delay others.

Effective evacuation requires a holistic approach that considers not just the physical escape routes, but also the human factors that influence their use.

Conclusion

The general principles of providing alternative escape routes, considering evacuation time, and understanding occupant behaviour are essential for the effective design and management of escape routes.

These principles emphasize the importance of redundancy, ensuring that there are always alternative means of escape available; of time, ensuring that occupants can evacuate before conditions become untenable; and of human behaviour, ensuring that escape routes are designed and managed in a way that aligns with how people actually behave in emergencies.

By incorporating these principles into their fire safety strategies, building designers and managers can create escape routes that are robust, efficient, and effective, providing the best possible chance of safe evacuation for all occupants.

7.7 Suitability of Escape Routes

For an escape route to be effective, it must be suitable for its intended use. This means it must be designed, constructed, and maintained to allow all occupants to evacuate safely and quickly in an emergency.

The layout of Escape Routes

The layout of escape routes is a crucial factor in their suitability. They should be designed to be as simple and direct as possible, leading occupants clearly and unambiguously to a place of safety [313].

Some key considerations for the layout of escape routes include:

1. **Avoidance of Dead Ends**: Escape routes should not lead to dead ends or areas from which there is no further escape. If dead ends are unavoidable, they should be as short as possible and provide access to an alternative escape route.
2. **Avoidance of Inner Rooms**: Inner rooms (rooms from which the only escape is through another room) should be avoided where possible. Where they are necessary, the outer room should be a low fire risk area and the travel distance from any point in the inner room to the exit from the outer room should be as short as possible.
3. **Provision of Adequate Exits**: There should be sufficient exits of adequate width to allow all occupants to evacuate safely within a reasonable time. The number and width of exits should be based on the occupancy of the building.
4. **Distribution of Exits**: Exits should be distributed as evenly as possible around the building to minimize travel distances and provide alternative escape routes from all areas.
5. **Protection of Escape Routes**: Escape routes should be protected from the effects of fire and smoke as far as reasonably practicable. This may involve providing fire resisting construction, fire doors, smoke control systems, and clear signage and lighting.

The specific layout requirements for escape routes will depend on the type and size of the building, its occupancy, and the findings of the fire risk assessment.

Travel Distance

Travel distance is a key factor in the suitability of escape routes. It is the distance that occupants have to travel from any point in a building to reach a place of relative safety, such as a protected stairway or final exit [314].

The maximum allowable travel distance depends on several factors:

1. **The Use of the Building**: Different occupancies have different travel distance limits based on the associated fire risks. For example, offices typically allow longer travel distances than industrial buildings.
2. **The Number of Escape Routes**: Buildings with more than one escape route in different directions can have longer travel distances than those with a single direction of escape.
3. **The Presence of Sprinklers**: Buildings with comprehensive automatic sprinkler protection may be allowed longer travel distances.

The travel distance should be measured along the actual route that a person would need to take to reach a place of relative safety, taking into account any obstructions or restrictions [315].

In general, the maximum travel distances for different building types are [316]:

- Offices: 18m in a single direction, 45m in more than one direction
- Shops: 18m in a single direction, 45m in more than one direction
- Industrial: 25m in a single direction, 45m in more than one direction
- Residential: 9m in a single direction, 18m in more than one direction

These are general guidelines and the actual limits may vary depending on the specific characteristics of the building and the findings of the fire risk assessment.

Width and Capacity of Escape Routes

The width and capacity of escape routes are critical factors in ensuring that all occupants can evacuate safely and quickly.

The required width of escape routes depends on the number of people using them and the type of building. As a general rule [317]:

- Doors: Minimum 750mm clear opening width
- Corridors: Minimum 1050mm for up to 200 people

- Stairs: Minimum 1050mm for up to 200 people

These widths should be increased by 75mm for every additional 15 people (or part thereof) above 200.

The capacity of an escape route is the number of people that can pass through it in a given time. This depends on the width of the route and the speed of movement, which can be affected by factors such as the age and mobility of the occupants, the presence of stairs or ramps, and the density of the crowd [318].

As a guide, a 1050mm wide corridor can accommodate up to 200 people in 2-3 minutes, while a 1050mm wide staircase can accommodate up to 150 people in the same time [319].

It's important to note that these are minimum requirements and the actual widths and capacities needed may be greater depending on the specifics of the building and its occupancy. The fire risk assessment should consider factors such as the likely rate of evacuation, the presence of bottlenecks or congestion points, and the potential for contraflow (people moving in opposite directions).

Height of Escape Routes

The height of escape routes is another important factor in their suitability. Low ceilings or obstructions can impede movement and lead to injury, particularly in dense crowd situations.

The minimum clear height for escape routes is generally 2 meters [320]. However, this may need to be increased in certain circumstances, such as:

1. **High-density occupancies**, such as theatres or stadiums, where a higher ceiling can provide a greater sense of space and reduce the risk of crowd crush.
2. **Industrial buildings** where large equipment or goods may need to be moved during an evacuation.
3. **Heritage buildings** with pre-existing low ceilings, where additional measures (such as enforced one-way flow or reduced occupancy) may be necessary.

Any changes in level, such as steps or ramps, should be clearly marked and adequately lit to prevent trips and falls.

Flooring and Surfaces

The flooring and surfaces of escape routes should be chosen and maintained to minimize the risk of slips, trips, and falls, which can seriously impede evacuation.

Some key considerations include [321]:

1. **Slip Resistance**: Flooring should have adequate slip resistance, particularly in areas prone to wetness or spillages. The slip resistance should be maintained throughout the life of the building.
2. **Evenness**: Floors should be as even as possible, with no abrupt changes in level or tripping hazards such as raised thresholds.
3. **Maintenance**: Floors should be regularly inspected and maintained to ensure they remain in good condition. Damage, wear, or contamination that could increase the risk of slipping should be addressed promptly.
4. **Lighting**: Adequate lighting should be provided to ensure that any potential hazards are clearly visible.

In some specialized environments, such as laboratories or factories, additional considerations may apply, such as the need for chemical-resistant or electrostatic-dissipative flooring.

Lifts and Escalators

Lifts (elevators) and escalators are not usually considered suitable as escape routes. This is because they can fail in an emergency due to power outages or fire damage, and because they can contribute to the spread of fire and smoke if not properly designed and maintained [322].

However, in some circumstances, lifts may be used as part of an evacuation strategy [323]:

1. **Firefighting Lifts**: Specially designed and protected lifts that can be used by firefighters to transport equipment and rescue people.
2. **Evacuation Lifts**: Lifts that are designed to be used for the evacuation of people with disabilities or reduced mobility. These must have backup power supplies and fire protection.
3. **Lifts in Phased Evacuation**: In very tall buildings, lifts may sometimes be used in the early stages of a phased evacuation to move occupants from the affected floor to a safe floor below.

The use of lifts for evacuation should only be considered after a thorough risk assessment and in consultation with the fire and rescue service. It should never be relied upon as the sole means of escape.

Escalators should never be used as part of an escape route unless they have been specifically designed for that purpose and have adequate fire protection [324].

External Escape Routes

In some buildings, particularly those with limited internal space, external escape routes such as external staircases or walkways may be used.

External escape routes can have some advantages, such as providing a source of fresh air and allowing smoke to dissipate. However, they also have some specific requirements [325]:

1. **Fire Resistance**: The structure of the external escape route should have adequate fire resistance to protect evacuees from the heat and flames of a fire inside the building.
2. **Protection from Falling**: There should be adequate guards and railings to prevent people from falling from the escape route.
3. **Width and Capacity**: The width and capacity of the escape route should be sufficient for the number of people likely to use it, taking into account the possibility of slower movement due to the external environment (e.g., wind, rain).
4. **Lighting**: Adequate emergency lighting should be provided to ensure the escape route can be safely used at all times.
5. **Maintenance**: The escape route should be regularly inspected and maintained to ensure it remains in good condition and free from obstruction.

External escape routes can be a useful addition to a building's evacuation strategy, but they should not be the sole means of escape. They should be carefully designed and maintained to ensure they are safe and effective to use in an emergency.

Conclusion

The suitability of escape routes is a critical factor in ensuring the safe evacuation of a building in an emergency. Suitable escape routes are those that are designed, constructed, and maintained to allow all occupants to evacuate quickly and safely.

This involves consideration of a wide range of factors, including the layout, travel distance, width and capacity, height, flooring, use of lifts and escalators, and the potential for external escape routes.

The specific requirements for escape routes will depend on the unique characteristics of each building and its occupancy. The fire risk assessment should carefully consider all aspects of the escape routes to ensure they are suitable and sufficient.

Ultimately, providing suitable escape routes is about more than just compliance with regulations. It's about ensuring that in the event of a fire, everyone can get out safely. This requires ongoing vigilance, maintenance, and a commitment to putting life safety first.

7.8 Fire-Resisting Construction

Fire-resisting construction is a key part of a building's passive fire protection. It involves using materials and construction methods that can withstand the effects of fire for a specified period of time, helping to contain the fire and prevent its spread.

The Role of Fire-Resisting Construction

Fire-resisting construction serves several important functions in a building's fire safety strategy [326]:

1. **Compartmentation**: Fire-resisting walls, floors, and doors can divide a building into separate fire compartments. This contains the fire and its effects to the compartment of origin, preventing spread to other parts of the building.
2. **Protection of Escape Routes**: Fire-resisting construction is used to protect escape routes such as corridors and staircases. This ensures that the routes remain passable and safe to use during the evacuation period.
3. **Protection of Structural Elements**: Fire-resisting construction can be used to protect load-bearing structural elements such as columns and beams. This maintains the stability and integrity of the building, preventing collapse that could endanger occupants and firefighters.
4. **Separation of Occupancies**: In mixed-use buildings, fire-resisting construction is used to separate different occupancies, such as residential and commercial areas. This prevents a fire in one occupancy from spreading to another.

The specific requirements for fire-resisting construction will depend on the size, use, and risk profile of the building, and will be specified in the relevant building regulations and fire safety standards.

Fire Resistance Ratings

The fire resistance of a construction element is quantified by its fire resistance rating. This is the period of time for which the element can withstand the effects of a standard fire test without loss of its fire separating function or load-bearing capacity (if load-bearing) [327].

Fire resistance ratings are usually expressed in terms of minutes or hours, such as 30 minutes, 60 minutes, 90 minutes, or 120 minutes. The required rating for a particular element will depend on its function and location in the building.

For example [328]:

- A fire door separating a hotel room from a corridor may require a 30-minute rating.
- A wall separating two retail units may require a 60-minute rating.
- A load-bearing column in a tall office building may require a 120-minute rating.

The fire resistance rating is achieved through the use of specific materials, construction methods, and design features, which are tested and certified to meet the required standard.

Materials and Methods

There are several types of materials and construction methods commonly used to achieve fire resistance [329]:

1. **Concrete**: Concrete has excellent fire-resisting properties due to its low thermal conductivity and non-combustibility. It can be used for walls, floors, columns, and beams.
2. **Masonry**: Brick and block walls can provide good fire resistance, particularly when combined with fire-resistant plaster or render.
3. **Fire-Rated Gypsum Board**: Also known as drywall, this is a type of plasterboard that includes fire-resistant additives. It's commonly used to create fire-resistant partitions and to protect structural steel.
4. **Intumescent Paint**: This is a special paint that swells when exposed to high temperatures, creating an insulating layer. It's often used on structural steel to maintain its load-bearing capacity in a fire.
5. **Spray-On Fire Resistive Material (SFRM)**: This is a spray-applied insulating material that can be used to protect structural steel and other elements.
6. **Fire Stopping**: This involves using fire-resistant materials to seal gaps and penetrations in fire-resisting construction, such as around pipes and cables. This maintains the integrity of the fire barrier.

The choice of materials and methods will depend on factors such as the required fire resistance rating, the type of construction, aesthetics, cost, and compatibility with other building systems.

Openings and Penetrations

Openings and penetrations in fire-resisting construction, such as doors, windows, vents, and service ducts, can create weak points that allow the passage of fire and smoke. Therefore, these must be carefully designed and protected to maintain the overall fire resistance of the construction.

Some common methods of protecting openings and penetrations include [330]:

1. **Fire Doors**: These are doors that are constructed and tested to provide a specific fire resistance rating. They must be fitted with appropriate hardware, such as self-closing devices and smoke seals, and must be regularly inspected and maintained.
2. **Fire Shutters**: These are rolling or sliding shutters that automatically close in the event of a fire to seal off an opening.
3. **Fire-Resistant Glazing**: This is a type of glass that can provide a degree of fire resistance. It may be used in doors, windows, or partitions.
4. **Fire Dampers**: These are devices fitted in ventilation ducts that automatically close in the event of a fire to prevent the spread of fire and smoke.
5. **Fire Stopping**: As mentioned earlier, this involves using fire-resistant materials to seal gaps around penetrations.

The specific requirements for protecting openings and penetrations will be specified in the relevant building regulations and fire safety standards.

Maintenance and Modification

To ensure that fire-resisting construction continues to perform its intended function, it must be regularly inspected and maintained. This involves [331]:

1. **Inspection**: Regular checks for any damage, deterioration, or unauthorized modifications that could compromise the fire resistance.
2. **Repair**: Prompt repair of any damage using appropriate fire-resistant materials and methods.
3. **Replacement**: Replacement of elements that have reached the end of their serviceable life or that no longer provide the required fire resistance.
4. **Record Keeping**: Maintaining records of all inspections, repairs, and modifications.

Any modifications to fire-resisting construction, such as creating new openings or altering the use of a space, must be carefully assessed to ensure they do not compromise the fire resistance. This may require consultation with a fire safety professional and obtaining approval from the relevant authorities.

Conclusion

Fire-resisting construction is a vital component of a building's passive fire protection. It helps to contain fires, protect escape routes and structural elements, and prevent the spread of fire and smoke.

The effectiveness of fire-resisting construction depends on the correct specification and installation of materials and systems, as well as ongoing inspection and maintenance to ensure integrity over the life of the building.

As with all aspects of fire safety, the requirements for fire-resisting construction should be based on a comprehensive fire risk assessment that considers the unique features and risk profile of the building. The goal is to provide a robust and reliable system of fire separation that will perform as intended in the event of a fire, helping to protect life, property, and business continuity.

7.9 Number of People Using the Premises

The number of people using a building is a key factor in determining the fire safety measures required, particularly in relation to escape routes. The more people in a building, the greater the potential for loss of life in the event of a fire, and the more stringent the safety measures must be.

Occupancy Levels

The first step in considering the implications of the number of people using a building is to determine the occupancy levels. This is the maximum number of people expected to be in the building at any one time.

Occupancy levels are typically determined by the floor area and the use of the building [332]. For example:

- An office might have an occupancy of one person per 6 square meters.
- A shop might have an occupancy of one person per 2 square meters.
- A nightclub might have an occupancy of one person per 0.3 square meters.

These are general figures and the actual occupancy for a particular building should be based on its specific use and layout.

It's important to consider not just the normal occupancy but also any potential peak occupancies, such as during a sale in a shop or a special event in a venue.

Implications for Escape Routes

The number of people using a building has a direct impact on the design and capacity of the escape routes. In general, the more people, the more escape routes and the wider they need to be.

Specific considerations include [333]:

1. **Number of Escape Routes**: There should be sufficient escape routes to evacuate all occupants safely. The number required will depend on the occupancy and the layout of the building.
2. **Width of Escape Routes**: The escape routes must be wide enough to accommodate the number of people likely to use them. The width is calculated based on the occupancy and a standard flow rate (e.g., 40 people per minute per unit width of 525mm).
3. **Travel Distances**: The maximum travel distances to an escape route must be appropriate for the number of people. Higher occupancies may require shorter travel distances to prevent congestion.
4. **Exit Capacity**: The number and width of exits must be sufficient to evacuate all occupants within a reasonable time. This is typically 2.5 to 3 minutes for normal risk buildings.
5. **Protected Routes**: With higher occupancies, there is a greater need for protected escape routes (corridors and staircases) to prevent the spread of fire and smoke.

The specific requirements for escape routes based on occupancy will be set out in the relevant building regulations and fire safety standards.

Implications for Fire Safety Systems

The number of people using a building also influences the type and extent of fire safety systems required. For example [334]:

1. **Fire Detection and Alarm**: Higher occupancies may require a more extensive and sophisticated alarm system to ensure that everyone can be alerted quickly in the event of a fire.
2. **Emergency Lighting**: Higher occupancies require more extensive emergency lighting to guide people to the escape routes.
3. **Firefighting Equipment**: The type and quantity of firefighting equipment, such as extinguishers and hose reels, will depend on the occupancy and the fire risk.
4. **Sprinklers**: Higher occupancies, particularly in buildings with sleeping risk such as hotels, may require sprinkler systems to control the spread of fire.
5. **Smoke Control**: Higher occupancies may require smoke control systems to keep escape routes clear of smoke.

The fire risk assessment should consider the number of people using the building and ensure that the fire safety systems are adequate and appropriate.

Management Implications

The number of people using a building also has implications for its management and the emergency procedures [335]:

1. **Staffing**: There must be sufficient trained staff to manage the evacuation of all occupants. This may include fire wardens, security staff, and first-aiders.
2. **Emergency Plans**: The emergency plans and evacuation procedures must be designed to cope with the maximum occupancy. This includes designating assembly points that can accommodate everyone.
3. **Training**: All staff must be trained in the emergency procedures and their specific roles. Higher occupancies may require more frequent and extensive training.
4. **Fire Drills**: Regular fire drills should be conducted to test the procedures and familiarize occupants with the escape routes. The frequency and timing of drills should reflect the occupancy.
5. **Crowd Management**: In venues with high occupancies, such as stadiums or nightclubs, specific crowd management strategies may be necessary to prevent overcrowding and ensure safe evacuation.

The management of the building must ensure that the occupancy levels are not exceeded and that the fire safety measures remain adequate for the number of people.

Conclusion

The number of people using a building is a critical determinant of the fire safety measures required. It directly influences the design and capacity of escape routes, the type and extent of fire safety systems, and the management procedures.

As the occupancy increases, so does the potential for loss of life in the event of a fire. Therefore, it's essential that the fire safety measures are scaled appropriately to the number of people.

This requires a thorough understanding of the building's occupancy characteristics, including normal and peak levels, the distribution of people, and any specific risks. This understanding should be based on a comprehensive fire risk assessment and should be regularly reviewed to ensure it remains valid.

Ultimately, the fire safety measures must be designed to ensure that all occupants can evacuate safely in the event of a fire, regardless of their number. This requires a robust and integrated approach that considers all aspects of fire safety, from the physical measures to the management procedures. Only by taking a holistic view can we ensure that the number of people using a building does not compromise their safety in the event of a fire.

7.10 Mobility Impairment

Individuals with mobility impairments face unique challenges when it comes to evacuating a building in the event of a fire. They may be unable to use stairs, move quickly, or navigate obstacles. It's crucial that building managers have robust plans in place to ensure their safe evacuation.

Types of Mobility Impairment

Mobility impairments can take many forms and can be temporary or permanent. Some common types include [336]:

1. **Wheelchair Users**: People who use wheelchairs for mobility due to conditions such as spinal cord injuries, cerebral palsy, or muscular dystrophy.
2. **Mobility Aid Users**: People who use aids such as crutches, canes, or walkers due to conditions such as arthritis, stroke, or amputation.
3. **Respiratory Conditions**: People with conditions such as asthma or chronic obstructive pulmonary disease (COPD) who may have difficulty moving quickly or climbing stairs.

4. **Pregnancy**: Pregnant women, especially in the later stages of pregnancy, may have reduced mobility and stamina.
5. **Temporary Injuries**: People with broken legs, sprains, or other temporary injuries that affect their mobility.

Each type of mobility impairment presents its own challenges for evacuation and requires specific considerations in the emergency plan.

Personal Emergency Evacuation Plans (PEEPs)

The cornerstone of evacuating individuals with mobility impairments is the Personal Emergency Evacuation Plan (PEEP). A PEEP is a bespoke plan developed for an individual who may need assistance to evacuate [337].

The PEEP should be developed in consultation with the individual and should consider [338]:

1. **The Nature of Their Impairment**: The specific mobility issues they face and how these might affect their ability to evacuate.
2. **The Assistance They Require**: This could range from guidance to the escape routes to physical assistance to transfer from a wheelchair to an evacuation chair.
3. **The Evacuation Options Available**: This could include using lifts (if safe to do so), evacuation chairs on stairways, or horizontal evacuation to a refuge area.
4. **The Personnel Assigned to Assist**: Specific staff members should be designated and trained to provide the necessary assistance.
5. **The Evacuation Procedure**: The step-by-step actions to be taken in an emergency, including how to raise the alarm, where to go, and what to do if the primary escape route is blocked.

PEEPs should be regularly reviewed and updated to ensure they remain effective. They should be practiced during fire drills to familiarize both the individual and the assisting staff with the procedure.

Evacuation Equipment

There are several types of equipment that can aid the evacuation of people with mobility impairments:

1. **Evacuation Chairs**: These are chairs designed to transport people down stairs. They usually have tracks or wheels that allow them to glide down stairs, and they may have features like restraints and head support [339].
2. **Stair Climbers**: These are motorized devices that can carry a wheelchair up or down stairs. They can be operated by a single person and can navigate narrow or curved stairs [340].
3. **Refuge Areas**: These are designated safe areas where people with mobility impairments can wait for assistance. They are typically in fire-protected stairwells or other areas that offer some protection from fire and smoke [341].
4. **Firefighter Lifts**: These are lifts that are designed to be used by firefighters during an emergency. They have features like a separate power supply and fire-resistant doors [342].

The specific equipment needed will depend on the building and the individual needs of the occupants. It's essential that the equipment is regularly maintained and that staff are trained in its use.

Building Design Considerations

The design of the building can significantly impact the ease and safety of evacuating people with mobility impairments. Some key considerations include [343]:

1. **Accessible Escape Routes**: Escape routes should be designed to be as accessible as possible, with features like wide doorways, level thresholds, and handrails on both sides of stairs.
2. **Refuge Areas**: The building should have designated refuge areas on each floor, preferably in fire-protected stairwells. These areas should be large enough to accommodate wheelchairs and should have communication systems to allow occupants to contact building management or emergency services.
3. **Evacuation Lifts**: In tall buildings, consideration should be given to installing evacuation lifts that can be used to transport people with mobility impairments during an emergency.

4. **Signage**: Clear signage should be provided to direct people to accessible escape routes and refuge areas.
5. **Fire Alarm Systems**: The fire alarm system should include features like visual alarms (flashing lights) and vibrating pagers for people with hearing impairments.

Incorporating these features into the building design can greatly facilitate the safe evacuation of people with mobility impairments.

Staff Training

Staff play a crucial role in assisting the evacuation of people with mobility impairments. They should receive regular training on [344]:

1. **Disability Awareness**: Understanding the different types of mobility impairments and how they can affect a person's ability to evacuate.
2. **Evacuation Procedures**: The specific procedures for assisting people with mobility impairments, including the use of equipment like evacuation chairs.
3. **Communication Skills**: How to communicate clearly and calmly with someone who may be distressed or anxious.
4. **Manual Handling**: Safe techniques for assisting people to transfer from a wheelchair to an evacuation chair or other equipment.

Training should be hands-on and should include practice sessions with the actual equipment. It should be refreshed regularly to ensure skills are maintained.

Conclusion

Evacuating people with mobility impairments requires careful planning, the right equipment, and well-trained staff. The key is to have a person-centered approach, with each individual's needs assessed and planned for through the PEEP process.

Building designers and managers must ensure that the building is as accessible as possible and that refuge areas and evacuation equipment are provided. Staff must be trained to use the equipment and to assist individuals safely and calmly.

Ultimately, the measure of success is whether every occupant, regardless of their mobility, can evacuate safely in the event of a fire. This requires a commitment from everyone involved - designers, managers, and staff - to prioritize accessibility and inclusivity in all aspects of fire safety planning.

By taking this approach, we can create buildings that are not only safe but also welcoming and equitable for all. This, surely, should be the goal of every modern fire safety strategy.

This concludes Unit 7: Emergency Escape Routes. The next unit will cover Emergency Escape Lighting, Signs, and Notices.

UNIT 8: EMERGENCY ESCAPE LIGHTING, SIGNS AND NOTICES

Emergency escape lighting and signs are essential components of a building's fire safety strategy. They ensure that occupants can safely find their way out of the building in the event of a fire, even if the normal lighting fails.

Emergency Escape Lighting

Emergency escape lighting is lighting that comes on automatically if the power to the normal lighting fails. Its purpose is to illuminate escape routes and help occupants find their way out of the building safely [345].

Types of Emergency Escape Lighting

There are two main types of emergency escape lighting [346]:

1. **Maintained**: These are luminaires that are on all the time. They have two power supplies, the normal supply and a battery backup. If the normal power fails, the battery takes over.
2. **Non-Maintained**: These luminaires only come on when the normal power fails. They are powered by a battery and are inactive when the normal lights are on.

Both types of luminaires must provide adequate illumination for a sufficient duration to allow all occupants to evacuate safely. The specific requirements will be set out in the relevant building regulations and standards.

Location of Emergency Escape Lighting

Emergency escape lighting should be provided in the following locations [347]:

1. **Escape Routes**: All escape routes, including corridors, stairways, and exit doors, should be provided with emergency lighting.
2. **Open Areas**: Large open areas like offices, warehouses, and retail spaces should have emergency lighting to guide occupants to the escape routes.
3. **High Risk Areas**: Areas that pose a particular risk in the event of a power failure, such as plant rooms or kitchens, should have emergency lighting.
4. **Changes of Level**: Any changes of level, such as stairs or ramps, should be well illuminated to prevent trips and falls.
5. **Intersections**: Intersections of corridors or escape routes should be provided with emergency lighting to help occupants choose the correct route.
6. **Outside the Final Exit**: Emergency lighting should be provided outside the final exit to help occupants move away from the building to a place of safety.

The exact location and spacing of luminaires will depend on the layout of the building and the requirements of the relevant standards.

Duration of Emergency Escape Lighting

Emergency escape lighting must operate for long enough to allow all occupants to evacuate safely. The minimum duration required will depend on the size and complexity of the building [348]:

- For small, simple buildings, a duration of 1 hour may be sufficient.
- For larger, more complex buildings, a duration of 3 hours is typically required.

These are minimum durations and longer durations may be necessary depending on the specific characteristics of the building and its occupants.

Testing and Maintenance

Emergency escape lighting must be regularly tested and maintained to ensure it will function correctly in an emergency. The typical testing regime includes [349]:

1. **Daily**: A visual check to ensure all luminaires are in place and appear to be functioning.
2. **Monthly**: A short functional test by simulating a power failure and checking that all luminaires operate correctly.
3. **Annually**: A full duration test to ensure the lights can operate for the full required duration.

Any faults found during testing must be rectified immediately. Luminaires should also be cleaned and have their batteries replaced as per the manufacturer's instructions.

Emergency Escape Signs

Emergency escape signs, also known as escape route signs or running man signs, are signs that indicate the route to the nearest emergency exit. They are a crucial part of the emergency escape lighting system.

Types of Emergency Escape Signs

There are two main types of emergency escape signs [350]:

1. **Escape Route Signs**: These signs indicate the direction of the nearest emergency exit. They feature a running man symbol and an arrow pointing in the direction of the exit.
2. **Emergency Exit Signs**: These signs mark the emergency exits themselves. They feature a running man symbol and the words "EXIT" or "FIRE EXIT".

Both types of signs must be clearly visible and understandable, even in smoky conditions. They should be illuminated, either internally (by the sign's own lighting) or externally (by the emergency escape lighting).

Location of Emergency Escape Signs

Emergency escape signs should be provided at all key decision points along the escape routes. This includes [351]:

1. **Intersections**: Where escape routes intersect or divide.
2. **Changes of Direction**: Where the escape route changes direction.
3. **Final Exits**: At all final exits from the building.

Signs should be positioned so that a person escaping will always have the next sign in sight. The maximum distance between signs should be determined based on the size of the signs and the layout of the building.

Design of Emergency Escape Signs

The design of emergency escape signs is strictly regulated to ensure consistency and clarity. Signs must comply with the relevant standards, such as BS 5499 in the UK [352].

Key design requirements include:

1. **Colour**: Signs must have a green background with white symbols and text.
2. **Symbols**: The running man symbol must be used, along with arrows indicating the direction of escape.
3. **Text**: The words "EXIT" or "FIRE EXIT" must be used, with letter heights based on the viewing distance.
4. **Illumination**: Signs must be illuminated to a sufficient level to be clearly visible in all conditions.

The specific design requirements will be detailed in the relevant standards and must be strictly adhered to.

Testing and Maintenance

Like emergency escape lighting, emergency escape signs must be regularly tested and maintained. This includes [353]:

1. **Daily**: A visual check to ensure all signs are in place, clean, and legible.
2. **Monthly**: A check that all illuminated signs are functioning correctly.
3. **Annually**: A check that all signs are still correctly positioned and are compliant with current standards.

Any faults found during testing must be rectified immediately. Signs should also be cleaned regularly to ensure they remain clearly visible.

Other Safety Signs and Notices

In addition to emergency escape signs, there are several other types of safety signs and notices that should be provided in buildings:

1. **Fire Action Notices**: These provide instructions on what to do in the event of a fire. They should be positioned next to fire alarm call points and in other prominent locations [354].
2. **Fire Door Signs**: These indicate which doors are fire doors and must be kept shut. They should be provided on both sides of all fire doors [355].

3. **Fire Extinguisher Signs**: These indicate the location and type of fire extinguishers. They should be positioned above or adjacent to the extinguishers [356].
4. **No Smoking Signs**: These remind occupants that smoking is not permitted. They should be provided at entrances and in other prominent locations, especially in areas where smoking is a particular risk [357].

All safety signs and notices must comply with the relevant standards in terms of their design, colour, and symbols used. They must be clearly visible and understandable by all occupants.

Conclusion

Emergency escape lighting and signs are critical elements of a building's fire safety strategy. They ensure that occupants can safely find their way out of the building in an emergency, even if the normal lighting fails.

Building managers must ensure that emergency lighting and signs are correctly positioned, are of the correct design, and are properly maintained and tested. This requires a thorough understanding of the relevant standards and a commitment to ongoing maintenance and testing.

The ultimate goal is to provide a clear and reliable means for occupants to escape to safety. By investing in high-quality emergency lighting and signs, and by maintaining them diligently, building managers can play a vital role in protecting the lives of their occupants.

Remember, in an emergency, it's not just about having an escape route - it's about being able to find it and follow it safely. That's the critical function that emergency lighting and signs fulfill.

This concludes Unit 8: Emergency Escape Lighting, Signs and Notices. The next unit will cover Rerecord, Plan, Inform, Instruct and Train.

UNIT 9: RERECORD, PLAN, INFORM, INSTRUCT AND TRAIN

Fire safety management isn't just about having the right physical measures in place - it's also about ensuring that people know what to do in the event of a fire. This unit focuses on the critical human elements of fire safety: recording information, planning for emergencies, informing and instructing occupants, and training staff.

Record Keeping

Accurate record keeping is an essential part of fire safety management. Records provide evidence of compliance with legal requirements and can be used to monitor and improve fire safety performance over time.

Key records that should be kept include [358]:

1. **Fire Risk Assessment**: The building's fire risk assessment should be regularly reviewed and updated. All versions should be dated and kept as a record.
2. **Maintenance and Testing**: Records should be kept of all maintenance and testing of fire safety equipment, including emergency lighting, fire alarms, and extinguishers.
3. **Training**: Records should be kept of all fire safety training provided to staff, including the date, content, and attendees.
4. **Drills**: Records should be kept of all fire drills, including the date, time, scenario, and any issues identified.
5. **Incidents**: Any fire incidents or false alarms should be recorded, along with the cause and any actions taken.

Records should be kept in a secure but accessible location and should be available for inspection by the fire and rescue service or other enforcing authority.

Emergency Plans

Every building should have a comprehensive emergency plan that details what to do in the event of a fire. The plan should be based on the findings of the fire risk assessment and should be tailored to the specific characteristics of the building and its occupants.

Key elements of an emergency plan include [359]:

1. **Evacuation Procedures**: Step-by-step instructions on how to evacuate the building safely, including primary and secondary escape routes and assembly points.
2. **Roles and Responsibilities**: Clear definition of who is responsible for what during an emergency, including who will call the fire service, who will assist with evacuation, and who will liaise with emergency services.
3. **Communication**: How occupants will be alerted to a fire (e.g., fire alarms) and how they will be kept informed during an incident.
4. **Equipment**: The location and use of fire safety equipment, including extinguishers, fire blankets, and evacuation chairs.
5. **Specific Needs**: Arrangements for evacuating individuals with specific needs, such as those with mobility impairments or language barriers.
6. **Business Continuity**: Plans for how critical business functions will be maintained or recovered after a fire.

The emergency plan should be clearly documented and should be regularly reviewed and updated to ensure it remains effective.

Informing and Instructing Occupants

All occupants of a building, whether they are staff, residents, or visitors, need to be provided with clear information and instruction about what to do in the event of a fire.

This should include [360]:

1. **Fire Action Notices**: Clear, concise instructions on what to do if you discover a fire or hear the fire alarm. These should be displayed prominently throughout the building.
2. **Evacuation Plans**: Floor plans showing escape routes, fire equipment, and assembly points. These should be displayed in key locations.
3. **Induction Training**: Basic fire safety information provided to all new staff or residents as part of their induction.
4. **Visitor Information**: Brief fire safety information provided to all visitors on arrival, including what to do if the fire alarm sounds.

The information provided should be easy to understand and should be available in alternative formats (such as different languages or large print) where necessary.

Staff Training

Staff play a crucial role in fire safety, particularly in terms of assisting with evacuation and using fire equipment. All staff should receive regular fire safety training commensurate with their roles and responsibilities.

Training should include [361]:

1. **Basic Fire Awareness**: The nature of fire, how it spreads, and the main fire hazards in the building.
2. **Evacuation Procedures**: How to evacuate the building safely and efficiently, including the use of different escape routes and assisting others.
3. **Fire Equipment**: How to use fire extinguishers, fire blankets, and other equipment safely and effectively.
4. **Specific Roles**: Additional training for staff with specific roles, such as fire wardens or those assisting with evacuations.

Training should be practical and hands-on where possible, and should be refreshed regularly (at least annually) to maintain skills and awareness.

Fire Drills

Fire drills

Fire drills are an essential part of fire safety training and preparation. They involve simulating an emergency situation and practicing the full evacuation of the building.

The main purposes of fire drills are [362]:

1. **Testing Procedures**: To test that the evacuation procedures work in practice and to identify any areas for improvement.
2. **Familiarising Occupants**: To familiarize all occupants with the evacuation procedures and escape routes.
3. **Training Staff**: To give staff practical experience in managing an evacuation and using fire equipment.
4. **Monitoring Performance**: To time how long the evacuation takes and monitor whether it improves over time.

Fire drills should be carried out regularly, at least annually and more frequently in high-risk environments. They should be carefully planned and should simulate different scenarios, such as different fire locations or blocked escape routes.

After each drill, a debrief should be held to discuss what went well and what could be improved. Any necessary changes to the evacuation procedures should be made and communicated to all occupants.

Personal Emergency Evacuation Plans (PEEPs)

Personal Emergency Evacuation Plans (PEEPs) are individual plans for people who may need assistance to evacuate in an emergency. This could include people with mobility impairments, visual or hearing impairments, cognitive disabilities, or health conditions.

PEEPs should be developed in consultation with the individual and should detail [363]:

1. **Evacuation Needs**: The specific assistance the individual needs to evacuate safely.
2. **Evacuation Routes**: The most suitable escape routes for the individual, considering their abilities and the layout of the building.
3. **Assistance Required**: Who will provide the necessary assistance, and how they will be alerted and coordinated.
4. **Equipment**: Any equipment needed, such as evacuation chairs or communication devices.
5. **Refuge Areas**: The location of any refuge areas where the individual can wait safely for further assistance.

PEEPs should be regularly reviewed and updated, and should be practiced during fire drills to ensure they are effective.

Conclusion

Recording, planning, informing, instructing, and training are vital components of effective fire safety management. They ensure that everyone in the building knows what to do in the event of a fire and has the knowledge and skills to evacuate safely.

Building managers must prioritize these activities and allocate sufficient time and resources to them. This includes keeping accurate records, developing comprehensive emergency plans, providing clear information and instruction, delivering regular staff training, and conducting realistic fire drills.

The ultimate test of these measures is whether they work in practice. Can every occupant evacuate safely and efficiently in the event of a real fire? Only by continually reviewing, practicing, and improving these measures can building managers have confidence in answering yes to this critical question.

Remember, fire safety is not a one-time event - it's an ongoing process of education, preparation, and vigilance. By embedding this process into the culture of the organization, building managers can create a safer environment for everyone.

This concludes Unit 9: Rerecord, Plan, Inform, Instruct and Train. This also marks the end of the main content of the book. The following sections will include a set of multiple-choice questions to test your understanding, followed by the answers.

Certificate of Completion

Congratulations on completing the Fire Safety course by Barking and Dagenham College. Your dedication to learning about this critical subject is commendable and will serve you well in your future endeavours.

Throughout this course, you have gained a comprehensive understanding of fire safety principles and practices. You have learned about the nature of fire, how it starts and spreads, and the various measures that can be taken to prevent and control it.

You have explored the legal framework surrounding fire safety, including key pieces of legislation such as the Regulatory Reform (Fire Safety) Order 2005. You now understand your legal duties and responsibilities when it comes to fire safety.

You have learned how to conduct a comprehensive fire risk assessment, identifying hazards, evaluating risks, and implementing appropriate control measures. You understand the importance of regularly reviewing and updating this assessment.

You have gained knowledge of fire protection measures, including fire detection and warning systems, firefighting equipment, and emergency lighting. You understand how these systems work, where they should be located, and how they should be maintained and tested.

You have learned about the critical importance of emergency escape routes and how to design, manage, and maintain them effectively. You understand the specific needs of individuals with mobility impairments and how to develop Personal Emergency Evacuation Plans (PEEPs).

Finally, you have understood the vital role of record keeping, planning, information, instruction, and training in fire safety management. You recognize that fire safety is an ongoing process that requires continuous effort and vigilance.

The knowledge and skills you have acquired will be invaluable in your role, whether you are a building manager, a health and safety professional, or simply someone with a responsibility for fire safety. You are now equipped to make a real difference in protecting lives and property from the devastating effects of fire.

But remember, learning about fire safety is not a one-time event. Regulations change, new technologies emerge, and best practices evolve. It's crucial that you commit to continuous learning and staying up-to-date with developments in this field.

Once again, congratulations on your achievement. Wear your certificate with pride, but more importantly, apply your knowledge with diligence and care. The safety of those around you is in your hands.

ASSESSMENT QUESTIONS

1. What three elements are needed for a fire to start? a) Fuel, Oxygen, Heat b) Fuel, Nitrogen, Heat c) Oxygen, Nitrogen, Smoke d) Fuel, Oxygen, Smoke
2. What is the main purpose of the Regulatory Reform (Fire Safety) Order 2005? a) To regulate the sale of fire extinguishers b) To place the responsibility for fire safety on building occupants c) To place the responsibility for fire safety on the building owner or manager d) To mandate the installation of sprinklers in all buildings
3. What is the first step in conducting a fire risk assessment? a) Identify people at risk b) Evaluate risks and decide on precautions c) Record your findings and implement them d) Identify fire hazards

4. Which of the following is NOT a common fire hazard in workplaces? a) Overloaded electrical sockets b) Potted plants c) Combustible materials near heat sources d) Smoking
5. What is the main purpose of a fire detection and warning system? a) To suppress fires b) To provide a means of escape c) To alert occupants of a fire and enable their swift evacuation d) To prevent fires from starting
6. Which type of fire extinguisher is suitable for use on electrical fires? a) Water extinguisher b) Foam extinguisher c) Wet chemical extinguisher d) Carbon dioxide (CO_2) extinguisher
7. How often should fire extinguishers be serviced? a) Every 6 months b) Every 12 months c) Every 18 months d) Every 24 months
8. What is the recommended minimum width for an escape route serving up to 110 people? a) 750mm b) 1050mm c) 1200mm d) 1500mm
9. What is the maximum travel distance allowed in a single direction in a normal fire risk building? a) 9m b) 18m c) 25m d) 45m
10. What is the main purpose of emergency escape lighting? a) To illuminate escape routes during normal conditions b) To indicate the location of fire extinguishers c) To illuminate escape routes in the event of a mains power failure d) To provide lighting for firefighters
11. What colour are fire exit signs? a) Red with white text b) Green with white text c) Blue with white text d) Yellow with black text
12. Which of the following is NOT a requirement for fire doors? a) They should be fitted with self-closing devices b) They should be kept locked at all times c) They should be kept closed when not in use d) They should be fitted with intumescent strips and cold smoke seals
13. How often should emergency escape lighting be tested? a) Daily b) Weekly c) Monthly d) Yearly
14. What is the main purpose of a fire drill? a) To test the speed of the fire brigade's response b) To test and practice the building's evacuation procedures c) To test the integrity of the building's fire compartments d) To test the reliability of the fire detection system
15. What does PEEP stand for in fire safety? a) Personal Evacuation and Egress Plan b) Personnel Emergency Escape Protocol c) Personal Emergency Evacuation Plan d) Premises Evacuation and Escape Procedure

Answers:

1. a) Fuel, Oxygen, Heat
2. c) To place the responsibility for fire safety on the building owner or manager
3. d) Identify fire hazards
4. b) Potted plants
5. c) To alert occupants of a fire and enable their swift evacuation
6. d) Carbon dioxide (CO_2) extinguisher
7. b) Every 12 months
8. a) 750mm
9. b) 18m
10. c) To illuminate escape routes in the event of a mains power failure
11. b) Green with white text
12. b) They should be kept locked at all times
13. c) Monthly
14. b) To test and practice the building's evacuation procedures
15. c) Personal Emergency Evacuation Plan

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Appendix 1: Site MAP

[Insert a site map of Barking and Dagenham College here, showing the location of buildings, fire assembly points, fire equipment, and access routes for fire appliances.]

Appendix 2: Fire Safety Checklist

Answer: (Y/N)

1. Fire Risk Assessment
    - Has a fire risk assessment been completed within the last 12 months?
    - Have all actions from the risk assessment been completed?
2. Fire Detection and Alarm Systems
    - Is there an appropriate fire detection and alarm system in place?
    - Is the system tested weekly?
    - Is the system maintained by a competent person every 6 months?
3. Emergency Lighting
    - Is emergency lighting provided on all escape routes?
    - Is the emergency lighting tested monthly?
    - Is the emergency lighting system maintained by a competent person annually?
4. Fire Extinguishers
    - Are appropriate fire extinguishers provided throughout the premises?
    - Are fire extinguishers checked monthly?
    - Are fire extinguishers serviced annually by a competent person?
5. Escape Routes
    - Are all escape routes clear and unobstructed?
    - Are fire doors in good condition and kept closed?
    - Is emergency escape lighting working on all routes?
    - Are escape route signs and notices clear and legible?
6. Fire Safety Training
    - Have all staff received fire safety training?
    - Are fire marshals/wardens appointed and trained?
    - Are fire evacuation drills conducted at least annually?
7. PEEPs
    - Are Personal Emergency Evacuation Plans (PEEPs) in place for all relevant persons?
    - Are PEEP arrangements reviewed regularly?
8. Fire Safety Management
    - Is there a Fire Safety Policy in place?
    - Are fire safety responsibilities clearly defined?
    - Are fire safety records maintained?
    - Is the Fire Safety Policy reviewed annually?

Appendix 3: Example Fire Risk Assessment

This is an example only. A full fire risk assessment should be conducted by a competent person.

1. Building Information

- Building Name: Barking and Dagenham College Main Building
- Address: Dagenham Road, Romford RM7 0XU
- Number of Floors: 3
- Primary Use: Education
- Number of Occupants: 1000+

2. Fire Hazards

- Sources of Ignition: Electrical equipment, cooking appliances, smoking materials
- Sources of Fuel: Paper, textiles, flammable liquids in science labs
- Work Processes: Science experiments, cooking in food tech rooms

3. People at Risk

- Students, including those with disabilities
- Staff
- Visitors
- Contractors

4. Means of Escape

- Multiple escape routes from each floor
- Fire exits clearly marked
- Emergency lighting provided on all routes
- Some routes obstructed by furniture - action required

5. Fire Safety Systems

- Fire alarm system with smoke detectors throughout
- Fire extinguishers provided on all floors
- Some fire doors wedged open - action required
- No sprinkler system

6. Fire Safety Management

- Fire safety policy in place
- Fire marshals appointed on each floor
- Fire drills conducted termly
- Some staff require refresher training - action required

7. Actions Required

1. Remove obstructions from escape routes - action by Facilities Manager, within 1 week
2. Ensure all fire doors are kept closed - action by all staff, immediately
3. Provide refresher training for identified staff - action by HR, within 1 month
4. Consider installation of sprinkler system - action by Health & Safety Manager, within 3 months

Signed: A. Smith *Position:* Fire Safety Manager *Date:* 01/06/2024

Appendix 4: Example PEEP

This is an example only. PEEPs should be personalized to the individual's needs.

Name: John Doe **Position:** Student **Location:** Room 101, First Floor, Main Building

Awareness of Procedure John is fully aware of the evacuation procedures and his PEEP. He has been involved in its development.

Alarm System John can hear the fire alarm. No adaptations are necessary.

Assistance Required John uses a wheelchair. He will require assistance to evacuate safely.

Assistants The following people have been designated to assist John:

1. Jane Smith (Tutor) - primary assistant
2. Bob Johnson (Learning Support Assistant) - backup assistant

Evacuation Procedure When the alarm sounds, the following procedure will be followed:

1. Jane (or Bob) will go to John's location.
2. Jane (or Bob) will assist John into the evacuation chair, located next to Room 101.
3. Jane (or Bob) will wheel John to the nearest safe refuge point (located at the top of the south stairwell).
4. Jane (or Bob) will stay with John at the refuge point.
5. The fire marshal will be informed that John is at the refuge point.
6. If it is safe to do so, Jane (or Bob) will assist John down the stairs and out of the building to the assembly point.
7. If it is not safe, John will remain at the refuge point with Jane (or Bob) until the fire service arrives.

Equipment Provided

- Evacuation chair, located next to Room 101.
- Refuge point with intercom, located at the top of the south stairwell.

Safe Route(s) The primary route is via the south stairwell to the rear exit. If this is blocked, the secondary route is via the north stairwell to the front exit.

Signed: John Doe (Student) *Signed:* Jane Smith (Tutor) *Date:* 01/09/2023

Glossary of Terms

- **ASET**: Available Safe Egress Time
- **BS**: British Standard
- **DSEAR**: Dangerous Substances and Explosive Atmospheres Regulations 2002
- **EAV**: Exposure Action Value
- **ELV**: Exposure Limit Value
- **FIA**: Fire Industry Association
- **HAVS**: Hand-Arm Vibration Syndrome
- **HSE**: Health and Safety Executive
- **ICP**: Insulated Core Panel
- **MHSW**: Management of Health and Safety at Work Regulations 1999
- **PADS**: Portable Appliance Discharge Station
- **PAT**: Portable Appliance Testing
- **PEEP**: Personal Emergency Evacuation Plan
- **PPE**: Personal Protective Equipment
- **RIDDOR**: Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 2013
- **RSET**: Required Safe Egress Time
- **RRO**: Regulatory Reform (Fire Safety) Order 2005
- **TFS**: Total Flooding System

Index

This concludes a College's Site Fire Safety. Remember, fire safety is a continuous process of risk assessment, prevention, protection, and preparedness. By understanding the principles covered in this book and applying them diligently, you can play a vital role in safeguarding lives and property from the devastation of fire.